



BSI Standards Publication

Eurocode 3 — Design of steel structures

Part 1-1: General rules and rules for buildings

National foreword

This British Standard is the UK implementation of EN 1993-1-1:2022. It supersedes [BS EN 1993-1-1:2005+A1:2014](#), which will be withdrawn on 30 March 2028.

The UK participation in its preparation was entrusted to Technical Committee CB/203, Design & execution of steel structures.

A list of organizations represented on this committee can be obtained on request to its committee manager.

National choice is allowed in this standard where explicitly stated within notes. The National Annex to this standard contains the national choices to be used for buildings and civil engineering works constructed in the UK.

The first generation of EN Eurocodes was published between 2002 and 2007, with conflicting British Standards withdrawn in 2010. This document forms part of the second generation of EN Eurocodes.

The second generation of EN Eurocodes is expected to be published between 2023 and 2026. These documents are being published as soon as they are available. This is being done to enable users to prepare for the transition from the first generation to second generation of EN Eurocodes.

UK adoptions of the first generation of EN Eurocodes will be withdrawn by BSI on 30 March 2028. Until that date, the first generation documents should be considered as the applicable standards for buildings and civil engineering works constructed in the UK unless otherwise specified by the relevant authority or in the specification for a particular project.

This standard is intended to be used with its National Annex and other referenced documents, including other second generation Eurocodes, as an interdependent suite of documents.

While the use of provisions in this standard in conjunction with first generation Eurocodes is not precluded, it should be undertaken with care and should only be done when users are satisfied that it will not result in a lower level of reliability than the minimum level set in the first generation Eurocodes and associated UK National Annexes.

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Stahlbauten - Teil 1-1: Allgemeine Bemessungsregeln
und Regeln für den Hochbau

This European Standard was approved by CEN on 24 July 2022.

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EUROPEAN COMMITTEE FOR STANDARDIZATION
COMITÉ EUROPÉEN DE NORMALISATION
EUROPÄISCHES KOMITEE FÜR NORMUNG

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European foreword

This document (EN 1993-1-1:2022) has been prepared by Technical Committee CEN/TC 250 “Structural Codes”, the secretariat of which is held by BSI. CEN/TC 250 is responsible for all Structural Eurocodes and has been assigned responsibility for structural and geotechnical design matters by CEN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by September 2027 and conflicting national standards shall be withdrawn at the latest by March 2028.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.

This document supersedes EN 1993-1-1:2005 and its amendments and corrigenda.

The first generation of EN Eurocodes was published between 2002 and 2007. This document forms part of the second generation of the Eurocodes, which have been prepared under Mandate M/515 issued to CEN by the European Commission and the European Free Trade Association.

The Eurocodes have been drafted to be used in conjunction with relevant execution, material, product and test standards, and to identify requirements for execution, materials, products and testing that are relied upon by the Eurocodes.

The Eurocodes recognize the responsibility of each Member State and have safeguarded their right to determine values related to regulatory safety matters at national level through the use of National Annexes.

The main changes compared to the previous edition are listed below:

- the scope of EN 1993-1-1 was extended to steel grades up to S700;
- the scope was extended to the design of elliptical hollow sections;
- the methods for the structural analysis were clarified and summarized in a flowchart;
- a new method for the design of semi-compact sections (Class 3) has been implemented;
- the effects of torsion on the resistance of cross-sections and members have been improved;
- a new method for the verification of beams to lateral torsional buckling has been introduced;
- the simplified method for lateral torsional buckling has been fully revised;
- the design of uniform members with mono-symmetric cross-sections was explicitly covered;
- a simplified design approach has been introduced for fatigue;
- an informative annex provides statistical data of material and dimensional properties as used for the calibration of the partial factors.

Any feedback and questions on this document should be directed to the users’ national standards body. A complete listing of these bodies can be found on the CEN website.

According to the CEN-CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Republic of North Macedonia, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Türkiye and the United Kingdom.

0 Introduction

0.1 Introduction to the Eurocodes

The Structural Eurocodes comprise the following standards generally consisting of a number of Parts:

- EN 1990 Eurocode: Basis of structural and geotechnical design
- EN 1991 Eurocode 1: Actions on structures
- EN 1992 Eurocode 2: Design of concrete structures
- EN 1993 Eurocode 3: Design of steel structures
- EN 1994 Eurocode 4: Design of composite steel and concrete structures
- EN 1995 Eurocode 5: Design of timber structures
- EN 1996 Eurocode 6: Design of masonry structures
- EN 1997 Eurocode 7: Geotechnical design
- EN 1998 Eurocode 8: Design of structures for earthquake resistance
- EN 1999 Eurocode 9: Design of aluminium structures
- New parts are under development, e.g. Eurocode for design of structural glass

0.2 Introduction to EN 1993 (all parts)

EN 1993 (all parts) applies to the design of buildings and civil engineering works in steel. It complies with the principles and requirements for the safety and serviceability of structures, the basis of their design and verification that are given in EN 1990 – Basis of structural and geotechnical design.

EN 1993 (all parts) is concerned only with requirements for resistance, serviceability, durability and fire resistance of steel structures. Other requirements, e.g. concerning thermal or sound insulation, are not covered.

EN 1993 is subdivided in various parts:

EN 1993-1, *Design of steel structures — Part 1: General rules and rules for buildings*;

EN 1993-2, *Design of steel structures — Part 2: Steel bridges*;

EN 1993-3, *Design of steel structures — Part 3: Towers, masts and chimneys*;

EN 1993-4, *Design of steel structures — Part 4: Silos and tanks*;

EN 1993-5, *Design of steel structures — Part 5: Piling*;

EN 1993-6, *Design of steel structures — Part 6: Crane supporting structures*;

EN 1993-7, *Design of steel structures — Part 7: Design of sandwich panels* (under preparation).

EN 1993-1 in itself does not exist as a physical document, but comprises the following 14 separate parts, the basic part being EN 1993-1-1:

EN 1993-1-1, *Design of steel structures — Part 1-1: General rules and rules for buildings*;

EN 1993-1-2, *Design of steel structures — Part 1-2: Structural fire design*;

EN 1993-1-3, *Design of steel structures — Part 1-3: Cold-formed members and sheeting*;

NOTE Cold-formed hollow sections supplied according to EN 10219 (all parts) are covered in EN 1993-1-1.

EN 1993-1-4, *Design of steel structures — Part 1-4: Stainless steel structures*;

EN 1993-1-5, *Design of steel structures — Part 1-5: Plated structural elements*;

EN 1993-1-6, *Design of steel structures — Part 1-6: Strength and stability of shell structures*;

EN 1993-1-7, *Design of steel structures — Part 1-7: Plate assemblies with elements under transverse loads*;

EN 1993-1-8, *Design of steel structures — Part 1-8: Design of joints*;

EN 1993-1-9, *Design of steel structures — Part 1-9: Fatigue*;

EN 1993-1-10, *Design of steel structures — Part 1-10: Material toughness and through-thickness properties*;

EN 1993-1-11, *Design of steel structures — Part 1-11: Design of structures with tension components*;

EN 1993-1-12, *Design of steel structures — Part 1-12: Additional rules for steel grades up to S960*;

EN 1993-1-13, *Design of steel structures — Part 1-13: Rules for beams with large web openings*;

EN 1993-1-14, *Design of steel structures — Part 1-14: Design assisted by finite element analysis* (under preparation).

All parts numbered EN 1993-1-2 to EN 1993-1-14 treat general topics that are independent from the structural type such as structural fire design, cold-formed members and sheeting, stainless steels, plated structural elements, etc.

All parts numbered EN 1993-2 to EN 1993-7 treat topics relevant for a specific structural type such as steel bridges, towers, masts and chimneys, silos and tanks, piling, crane supporting structures, etc. EN 1993-2 to EN 1993-7 refer to the generic rules in EN 1993-1 and supplement, modify or supersede them, where relevant.

0.3 Introduction to EN 1993-1-1

EN 1993-1-1 gives general design rules for steel structures. It also includes supplementary design rules for steel buildings. The focus in EN 1993-1-1 is on design methods and design rules for individual members (beams, columns and beam-columns) and skeletal structures (frames) regarding resistance and stability.

0.4 Verbal forms used in the Eurocodes

The verb “shall” expresses a requirement strictly to be followed and from which no deviation is permitted in order to comply with the Eurocodes.

The verb “should” expresses a highly recommended choice or course of action. Subject to national regulation and/or any relevant contractual provisions, alternative approaches could be used/adopted where technically justified.

The verb “may” expresses a course of action permissible within the limits of the Eurocodes.

The verb “can” expresses possibility and capability; it is used for statements of fact and clarification of concepts.

0.5 National Annex for EN 1993-1-1

National choice is allowed in this standard where explicitly stated within notes. National choice includes the selection of values for Nationally Determined Parameters (NDPs).

The national standard implementing EN 1993-1-1 can have a National Annex containing all national choices to be used for the design of buildings and civil engineering works to be constructed in the relevant country.

When no national choice is given, the default choice given in this standard is to be used.

When no national choice is made and no default is given in this standard, the choice can be specified by a relevant authority or, where not specified, agreed for a specific project by appropriate parties.

National choice is allowed in EN 1993-1-1 through the following clauses:

4.4.3 (2)	5.1 (1)	5.2.1 (1)	5.2.2 (1)
7.2.1 (4)	7.2.2 (9)	7.3.3.1 (1)	7.3.3.2 (1)
7.4.1(3)	8.1 (1)	8.2.8 (3)	8.3.2.3 (1)
8.3.3 (2)	8.3.4 (1)	9.2 (2)B	A.4 (2)
A.4 (3)	A.4 (5)		

National choice is allowed in EN 1993-1-1 on the application of the following informative annexes:

Annex E

The National Annex can contain, directly or by reference, non-contradictory complementary information for ease of implementation, provided it does not alter any provisions of the Eurocodes.

1 Scope

1.1 Scope of EN 1993-1-1

(1) EN 1993-1-1 gives basic design rules for steel structures.

(2) It also gives supplementary provisions for the structural design of steel buildings. These supplementary provisions are indicated by the letter “B” after the paragraph number, thus ()B.

1.2 Assumptions

(1) The assumptions of EN 1990 apply to EN 1993-1-1.

(2) EN 1993 is intended to be used in conjunction with EN 1990, EN 1991 (all parts), the parts of EN 1992 to EN 1999 where steel structures or steel components are referred to within those documents, EN 1090-2, EN 1090-4 and ENs, EADs and ETAs for construction products relevant to steel structures.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

NOTE See the Bibliography for a list of other documents cited that are not normative references, including those referenced as recommendations (i.e. through ‘should’ clauses) and permissions (i.e. through ‘may’ clauses).

EN 1090-2, *Execution of steel structures and aluminium structures — Part 2: Technical requirements for steel structures*

EN 1090-4, *Execution of steel structures and aluminium structures — Part 4: Technical requirements for cold-formed structural steel elements and cold-formed structures for roof, ceiling, floor and wall applications*

EN 1990:—¹, *Eurocode — Basis of structural and geotechnical design*

EN 1991 (all parts), *Eurocode 1 — Actions on structures*

EN 1993-1 (all parts), *Eurocode 3: Design of steel structures*

3 Terms, definitions and symbols

For the purposes of this document, the terms and definitions given in EN 1990 and the following terms, definitions and symbols apply.

3.1 Terms and definitions

3.1.1

frame

whole or a portion of a structure, comprising an assembly of directly connected structural elements, designed to act together to resist load

Note 1 to entry: This term refers to both moment-resisting frames and triangulated frames; it covers both plane frames and three-dimensional frames.

¹ Under preparation. Stage at the time of publication: FprEN 1990:2022.

3.1.2

sub-frame

frame that forms part of a larger frame, but is treated as an isolated frame in a structural analysis

3.1.3

semi-continuous framing

framing in which the structural properties of the members and joints need explicit consideration in the global analysis

3.1.4

continuous framing

framing in which only the structural properties of the members need to be considered in the global analysis

3.1.5

simple framing

framing in which the joints are not designed to resist moments

3.1.6

system length

distance in a given plane between two adjacent points at which a member is braced against lateral displacement in this plane, or between one such point and the end of the member

3.1.7

buckling length

system length of an otherwise similar member with pinned ends, which has the same critical buckling load as a given member or segment of member

3.1.8

equivalent member

simply supported single span member of uniform cross-section with constant compressive axial force used for buckling verification

Note 1 to entry: Its length, cross-section and axial force are equal to the appropriate buckling length, cross-section and axial force at the investigated position in the structure.

3.1.9

shear lag effect

non-uniform stress distribution in wide flanges due to shear deformation

3.1.10

capacity design

design method for achieving the plastic deformation capacity of a member by providing additional strength in its connections and in other parts connected to it

3.1.11

uniform built-up member

built-up member made of parallel chords with nominally constant cross-section along their whole length, connected by regularly spaced lacings or battens

3.1.12

uniform member

member with a nominally constant cross-section along its whole length

3.1.13
fork end condition

support condition where the beam is supported vertically and/or laterally but free to rotate on plan

Note 1 to entry: More detailed explanations are given in prCEN/TR 1993-1-103², *Eurocode 3 – Design of steel structures – Part 1-103: Elastic critical buckling of members*.

3.2 Symbols and abbreviations

3.2.1 Latin upper-case symbols

A	cross-sectional area
A_c	area of the equivalent compression flange
A_{ch}	cross-sectional area of one chord of a built-up column
A_{dia}	cross-sectional area of one diagonal of a built-up column
A_{eff}	effective area of a cross-section
A_f	area of one flange
A_i	cross-sectional area for the calculation of the characteristic resistance to an axial force
A_{net}	net area of a cross-section
A_p	cross-sectional area of one post (or transverse element) of a built-up column
A_t	area of the tension flange
$A_{t,net}$	net area of the tension flange
A_v	shear area
A_w	area of a web
A_0	original cross-sectional area
B_{Ed}	design value of the bimoment
B_{Rd}	design value of the bimoment resistance
B_{Rk}	characteristic value of the bimoment resistance
C_D	rotational stiffness provided by stabilizing continuum and connections
$C_{D,A}$	rotational stiffness of the connection between the beam and the stabilizing continuum
$C_{D,B}$	rotational stiffness deduced from an analysis of the distortional deformations of the beam cross-sections
$C_{D,C}$	rotational stiffness provided by the stabilizing continuum to the beam assuming a stiff connection to the member
C_{my}, C_{mz}, C_{mLT}	equivalent uniform moment factors
E	modulus of elasticity

² Under preparation.

$EI \left \eta''_{cr,m} \right $	absolute value of the bending moment due to $\eta_{cr,m}$ at the critical cross-section m
$F_{cr,ns}$	minimum elastic critical flexural buckling load for either the in-plane or out-of-plane member (non-sway) buckling mode
$F_{cr,sw}$	minimum elastic critical in-plane flexural buckling load for a global (sway) buckling mode
F_d	design value of the loading on the structure
$F_{H,d}$	design value of the total horizontal load
$F_{V,d}$	design value of the total vertical load
$F_{z,Ed}$	design value of transverse force
$F_{z,Rd}$	design value of the resistance to transverse force
G	shear modulus
G_k	characteristic value of a permanent action
H	height of the structure
H_f	fictitious horizontal force
H_{st}	storey height
I	moment of inertia (second moment of area)
I_b	in-plane moment of inertia of a batten
I_{ch}	in-plane moment of inertia of a chord
I_{eff}	effective moment of inertia of a built-up member
I_T	torsion constant
I_w	warping constant
I_y, I_z	moment of inertia about y-y axis and z-z axis, respectively
$I_{z,fl}$	moment of inertia about z-z axis of one flange
I_1	equivalent moment of inertia of a battened built-up member
K_{st}	lateral rigidity of a storey
K_v	factor for considering the type of verification in evaluating torsional restraints
K_θ	factor for considering the moment distribution in evaluating torsional restraints
L	length (member length, span length, etc.)
L_c	length between two consecutive lateral restraints
L_{ch}	buckling length of chord in a built-up member
L_{cr}	buckling length
L_d	length of a diagonal in a built-up member
L_{st}	stable length of segment

$M_{b,Rd}$	design value of the buckling resistance of a member in bending
$M_{B,Rd}$	design value of the reduced cross-section resistance to bending moment making allowance for the presence of bimoment
$M_{B,V,Rd}$	design value of the reduced cross-section resistance to bending moment making allowance for the presence of shear force and bimoment
M_{cr}	elastic critical moment for lateral torsional buckling
$M_{c,Rd}$	design value of the bending moment resistance about one principal axis of a cross-section
M_{Ed}	design value of bending moment
M_{Ed}^I	design value of the maximum first order moment in the built-up member
$M_{el,Rd}$	design value of the elastic bending moment resistance
$M_{ep,Rd}$	design value of the elasto-plastic bending moment resistance
M_h	hogging moment at member ends
$M_{N,ep,Rd}$	design value of the elasto-plastic bending moment resistance making allowance for the presence of axial force
$M_{N,Rd}$	sign value of the reduced bending moment resistance making allowance for the presence of axial force
$M_{pl,Rd}$	design value of the plastic bending moment resistance
M_{Rk}	characteristic value of the bending moment resistance
$M_{Rk,m}$	characteristic value of the bending moment resistance of the critical cross-section m
M_s	sagging moment at mid-span of a member
$M_{V,Rd}$	design value of the reduced plastic bending moment resistance making allowance for the presence of shear force
$M_{y,Ed}$	design value of the bending moment about y-y axis
$M_{y,Rd}$	design value of the bending moment resistance about y-y axis
$M_{y,Rk}$	characteristic value of the bending moment resistance about y-y axis
$M_{z,Ed}$	design value of the bending moment about z-z axis
$M_{z,Rd}$	design value of the bending moment resistance about z-z axis
$M_{z,Rk}$	characteristic value of the bending moment resistance about z-z axis
M_0	free bending moment at mid-span (determined on an equivalent member with simply supported end conditions)
N	number of stress cycles during the design service life
$N_{b,Rd}$	design value of the buckling resistance of a member in compression
$N_{c,Rd}$	design value of the resistance to axial force of the cross-section for uniform compression
$N_{ch,Ed}$	design value of the axial force in a chord, in the middle of a built-up member

N_{cr}	elastic critical axial force for the relevant buckling mode based on the gross cross-sectional properties
$N_{cr,c,z}$	elastic critical axial force for flexural buckling of the equivalent compression flange
$N_{cr,m}$	elastic critical axial force at the cross-section m
$N_{cr,T}$	elastic critical axial force for torsional buckling
$N_{cr,TF}$	elastic critical axial force for torsional-flexural buckling
$N_{cr,V}$	effective critical axial force of a built-up member including the effect of its shear stiffness
$N_{cr,y}, N_{cr,z}$	elastic critical axial force for flexural buckling about y-y axis and z-z axis, respectively
N_{Ed}	design value of the compressive axial force
$N_{Ed,m}$	design value of the compressive axial force at the cross-section m
$N_{f,Ed}$	design axial force in the compressed flange of a stabilized member at a plastic hinge location
$N_{net,Rd}$	design value of the plastic resistance to axial force of the net cross-section
$N_{pl,Rd}$	design value of the plastic resistance to axial force of the gross cross-section
N_{Rd}	design value of the resistance to axial force
N_{Rk}	characteristic value of the resistance to axial force
$N_{Rk,m}$	characteristic value of the resistance to axial force at the cross-section m
$N_{t,Rd}$	design value of the resistance to tensile axial force
$N_{u,Rd}$	design value of the ultimate resistance to axial force of the net cross-section at holes for fasteners
P_k	characteristic value of a prestressing action imposed during erection
Q_m	local force applied at each stabilized member at the plastic hinge locations
R_{eH}	yield strength taken from the relevant product standard
R_d	design value of resistance
R_k	characteristic value of resistance
R_m	ultimate strength taken from the relevant product standard
S	static moment (first moment of area)
S_{Vb}	shear stiffness of a built-up member from the lacing or battened panel
S_v	shear stiffness provided by sheeting
T_{Ed}	design value of total internal torsional moment
$T_{t,Ed}$	design value of internal St. Venant torsional moment
T_{Rd}	design value of the resistance to torsional moment
$T_{w,Ed}$	design value of internal warping torsional moment

$V_{c,Rd}$	design value of the resistance to shear force
V_{Ed}	design value of the shear force
V_{Ed}^I	design value of the first order shear force in a built-up member
$V_{pl,Rd}$	design value of the plastic resistance to shear force
$V_{pl,T,Rd}$	design value of the reduced plastic shear resistance making allowance for the presence of a torsional moment
$W_{pl,B}$	plastic section modulus for the plastic resistance to bimoment
W_{eff}	effective section modulus
$W_{eff,y,min}$	minimum effective section modulus for bending about y-y axis
W_{el}	elastic section modulus
$W_{el,min}$	minimum elastic section modulus
$W_{el,y}, W_{el,z}$	elastic section modulus for bending about y-y axis and z-z axis, respectively
$W_{el,y,min}$	minimum elastic section modulus for bending about y-y axis
W_{ep}	elasto-plastic section modulus for Class 3 section
$W_{ep,y}, W_{ep,z}$	elasto-plastic section modulus for bending about y-y axis and z-z axis, respectively
W_{pl}	plastic section modulus
$W_{pl,y}, W_{pl,z}$	plastic section modulus for bending about y-y axis and z-z axis, respectively
X_d	design value of a basic variable
X_k	characteristic value of a basic variable
$X_{k,i}$	characteristic value of material property
$X_{k,th}$	target characteristic value of a basic variable
X_m	mean value of a basic variable
X_n	nominal value of a basic variable
X_{rep}	representative value of material properties
$X_{5\%}$	upper reference value
$X_{0,12\%}$	lower reference value
Z_{Ed}	required design Z-value resulting from the magnitude of strains from restrained metal shrinkage under the weld beads
Z_{Rd}	available design Z-value

3.2.2 Latin lower-case symbols

a	ratio of web area to gross section area
a_d	design value of geometrical properties
a_f	ratio of section area of the smaller sides to gross section area for rectangular hollow sections
a_w	ratio of section area of the larger sides to gross section area for rectangular hollow sections
b	width of a cross-section
c	width or depth of a part of a cross-section
d	external diameter of a circular hollow section
d_{ch}	distance along a chord of a built-up member between two consecutive lacings or battens
d_e	equivalent diameter for circular and elliptical hollow sections
d_w	web depth between fillets or weld toes
d_z	distance of intermediate lateral supports from the shear centre of the cross-section
d_0	diameter of bolt hole
e_N	shift of the centroid of the effective area A_{eff} relative to the centroid of the gross cross-section
$e_{N,y}$	shift of the centroid of the effective area A_{eff} relative to the centroid of the gross cross-section, resulting in a bending moment about y-y axis
$e_{N,z}$	shift of the centroid of the effective area A_{eff} relative to the centroid of the gross cross-section, resulting in a bending moment about z-z axis
e_0	equivalent bow imperfection
$e_{0,LT}$	equivalent bow imperfection for lateral torsional buckling
$e_{0,m}$	equivalent bow imperfection at the critical cross-section m
f_m	factor that accounts for the effect of the bending moment distribution between discrete lateral restraints
f_y	yield strength
$f_{y,red}$	reduced yield strength
f_u	ultimate tensile strength
h	depth of a cross-section
h_w	depth of a web
h_0	distance between centroids of chords of a built-up column
i	radius of gyration about the relevant axis, determined using the properties of the gross cross-section
i_{min}	minimum radius of gyration of single angles
i_y, i_z	radius of gyration for bending about y-y axis and z-z axis respectively
i_p	polar radius of gyration

i_0	minimum radius of gyration of the cross-section of a star-battened built-up member
k	correction factor for ultimate resistance of the net cross-section
k_c	correction factor for moment distribution
k_{ij}	interaction factors (generic symbol for k_{yy} , k_{yz} , k_{zy} and k_{zz})
k_{sw}	amplification factor for taking into account second order sway effects due to vertical loads
k_w	interaction factor for uniform members in bending, axial compression and torsion covering the effect of bi-moment
k_{yy} , k_{yz}	interaction factors for uniform members in bending and axial compression
k_{zy} , k_{zz}	
k_{zw}	interaction factor for uniform members in bending, axial compression and torsion covering the effect of bending moment about z-z
k_α	interaction factor for uniform members in bending, axial compression and torsion covering the effect of bending moment about y-y
k_σ	plate buckling factor
k_0	limiting value for $\alpha_{cr,ns}$
ℓ_y	effective length for the resistance to transverse force
m	number of columns in a row or members to be restrained
n	ratio of design axial force to design plastic resistance to axial force of the gross cross-section
n_p	number of planes of lacing or batten
n_y	ratio of design axial force to design value of the buckling resistance of a member in compression for buckling about y-y axis
n_z	ratio of design axial force to design value of the buckling resistance of a member in compression for buckling about z-z axis
n_1	number of holes extending in any diagonal or zig-zag line progressively across the member or part of the member
n_2	number of holes in a section where the holes are not staggered.
p_2	spacing of the centres of the same two holes measured perpendicular to the member axis
q_d	equivalent design load per unit length
r	radius of root fillet
r_e	external corner radius of a rectangular hollow section
r_1	radius of root fillet
r_2	toe radius
s	staggered pitch, the spacing of the centres of two consecutive holes measured parallel to the member axis
s_s	length of stiff bearing
t	thickness

t_{cor}	core thickness (i.e. nominal thickness minus the thickness of the zinc and other metallic coatings)
t_{d}	design thickness
t_{f}	flange thickness
t_{mc}	thickness of metallic coating
t_{nom}	nominal sheet thickness
tol	minus tolerance of the thickness in % from the relevant EN or EN ISO, EAD or ETA
t_{w}	web thickness

3.2.3 Greek upper-case symbols

ΔA_{net}	section area to be deducted for a section to take into account the holes
Δ_{f}	horizontal displacement at the top of the storey due to H_{f} relative to the bottom of the storey
ΔM_{Ed}	additional moment from shift of the centroid of the effective area A_{eff} relative to the centroid of the gross cross-section
$\Delta M_{\text{y,Ed}}$	additional moment about y-y axis due to the shift of the centroid
$\Delta M_{\text{z,Ed}}$	additional moment about z-z axis due to the shift of the centroid
Φ	value to determine the reduction factor χ for flexural buckling
Φ_{LT}	value to determine the reduction factor χ_{LT} for lateral torsional buckling

3.2.4 Greek lower-case symbols

α	imperfection factor
α_{c}	portion of a part of a cross-section in compression
α_{cr}	factor by which the design value of the loading would have to be increased to cause elastic instability
$\alpha_{\text{cr,ns}}$	factor by which the design value of the loading would have to be increased to cause elastic instability in the in-plane or out-of-plane member (non-sway) buckling mode
$\alpha_{\text{cr,op}}$	minimum amplifier for the design value of the in-plane loading to reach the elastic critical buckling load with regard to lateral or lateral torsional buckling
$\alpha_{\text{cr,sw}}$	factor by which the design value of the loading would have to be increased to cause elastic instability in a global, in-plane (sway) mode
α_{h}	factor describing the moment diagram; h = hogging
α_{H}	reduction factor for height H applicable to columns
α_{LT}	imperfection factor for lateral torsional buckling
α_{m}	reduction factor for the number of columns in a row or members to be restrained
α_{R}	resistance-side weighting factor for calibration of partial factors
α_{s}	factor describing the moment diagram; s = sagging

α_T	coefficient of linear thermal expansion
α_{TF}	imperfection factor for torsional-flexural buckling
$\alpha_{ult,k}$	minimum load amplifier of the design loads to reach the characteristic resistance of the most critical cross-section
α_y	parameter introducing the effect of biaxial bending
α_z	parameter introducing the effect of biaxial bending
$\alpha_{\eta,m}$	imperfection factor α for the relevant buckling curve at the critical cross-section m
β	reference relative bow imperfection
β_c	modification factor of equivalent compression flange
$\beta_{ep,y}, \beta_{ep,z}$	interpolation factor for bending about y-y axis and z-z axis, respectively
β_{LT}	reference relative bow imperfection for lateral torsional buckling
γ_{Mi}	partial factor (generic symbol)
γ_{M0}	partial factor for resistance of cross-sections
γ_{M1}	partial factor for resistance of members to instability assessed by member checks
γ_{M2}	partial factor for resistance of cross-sections in tension to fracture
γ_R	partial factor for the particular resistance
δ_q	in-plane deflection of a bracing system
ε	material parameter depending on f_y
ε_t	tensile strain
η	factor for shear area
$\eta_{cr}(x)$	shape of the relevant (first or higher) elastic critical buckling mode
η_i	conversion factor
$\eta_{init}(x)$	shape of the equivalent geometrical imperfection
θ	angle between axes of chord and lacing
$\bar{\lambda}$	relative slenderness
$\bar{\lambda}_{c,z}$	relative slenderness of the equivalent compression flange
$\bar{\lambda}_{c,z,mod}$	modified relative slenderness of the equivalent compression flange
$\bar{\lambda}_m$	relative slenderness related to the section m
$\bar{\lambda}_{LT}$	relative slenderness for lateral torsional buckling
$\bar{\lambda}_{LT,lim}$	limit of relative slenderness for susceptibility to lateral torsional buckling
$\bar{\lambda}_{LT,0}$	plateau length of the lateral torsional buckling curves

$\bar{\lambda}_{op}$	global relative slenderness of a structural component for out-of-plane buckling
$\bar{\lambda}_T$	relative slenderness for torsional buckling
$\bar{\lambda}_{TF}$	relative slenderness for torsional-flexural buckling
λ_1	reference slenderness to determine the relative slenderness
μ	efficiency factor in built-up members
ν	Poisson's ratio in elastic range
ρ	reduction factor to determine the design value of the reduced plastic bending moment resistance making allowance for the presence of shear forces
σ	stress
$\sigma_{com,Ed}$	maximum design value of compressive longitudinal stress
$\sigma_{eq,Ed}$	design value of equivalent longitudinal stress
$\sigma_{w,Ed}$	design value of the longitudinal stress due to bimoment B_{Ed}
$\sigma_{x,Ed}$	design value of the longitudinal stress
$\sigma_{z,Ed}$	design value of the transverse stress
τ_{Ed}	design value of the shear stress
$\tau_{t,Ed}$	design value of the shear stress due to St. Venant torsion
$\tau_{w,Ed}$	design value of the shear stress due to warping torsion
ϕ	initial sway imperfection
ϕ_0	basic value for initial sway imperfection
χ	reduction factor for relevant buckling mode
χ_{LT}	reduction factor for lateral torsional buckling
χ_{op}	reduction factor for the relative slenderness $\bar{\lambda}_{op}$
χ_T	reduction factor due to torsional buckling
χ_{TF}	reduction factor due to torsional-flexural buckling
χ_y	reduction factor due to flexural buckling about y-y axis
χ_z	reduction factor due to flexural buckling about z-z axis
ψ	ratio of end moments in a segment of beam, or stress ratio

3.3 Symbols for member axes

(1) The convention for member axes is:

x-x – along the member

y-y – axis of the cross-section

z-z – axis of the cross-section

(2) For steel members, the conventions used for cross-section axes are:

— generally:

y-y – cross-section axis parallel to the flanges

z-z – cross-section axis perpendicular to the flanges

— for angle sections:

y-y – cross-section axis parallel to the smaller leg

z-z – cross-section axis perpendicular to the smaller leg

— where necessary:

u-u – major principal axis (where this does not coincide with the y-y axis)

v-v – minor principal axis (where this does not coincide with the z-z axis)

(3) The symbols used for dimensions and axes of rolled steel sections are indicated in Figure 3.1.

(4) The convention for subscripts indicating the axes for moments is: “Use the axis about which the moment acts.”

NOTE All rules in EN 1993 relate to principal axis properties, which are generally defined by the axes y-y and z-z but for sections such as angles are defined by the axes u-u and v-v.

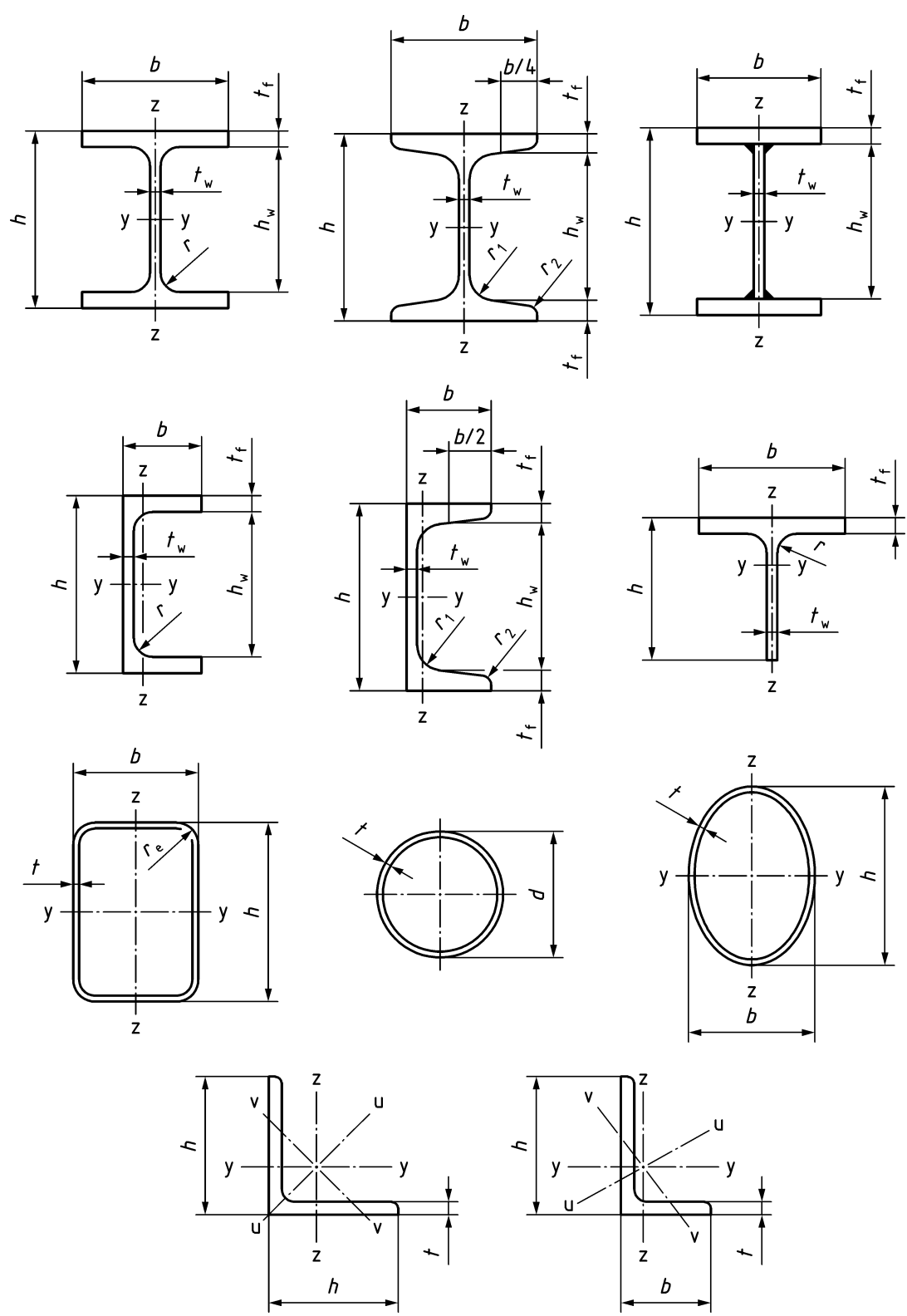


Figure 3.1 — Dimensions and axes of sections

4 Basis of design

4.1 General rules

4.1.1 Basic requirements

(1) The design of steel structures shall be in accordance with the general rules given in EN 1990 and EN 1991 (all parts) and the specific design provisions for steel structures given in the relevant parts of EN 1993-1 (all parts).

(2) Steel structures designed according to EN 1993 shall be executed according to EN 1090-2 and/or EN 1090-4 with construction materials and products used as specified in the relevant parts of EN 1993, or in the relevant material and product specifications.

4.1.2 Structural reliability

(1) To obtain the level of structural reliability of the completed works required according to EN 1990, an appropriate execution class shall be selected in accordance with Annex A for steel structures executed to EN 1090-2 and EN 1090-4.

(2) If different levels of reliability are required, they should be achieved by an appropriate choice of quality management measures in design and execution according to EN 1990, EN 1090-2 and EN 1090-4.

4.1.3 Robustness

(1) The provisions on robustness given in EN 1990:—, 4.4 should be followed.

(2) The general arrangement of the structure and the interaction and connection of its various parts should be such as to give appropriate robustness during construction and use.

4.1.4 Design service life for buildings

(1)B For the specification of the intended design service life of a permanent building, the provisions given in EN 1990 should be followed.

(2)B For structural elements that cannot be designed for the total design life of the building (e.g. components that need to be replaceable such as bearings), see Clause 6(5).

4.1.5 Durability

(1) Depending upon the type of action affecting the design service life (see EN 1990), steel structures shall be:

- designed for the effects of corrosion;
- designed for the effects of fatigue in accordance with Clause 10 and EN 1993-1-9;
- designed against brittle fracture in accordance with EN 1993-1-10;
- designed for wearing effects;
- designed for accidental actions (see EN 1991-1-7);
- inspected and maintained.

(2) Corrosion protection may be achieved by:

- suitable surface protection in accordance with EN ISO 12944 (all parts), EN ISO 1461;
- the use of weathering steel (see EN 10025-5);
- the use of stainless steel in accordance with EN 1993-1-4.

(3) For further consideration of durability, Clause 6 should be followed.

4.2 Principles of limit state design

(1) For steel structures, the ultimate limit state and serviceability limit state shall be considered for all aspects of the structure.

(2) For steel structures, all relevant design situations shall be considered, including relevant phases in the erection stage.

NOTE For the selection of design situations, see EN 1990.

4.3 Basic variables

4.3.1 Actions and environmental influences

(1) The characteristic values of actions for the design of steel structures, including any regional, climatic and accidental situations, shall be obtained from the relevant Parts of EN 1991.

(2) The actions to be considered during the erection stage may be obtained from EN 1991-1-6.

(3) Where the effects of predicted absolute and differential settlements need to be considered, estimates of imposed deformations should be taken into account.

(4) The effects of uneven settlements, imposed deformations and forms of prestressing imposed during erection should be taken into account using the nominal value of the prestressing action P_k as a permanent action and grouping them with other permanent actions G_k to form a single action ($G_k + P_k$).

NOTE For more information on load combinations, see EN 1990 and for prestress, see EN 1993-1-11.

(5) Fatigue actions not defined in EN 1991 (all parts) should be determined in accordance with EN 1990:—, Clause 8 and the fatigue action effects in accordance with EN 1993-1-9.

4.3.2 Material, geometrical and product properties

(1) The material properties for steels and other construction products and the geometrical properties to be used for design should be those specified in the relevant EN and EN ISO product standards, or according to a transparent and reproducible assessment that complies with all the requirements of a European Assessment Document (EAD) or European Technical Assessment (ETA), unless otherwise indicated, see Clause 5.

(2) The nominal values of material properties should be adopted as characteristic values in design calculations.

(3) For structural analysis and design, the nominal values of dimensions may be used.

NOTE See EN 1990:—, 6.3(1).

4.4 Verification by the partial factor method

4.4.1 Design values of actions

(1) For the design of steel structures, combination of actions and partial factors of actions shall be derived from EN 1990:—, Annex A.

4.4.2 Design values of material properties

(1) For the design of steel structures, the design value for a material property shall be obtained by dividing its characteristic value X_k or nominal value X_n by the relevant partial factor for materials, γ_M .

NOTE Usually the partial factor is explicitly incorporated in the design rules.

4.4.3 Design values of geometrical properties

(1) Geometrical properties for cross-sections and systems may be taken from EN and EN ISO product standards, EADs, ETAs or component specifications for the execution to EN 1090 (all parts) and treated as nominal values.

(2) The dimensions of hot rolled profiles may be taken from EN 10365.

NOTE The standard EN 10365 does not give the fillet radius. However, it can be calculated from the section area and the other dimensions, unless the National Annex gives other provisions.

(3) Design values of geometrical imperfections specified in this document are equivalent geometrical imperfections that take into account the effects of:

- geometrical imperfections as governed by geometrical tolerances in product standards or the execution standard;
- structural imperfections due to fabrication and erection;
- residual stresses;
- variation of the yield strength.

(4) The design rules should not be applied to steels with nominal material thicknesses t less than 3 mm, except as stated in (5).

(5) The design rules may be applied to components made of steels with nominal material thicknesses t less than 3 mm but higher than or equal to 1,5 mm provided that the design thickness t_d based on the steel core thickness t_{cor} is used instead of the nominal thickness t as follows:

$$t_d = t_{cor} \quad \text{if } tol \leq 5 \% \quad (4.1)$$

$$t_d = t_{cor} \frac{100 - tol}{95} \quad \text{if } tol > 5 \% \quad (4.2)$$

where

$$t_{cor} = t_{nom} - t_{mc}$$

tol is the minus tolerance of the thickness in % from the relevant EN or EN ISO, EAD or ETA;

t_{nom} is the nominal sheet thickness of the original sheet, inclusive of zinc and other metallic coating and not including organic coating;

t_{mc} is the thickness of the metallic coating.

4.4.4 Tolerances

(1) The dimensional and mass tolerances of rolled steel sections, structural hollow sections and plates should conform with the relevant EN or EN ISO product standard, EAD or ETA, unless more severe tolerances are specified by the relevant authority or agreed for a specific project by the relevant parties.

(2) The fabrication and erection tolerances of structural components should comply with EN 1090-2 and EN 1090-4.

4.4.5 Design resistances

(1) For steel structures, Formula (8.20) of EN 1990:— applies:

$$R_d = \frac{R_k}{\gamma_R} = \frac{R\{X_{rep}; a_d; \Sigma F_{Ed}\}}{\gamma_R} \quad (4.3)$$

where

F_{Ed} denotes the design values of actions, see 4.4.1;

$R\{\dots\}$ denotes the output of the resistance model;

R_k is the characteristic value of the particular resistance;

X_{rep} denotes the representative values of the material properties, defined as ηX_k . Unless defined differently for specific rules, the nominal values of the material properties as defined in 5.2.1 are used;

a_d denotes the design values of the geometrical properties, see 4.4.3;

γ_R is the partial factor for the particular resistance. In this document, γ_R will take the value of the appropriate partial factor γ_{Mi} , see 8.1.

4.5 Design assisted by testing

(1) Values for resistances R_k or R_d may be determined using design assisted by testing methods.

NOTE Guidance on design assisted by testing is given in EN 1990:—, Annex D. More specific provisions can be given in other parts of EN 1993.

5 Materials

5.1 General

(1) For the design of steel structures according to this document the material for fabrication should conform to the grades in Table 5.1 and 5.2 and the following product standards: EN 10025 (all parts), EN 10149 (all parts), EN 10210 (all parts) and EN 10219 (all parts).

NOTE Other steel materials and products can be set by the National Annex.

(2) If other steel materials than those mentioned in (1) are used, their properties (mechanical properties and weldability) shall be known and their mechanical properties shall conform to the conditions given in 5.2.2, 5.2.3 and 5.2.4 when tested in accordance to the relevant EN, ISO or EN ISO testing standards.

5.2 Structural steel

5.2.1 Material properties

(1) The nominal values of the yield strength f_y and the ultimate tensile strength f_u for structural steel should be obtained:

- a) either by adopting the values $f_y = R_{eH}$ and $f_u = R_m$ (as lower bound of the given range) directly from the product standard;

b) or by using the values given in Table 5.1 for steel conforming to EN 10025 (all parts), EN 10210 (all parts) and EN 10219 (all parts), and in Table 5.2 for steel conforming to EN 10149 (all parts), and considering the availability of the material in the thickness range according to the product standard.

NOTE 1 The choice of the approach can be set by the National Annex considering the effects on partial factors and their calibration according to Annex E and EN 1990. In case of option b), the partial factor γ_{M1} needs to be increased.

NOTE 2 Rules for the use of the steels according to Table 5.1 and Table 5.2 can be set by the National Annex.

Table 5.1 — Nominal values of yield strength f_y and ultimate tensile strength f_u for structural steels conforming to the following standards: EN 10025 (all parts), EN 10210 (all parts), and EN 10219 (all parts)

Steel grade ^a	Nominal thickness of the element			
	t mm			
	$t \leq 40$ mm		$40 \text{ mm} < t \leq 80$ mm	
	f_y N/mm ²	f_u N/mm ²	f_y N/mm ²	f_u N/mm ²
S235	235	360	215	360
S275	275	390	245	370
S355	355	490	325	470
S420	420	510	390	490
S460	460	540	410	510
S500	500	580	450	580
S550	550	600	500	600
S600	600	650	550	650
S620	620	700	560	660
S650	650	700	-	-
S690	690	770	630	710
S700	700	750	-	-
^a Principal symbols in EN 10027-1.				

Table 5.2 — Nominal values of yield strength f_y and ultimate tensile strength f_u for hot rolled flat products for cold-forming, according to EN 10149-2 and EN 10149-3

Standards, Grades and quality	Nominal thickness of the element			
	t mm			
	$t \leq 8$ mm		$8 \text{ mm} < t \leq 20$ mm	
	f_y N/mm ²	f_u N/mm ²	f_y N/mm ²	f_u N/mm ²
EN 10149-2^a				
S315MC	315	390	315	390
S355MC	355	430	355	430
S420MC	420	480	420	480
S460MC	460	520	460	520
S500MC	500	550	500 ^b	550 ^b
S550MC	550	600	550 ^b	600 ^b
S600MC	600	650	600 ^b	650 ^b
S650MC	650	700	630 ^b	700 ^b
S700MC	700	750	680 ^b	750 ^b
EN 10149-3^a				
S260NC	260	370	260	370
S315NC	315	430	315	430
S355NC	355	470	355	470
S420NC	420	530	420	530
^a Verification of the impact energy in accordance with EN 10149-1:2013, Clause 11, option 5 should be specified. ^b $t \leq 16$ mm				

5.2.2 Ductility requirements

- (1) A minimum ductility shall be provided and shall be expressed in terms of limiting values for:
- the ratio f_u/f_y ;
 - the elongation at failure on a gauge length of $5,65\sqrt{A_0}$ (where A_0 is the original cross-sectional area).

NOTE The limiting values are given below unless the National Annex gives different limiting values:

a) For plastic global analysis

- $f_u / f_y \geq 1,10$;

- elongation at failure not less than 15 %.
- b) For elastic global analysis
- $f_u / f_y \geq 1,05$;
 - elongation at failure not less than 12 %.

(2) Steels conforming to any of the grades listed in Table 5.1 and Table 5.2 may be assumed to satisfy the minimum ductility requirements for elastic global analysis.

(3) Steels conforming to one of the grades up to and including S460 listed in Table 5.1 and Table 5.2 may be assumed to satisfy the minimum ductility requirements for plastic global analysis.

5.2.3 Fracture toughness

(1) Material shall have sufficient fracture toughness to avoid brittle fracture of elements within the intended design service life of the structure. Checking for fracture toughness is required under all loading conditions.

NOTE For steels intended to be hot dip galvanized, see EN ISO 1461.

(2) Further checks for brittle fracture may be omitted if the conditions given in EN 1993-1-10 are satisfied.

5.2.4 Through-thickness properties

(1) The target through-thickness properties of the material should be determined by calculating the required design Z-value Z_{Ed} for the material according to EN 1993-1-10.

(2) Where material with improved through-thickness properties is required ($Z_{Ed} > 10$), a material quality Class should be selected according to EN 10164, i.e. Z 15, Z 25 or Z 35.

(3) The selection of material quality Class should be in accordance with Table 5.3.

NOTE Through-thickness properties are relevant for plates loaded perpendicular to their plane, e.g. for welded beam to column connections and welded end plates with tension in the through-thickness direction.

Table 5.3 — Choice of quality Class according to EN 10164

Target design value Z_{Ed} according to EN 1993-1-10	Quality Class according to EN 10164 (Available design Z-value Z_{Rd})
$Z_{Ed} \leq 10$	—
$10 < Z_{Ed} \leq 20$	Z 15
$20 < Z_{Ed} \leq 30$	Z 25
$Z_{Ed} > 30$	Z 35

5.2.5 Values of other material properties

(1) The material properties to be adopted in calculations should be taken as the following mean values for steel:

- modulus of elasticity $E = 210\,000 \text{ N/mm}^2$
- shear modulus $G = \frac{E}{2(1+\nu)} \approx 81\,000 \text{ N/mm}^2$
- Poisson's ratio in elastic range $\nu = 0,3$
- coefficient of linear thermal expansion $\alpha_T = 12 \times 10^{-6} \text{ K}^{-1}$ (for $T \leq 100 \text{ °C}$)

(2) The material parameter ε is used in several expressions in EN 1993 (all parts). It is defined as follows:

$$\varepsilon = \sqrt{\frac{235}{f_y}} \quad (5.1)$$

where

f_y is the yield strength in N/mm^2 .

5.3 Connecting devices

(1) For fasteners and welding consumables, the provisions given in EN 1993-1-8 should apply.

5.4 Other prefabricated products in buildings

(1)B Any semi-finished or finished structural product used in the structural design of buildings shall comply with the relevant EN or EN ISO product standard, EAD or ETA.

6 Durability

(1) The effects of deterioration of material, corrosion, mechanical wear or fatigue where relevant shall be taken into account by the choice of an appropriate corrosion protection system, by appropriate choice of material, (see EN 1993-1-4, EN 1993-1-10), and appropriate detailing, (see EN 1993-1-9) or by structural redundancy.

(2) Execution of protective treatment shall comply with EN 1090-2.

NOTE EN 1090-2 lists the factors affecting execution that need to be specified during design.

(3) Parts susceptible to corrosion, mechanical wear or fatigue should be designed such that inspection, maintenance and repair can be carried out satisfactorily and access is available for in-service inspection and maintenance.

(4) For elements that cannot be inspected, an appropriate corrosion allowance should be included.

(5) If a structure includes components that need to be replaceable (e.g. bearings), the possibility of their safe replacement should be verified as a transient design situation.

(6)B Corrosion protection does not need to be applied to interior building structures, if the corrosivity category is C1 according to EN ISO 12944-2:2017.

7 Structural analysis

7.1 Structural modelling for analysis

7.1.1 Basic assumptions

(1) Analysis shall be based upon calculation models of the structure that are appropriate for the limit state under consideration. The method used for the analysis shall be consistent with the design assumptions. EN 1990:—, 7.1.1 applies.

7.1.2 Joint modelling

(1) Where the effects of the behaviour of the joints on the distribution of internal forces and moments within a structure, and on the overall deformations of the structure, are significant (such as in the case of semi-continuous joints), they should be taken into account, see EN 1993-1-8.

(2) To identify whether the effects of joint moment-rotation behaviour need be taken into account in the analysis, a distinction may be made between three joint models as follows:

- simple joints, in which the joint may be assumed not to transmit bending moments;
- continuous joints, in which the joint may be assumed not to allow any relative rotation between connected members, and hence the joint behaviour has no effect on the analysis;
- semi-continuous joints, in which the joint should be assumed to allow relative rotation and bending moment transfer between connected members, and hence the behaviour of the joint needs to be taken into account in the analysis.

(3) For various types of joints the provisions given in EN 1993-1-8 should apply.

7.2 Global analysis

7.2.1 Consideration of second order effects

(1) The internal forces and moments may generally be determined using either:

- first-order analysis, using the undeformed geometry of the structure or
- second-order analysis, taking into account the influence of the deformation of the structure.

(2) The effects of the deformed geometry (second-order effects) should be considered if they increase the action effects significantly or modify significantly the structural behaviour.

(3) Second order analysis is not required if the conditions (7.1) and (7.2) are satisfied.

(4) If the condition in Formula (7.1) is satisfied, second order effects due to in-plane or out-of-plane member (non-sway) buckling may be neglected for the global analysis.

$$\alpha_{cr,ns} = \frac{F_{cr,ns}}{F_d} \geq k_0 \quad (7.1)$$

where

$F_{cr,ns}$ is the minimum elastic critical flexural buckling load for either the in-plane or out-of-plane member non-sway buckling mode. Torsional buckling, torsional-flexural and lateral torsional buckling are not considered;

F_d is the design value of the loading on the structure;

$\alpha_{cr,ns}$ is the factor by which the design value of the loading would have to be increased to cause elastic instability in the in-plane or out-of-plane member (non-sway) buckling mode.

NOTE 1 The value of k_0 is 25 unless the National Annex gives a different value.

NOTE 2 For frames, buckling modes can be classified as sway or non-sway, see Figure 7.1. Sway modes are characterized by significant relative displacements of the member ends, perpendicular to the undeformed member geometry.

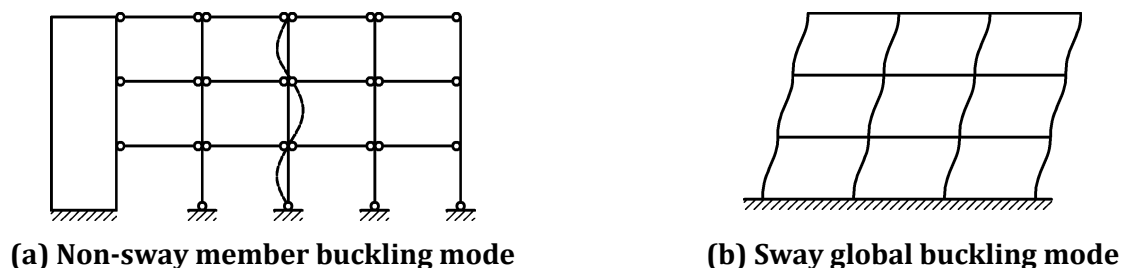


Figure 7.1 — Buckling modes of frames

NOTE 3 For the application of the condition in Formula (7.1) to plastic analysis, see 7.4.3.

(5) If the condition in Formula (7.2) is satisfied, second order effects due to in-plane sway buckling may be neglected for the global analysis and first order analysis may be used for the determination of the in-plane sway bending moments.

$$\alpha_{cr,sw} = \frac{F_{cr,sw}}{F_d} \geq 10 \quad (7.2)$$

where

$F_{cr,sw}$ is the minimum elastic critical in-plane flexural buckling load for a global sway buckling mode;

F_d is the design value of the loading on the structure;

$\alpha_{cr,sw}$ is the factor by which the design value of the loading would have to be increased to cause elastic instability in a global, in-plane (sway) mode.

NOTE 1 When the condition (7.2) is satisfied, the increase of the internal forces and moments due to sway second order effects is no more than 10 % of the internal forces and moments according to first order theory.

NOTE 2 For the application of the condition in Formula (7.2) to plastic analysis, see 7.4.3.

(6) Second order effects due to lateral torsional buckling may be neglected for the global analysis and the verifications in the following cases:

- for certain types of sections, e.g. structural hollow sections and equivalent welded box sections;
- when only weak-axis bending moments act on members with mono-symmetric or doubly symmetric sections;

- in case of sufficient restraint to the compression flange, see Annex D;
- when the limit slenderness for susceptibility to lateral torsional buckling of 8.3.2.3(1) is not exceeded.

(7) The effects of shear lag and plate buckling on the stiffness should be taken into account if they significantly influence the global analysis, see EN 1993-1-5.

(8) For rolled sections and welded sections with similar dimensions, the effects of shear lag and plate buckling on the stiffness may be neglected.

(9) The effects of the slip in bolt holes and similar deformations of connection devices like studs and anchor bolts should be taken into account if they significantly influence the global analysis.

(10)B Portal frames with rafter slopes not steeper than 1 : 2 (26°) and multi-storey beam-and-column type plane frames in buildings may be checked for in-plane, global (sway) mode failure with first order analysis if the condition in Formula (7.2) is satisfied for each storey. $\alpha_{cr,sw}$ may be calculated for a storey using Formula (7.3), see Figure 7.2, provided that the axial compression in the beams or rafters is not significant (see also (11)):

$$\alpha_{cr,sw} = \frac{K_{st} H_{st}}{\sum N_{Ed,i}} \quad (7.3)$$

where

K_{st} is the lateral rigidity of the storey. This may be calculated from a linear elastic analysis using Formula (7.4):

$$K_{st} = \frac{H_f}{\Delta_f} \quad (7.4)$$

H_f is a fictitious horizontal force applied at the top of the columns of the storey;

Δ_f is the horizontal displacement at the top of the storey due to H_f , relative to the bottom of the storey;

$\sum N_{Ed,i}$ is the sum of axial forces in the columns within the storey under consideration;

H_{st} is the height of the storey.

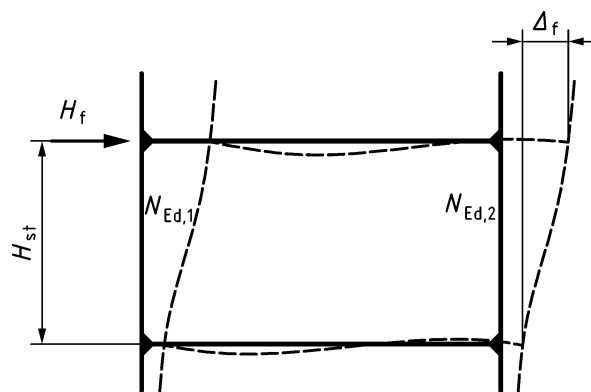


Figure 7.2 — Notations for 7.2.1(10)B

(11) For the application of (10)B, the axial compression in the beams or rafters should be assumed to be significant if:

$$\frac{N_{Ed}}{N_{cr}} \geq 0,1 \quad (7.5)$$

where

N_{Ed} is the design value of the compressive axial force;

N_{cr} is the in-plane elastic critical buckling load of the beam or rafter, considered as hinged at the ends of the system length measured along the beam or rafter.

7.2.2 Methods of analysis for ultimate limit state design checks

(1) The method of analysis (first or second order analysis combined with consideration of imperfections) should be compatible with the cross-section and member verification requirements in 8.2 and 8.3.

(2) According to the type of structure and the extent of the global analysis, imperfections and second order effects should be considered using one of the following approaches:

- a) entirely in the global analysis;
- b) partially in the global analysis and partially by verification of the buckling resistance of individual members according to 8.3;
- c) by verification of the buckling resistance of 'Equivalent Members' according to 8.3, using appropriate buckling lengths in accordance with the global buckling modes of the structure.

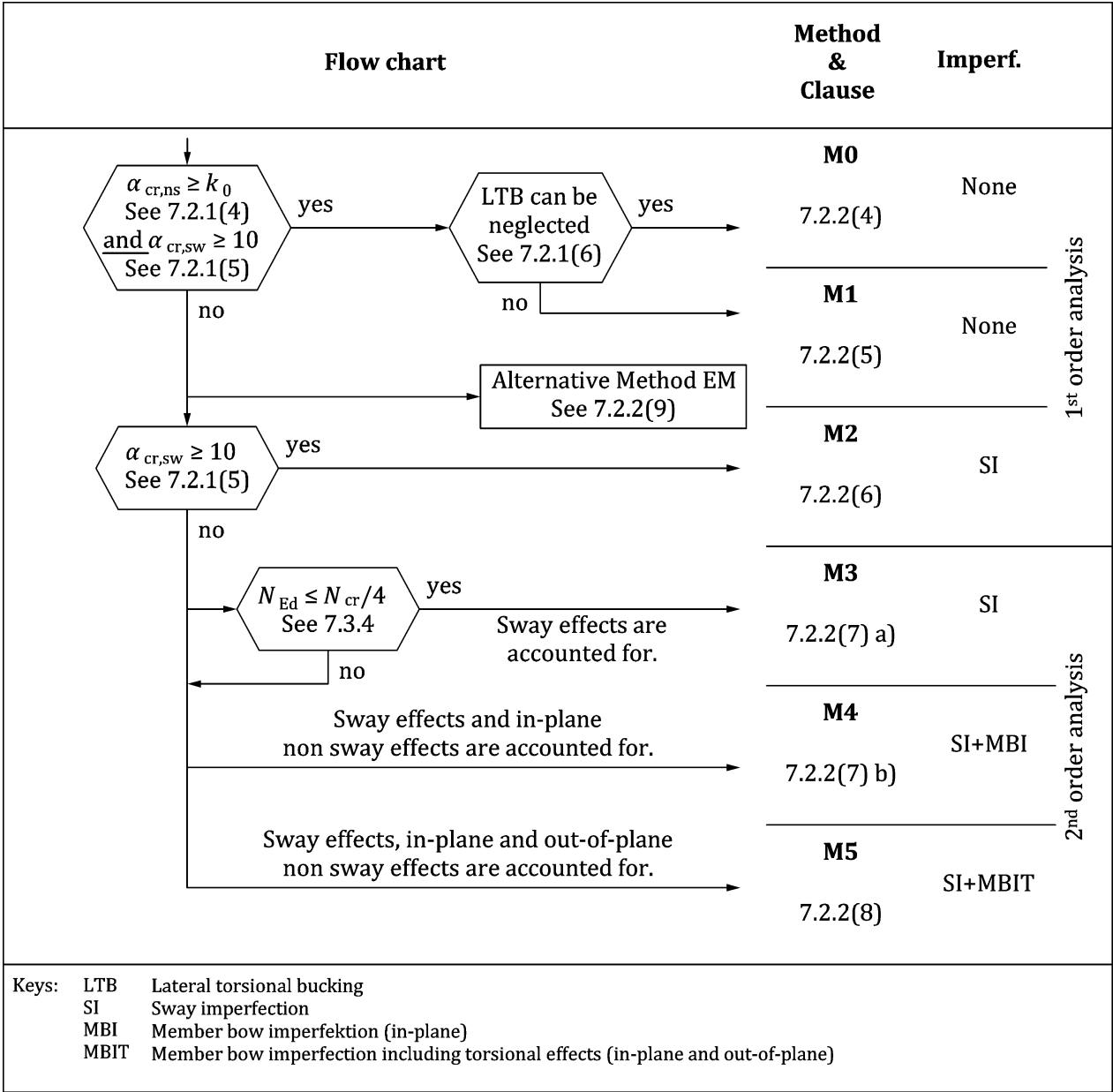
NOTE 1 For torsional and torsional-flexural buckling, see 8.3.1.4.

NOTE 2 Elastic critical buckling formulae are provided in prCEN/TR 1993-1-103:—³, *Eurocode 3 – Design of steel structures – Part 1-103: Elastic critical buckling of members*.

(3) Ultimate limit state design checks may be carried out using methods of analysis named hereafter M0, M1, M2, M3, M4, M5 or EM considering the application limitations defined in (4) to (9), see Figure 7.3 for an overview of the applicability of the methods. More complex methods may be used in place of less complex methods.

³ Under preparation.

NOTE Methods M0, M1, M2, M3, M4 and M5 are numerically ordered based on the complexity of the analysis. Method M0 is the least complex and method M5 is the most complex.



Key

LTB Lateral torsional buckling

EM Equivalent member method

SI Sway imperfection

MBI Member bow imperfection (in-plane)

MBIT Member bow imperfection considering lateral torsional buckling (in-plane and out-of-plane)

Figure 7.3 — Methods of structural analysis applicable to ultimate limit state design checks

(4) Method M0:

If second order effects may be neglected in the global analysis by satisfying the conditions in 7.2.1(4) and 7.2.1(5), and lateral torsional buckling may be neglected in accordance with 7.2.1(6):

- internal forces and moments may be determined from a first order global analysis;
- imperfections do not need to be included in the global analysis;
- the verification of the cross-sectional resistance should be carried out according to 8.2;
- the verification of the buckling resistance of individual members according to 8.3 may be omitted.

(5) Method M1:

If second order effects may be neglected in the global analysis by satisfying the conditions in 7.2.1(4) and 7.2.1(5), but lateral torsional buckling may not be neglected in accordance with 7.2.1(6):

- internal forces and moments may be determined from a first order global analysis;
- imperfections do not need to be included in the global analysis;
- the verification of the cross-sectional resistance should be carried out according to 8.2;
- the out-of-plane verification of the buckling resistance of individual members according to 8.3 is required.

(6) Method M2:

If second order effects due to member buckling may not be neglected (i.e. the condition in 7.2.1(4) is not satisfied) but second order effects due to in-plane, global (sway) buckling may be neglected (i.e. the condition in 7.2.1(5) is satisfied):

- internal forces and moments may be determined from a first order global analysis;
- the global analysis may neglect equivalent bow imperfections, but should consider global (sway) imperfections;
- the verification of the cross-sectional resistance should be carried out according to 8.2;
- the in-plane and out-of-plane verification of the buckling resistance of individual members according to 8.3 is required considering appropriate buckling lengths for the non-sway mode and corresponding bending moment diagrams.

(7) Method M3 and M4:

If second order effects due to member buckling may not be neglected (i.e. the condition in 7.2.1(4) is not satisfied) and second order effects due to in plane, global sway buckling may not be neglected (i.e. the condition in 7.2.1(5) is not satisfied), verification may be performed in accordance with one of the following methods:

a) Method M3:

If global (sway) imperfections are included in the global analysis and local bow imperfections may be neglected in the global analysis in accordance with 7.3.4:

- internal forces and moments should be determined from a second order global analysis;

- the verification of the cross-sectional resistance according to 8.2 should use the partial factor γ_{M1} instead of γ_{M0} ;
- the in-plane and out-of-plane verification of the buckling resistance of individual members according to 8.3 is required and should be based on second order internal forces and moments, however, the distribution of the internal forces and moments between the member ends may be determined based on first order theory;
- the in-plane buckling length for the non-sway mode may be used.

b) Method M4:

If all in-plane second order effects and both global (sway) imperfections and local bow imperfections are included in the global analysis:

- internal forces and moments should be determined from a second order global analysis;
- the verification of the cross-sectional resistance according to 8.2 should use the partial factor γ_{M1} instead of γ_{M0} ;
- the in-plane verification of the buckling resistance of individual members according to criterion in Formula (8.88) may be omitted;
- the out-of-plane verification of the buckling resistance of individual members according to criterion in Formula (8.89) is required.

The references to criteria in Formulae (8.88) and (8.89) apply when in-plane bending moments cause bending about the strong cross-sectional axis. Otherwise, the formulae have to be applied accordingly.

(8) Method M5:

If all in-plane and out-of-plane second order effects, including torsional effects, and global sway imperfections and in-plane and out-of-plane local bow imperfections are accounted for in the global analysis:

- internal forces and moments should be determined from a second order global analysis;
- the verification of the cross-sectional resistance according to 8.2 should use the partial factor γ_{M1} instead of γ_{M0} ;
- the verification of the buckling resistance of individual members according to 8.3 may be omitted.

NOTE If relevant, the out-of-plane local bow imperfection can either be from Table 7.1 or Table 7.2, whichever is larger.

(9) Method EM:

If second order effects may not be neglected in the global analysis (see 7.2.1(4) or/and in-plane, global sway buckling modes cannot be neglected, see 7.2.1(5)), the 'Equivalent Member method' may alternatively be used for verification of the buckling resistance of individual members:

- internal forces and moments may be determined from a first order global analysis;
- imperfections do not need to be included in the global analysis;
- the verification of the cross-sectional resistance should be carried out according to 8.2;

- the verification of the buckling resistance of individual members according to 8.3 may be carried out, considering the system effects (possible sway buckling modes) to determine the appropriate buckling length of each individual member.

NOTE Limitations on the use of Method EM can be set by the National Annex.

(10) The implications of neglecting second order effects in determining the internal forces and moments for Method EM should be considered for the design of associated joints and connected members, including verification of the out-of-plane buckling resistance of individual members.

(11) Second order effects may be calculated by using an analysis appropriate to the structure (including step-by-step or other iterative procedures). For frames where the first sway buckling mode is predominant, first order elastic analysis may be carried out with subsequent amplification of relevant action effects (e.g. bending moments) by appropriate factors.

(12)B For single storey frames designed on the basis of elastic global analysis, second order sway effects due to vertical loads may be calculated from a first order theory, by increasing the horizontal loads $F_{H,d}$ (e.g. horizontal action of wind) and equivalent loads $F_d \phi$ due to imperfections (see 7.3.2(5)) and other possible sway effects by the factor k_{sw} :

$$k_{sw} = \frac{1}{1 - \frac{1}{\alpha_{cr,sw}}} \quad (7.6)$$

provided that $\alpha_{cr,sw} \geq 3,0$,

where

$\alpha_{cr,sw}$ may be calculated according to Formula (7.3) in 7.2.1(10)B, provided that the axial compression in the beams or rafters is not significant as defined in 7.2.1(11)B.

NOTE 1 For $\alpha_{cr,sw} < 3,0$, a more accurate second order analysis applies.

NOTE 2 Formula (7.6) can be used also for other structures than buildings.

(13)B For multi-storey frames, second order sway effects may be calculated by means of the method given in 7.2.2(12)B provided that all storeys have a similar distribution of:

- vertical loads; and
- horizontal loads; and
- frame stiffness with respect to the applied storey shear forces.

NOTE For the limitation of the method, see 7.2.2(12)B.

7.3 Imperfections

7.3.1 Basis

(1) Appropriate allowances should be incorporated in the structural analysis to cover the effects of imperfections, including residual stresses and geometrical imperfections such as lack of verticality, lack of straightness, lack of flatness, lack of fit and eccentricities greater than the essential tolerances given in EN 1090-2 present in joints of the unloaded structure.

- (2) Equivalent geometrical imperfections should be used with values which reflect the possible effects of all types of imperfections, unless these effects are included in the resistance formulae for member design, see 7.3.3.
- (3) The assumed shape of sway and bow imperfections may be derived from the elastic buckling modes of a structure in the plane of buckling considered.
- (4) Both in-plane and out-of-plane buckling including torsional buckling with symmetric and asymmetric buckling shapes should be taken into account in the most unfavourable direction and form.
- (5) The following imperfections should be taken into account as relevant:
- sway imperfections for global analysis of frames (see 7.3.2 and 7.3.4);
 - bow imperfections for global and member analysis (see 7.3.3 and 7.3.4);
 - imperfections for analysis of bracing systems (see 7.3.5);
 - imperfections based on elastic critical buckling modes (see 7.3.6).

7.3.2 Sway imperfections for global analysis of frames

(1) For frames sensitive to buckling in a sway mode (see 7.2.1(5)), the effect of imperfections should be taken into account in the frame analysis by means of an equivalent imperfection in the form of an initial sway imperfection ϕ (see Figure 7.4).

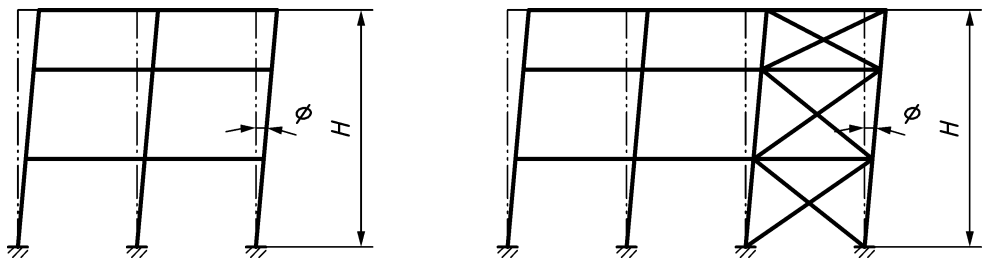


Figure 7.4 — Initial sway imperfections

(2) The initial sway imperfection ϕ may be determined from Formula (7.7):

$$\phi = \phi_0 \alpha_H \alpha_m \tag{7.7}$$

where

- ϕ_0 is the basic value:
- $\phi_0 = 1/400$ for verification of elastic resistance of cross-sections and members;
- $\phi_0 = 1/200$ for verification of plastic resistance of cross-sections and members;
- α_H is the reduction factor for height H applicable to columns:
- $$\alpha_H = \frac{2}{\sqrt{H}} \quad \text{but} \quad \alpha_H \leq 1,0$$
- H is the height of the structure in metres;

α_m is the reduction factor for the number of stabilised columns: $\alpha_m = \sqrt{0,5 \left(1 + \frac{1}{m}\right)}$;

m is the number of columns in a row including only those columns which carry a vertical load N_{Ed} not less than 50 % of the average value of all the columns in the vertical plane considered.

(3) The initial sway imperfection ϕ should apply in all relevant horizontal directions, but need only be considered in one direction at a time.

(4)B For building frames, sway imperfections may be disregarded if:

$$F_{H,d} \geq 0,15 F_{V,d} \quad (7.8)$$

where

$F_{H,d}$ is the design value of the total horizontal load;

$F_{V,d}$ is the design value of the total vertical load.

(5) The effects of sway imperfection may be replaced by systems of equivalent horizontal forces, introduced for each column, see Figure 7.5.

(6)B Where, in multi-storey beam-and-column building frames, equivalent forces are used, they should be applied at each floor and at roof level.

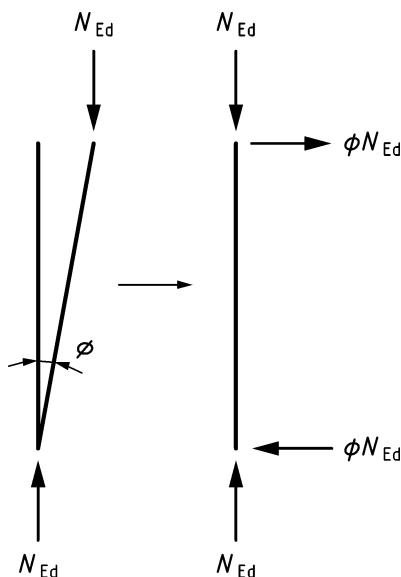


Figure 7.5 — Replacement of initial sway imperfection by equivalent horizontal forces

7.3.3 Equivalent bow imperfection for global and member analysis

7.3.3.1 Flexural buckling

(1) The equivalent bow imperfection, e_0 , of members for flexural buckling may be determined according to Formula (7.9).

$$e_0 = \frac{\alpha}{\varepsilon} \beta L$$

(7.9)

where

- L is the member length;
- α is the imperfection factor, depending on the relevant buckling curve according to Table 8.2;
- ε is the material parameter defined in 5.2.5(2);
- β is the reference relative bow imperfection according to Table 7.1.

Table 7.1 — Reference relative bow imperfection β

Buckling about axis	Elastic cross-section verification	Plastic cross-section verification
y-y	1/110	1/75
z-z	1/200	1/68

NOTE The values of e_0 are determined by Formula (7.9) unless the National Annex gives different values.

- (2) If the member is verified with reference to the plastic resistance of the cross-section:
- For buckling about the y-y axis of cross-sections, the linear plastic interaction given in 8.2.1(7) should be used.
 - For buckling about the z-z axis of cross-sections, the relevant plastic interaction according to 8.2.9 may be used but the design value of the plastic bending moment resistance $M_{pl,Rd}$ should be limited to $1,25 M_{el,Rd}$.

NOTE For circular and square hollow sections, either buckling about y-y axis or z-z axis can be considered.

(3) The effects of equivalent bow imperfection may be replaced by systems of equivalent horizontal forces, introduced for each member, see Figure 7.6.

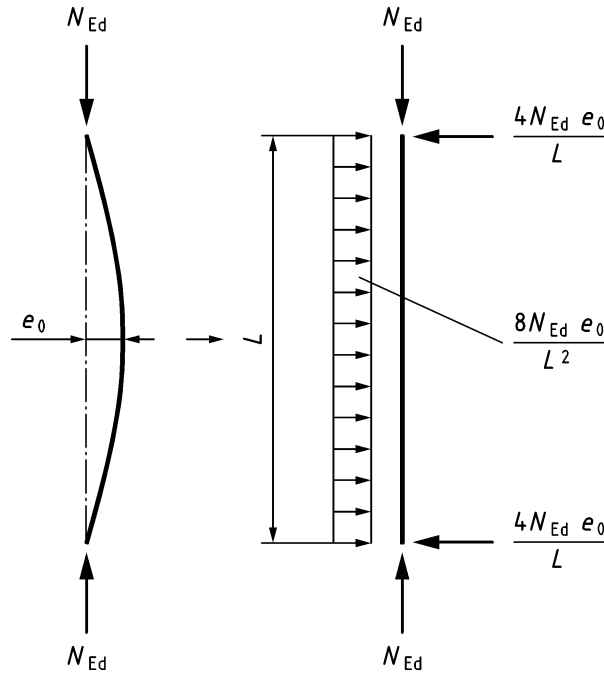


Figure 7.6 — Replacement of equivalent bow imperfection by equivalent forces

(4) For the design of uniform built-up members according to 8.4, the amplitude e_0 of the equivalent bow imperfection should be determined from Formula (7.10):

$$e_0 = \frac{L}{500} \quad (7.10)$$

where

L is the length of the built-up member.

7.3.3.2 Lateral torsional buckling

(1) For a second order analysis taking account of lateral torsional buckling of a member in bending, the equivalent bow imperfection for flexural buckling about the weak axis of the cross-section $e_{0,LT}$ may be determined according to Formula (7.11). In general, an additional torsional imperfection may be neglected.

$$e_{0,LT} = \beta_{LT} \frac{L}{\varepsilon} \quad (7.11)$$

where

L is the member length;

ε is the material parameter defined in 5.2.5(2);

β_{LT} is the reference relative bow imperfection for lateral torsional buckling according to Table 7.2.

Table 7.2 — Reference relative bow imperfection β_{LT} for lateral torsional buckling

Cross-section	Condition	Elastic cross-section verification	Plastic cross-section verification
rolled	$h/b \leq 2,0$	1/250	1/200
	$h/b > 2,0$	1/200	1/150
welded	$h/b \leq 2,0$	1/200	1/150
	$h/b > 2,0$	1/150	1/100

NOTE The values of $e_{0,LT}$ are determined by the Formula (7.11) unless the National Annex gives different values.

7.3.4 Combination of sway and equivalent bow imperfections for global analysis of frames

(1) When performing the global analysis for determining end forces and end moments to be used in member checks according to 8.3, equivalent bow imperfections may be neglected. However, for frames sensitive to second order effects, equivalent bow imperfections of members additionally to global sway imperfections should be introduced in the structural analysis of the frame for each compressed member if the following conditions are met:

- at least one moment resistant joint at one member end;
- $N_{Ed} > 0,25 N_{cr}$ (7.12)

where

N_{Ed} is the design value of the compressive axial force and

N_{cr} is the critical axial force determined for in-plane flexural buckling of the member considered as hinged at its ends.

NOTE Equivalent bow imperfections are taken into account in member checks.

7.3.5 Imperfections for analysis of bracing systems

7.3.5.1 Horizontal bracing systems

(1) In the analysis of bracing systems which are required to provide lateral stability within the length of beams or compression members, the effects of imperfections should be included by means of an equivalent geometrical imperfection of the members to be restrained, in the form of an initial bow imperfection calculated from Formula (7.13):

$$e_0 = \alpha_m L / 500$$

(7.13)

where

L is the span of the bracing system

$$\alpha_m = \sqrt{0,5 \left(1 + \frac{1}{m} \right)}$$

in which m is the number of members to be restrained.

(2) The effects of the initial bow imperfections of the members to be restrained by a bracing system, may be replaced by an equivalent load, q_d , as shown in Figure 7.7, calculated from Formula (7.14):

$$q_d = \sum N_{Ed} 8 \frac{e_0 + \delta_q}{L^2} \quad (7.14)$$

where

N_{Ed} is the design value of the compressive axial force

δ_q is the in-plane deflection of the bracing system due to q_d plus any external loads determined from first order analysis. δ_q may be taken as 0 if second order theory is used.

(3) Where the bracing system is required to stabilize the compression flange of a beam of constant height, the force N_{Ed} in Figure 7.7 may be obtained from Formula (7.15):

$$N_{Ed} = M_{Ed} / h \quad (7.15)$$

where

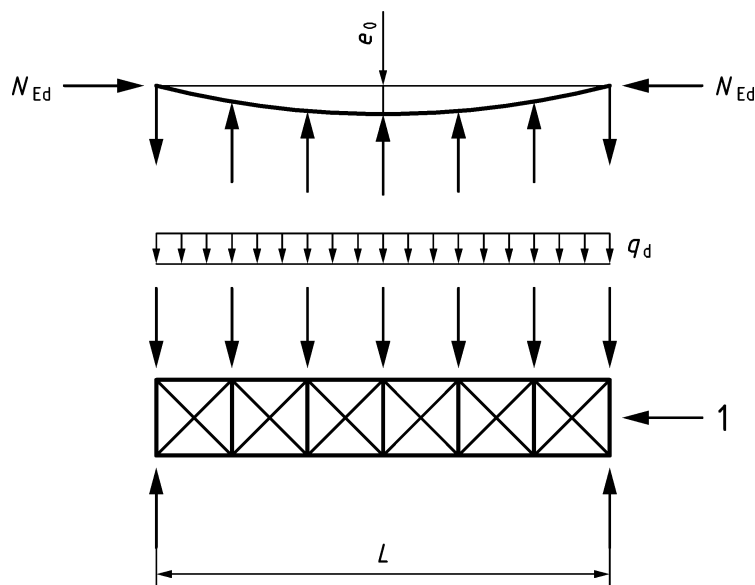
M_{Ed} is the maximum moment in the beam

h is the overall depth of the beam.

(4) Where a beam is subjected to external compression, N_{Ed} in Formula (7.15) should include an appropriate part of the compressive axial force.

NOTE The force N_{Ed} is assumed uniform within the span L of the bracing system. For non-uniform forces, this is slightly conservative.

(5) At points where beams or compression members are spliced, it should also be verified that the horizontal bracing system is able to resist a local force Q_m according to 7.3.5.2(2).



Key
1 bracing system

Figure 7.7 — Equivalent load acting on the bracing system

7.3.5.2 Vertical bracings

(1)B For the determination of horizontal forces to floor diaphragms, imperfection should be included. Configuration of imperfections as given in Figure 7.8 may be applied, where ϕ is determined by the Formula (7.16) assuming a single storey height H_{st} .

$$\phi = \alpha_m \alpha_H \phi_0 \tag{7.16}$$

where

$\phi_0 = \frac{1}{400}$ for non-spliced compression members;

$\phi_0 = \frac{1}{200}$ for spliced compression members;

α_m is the reduction factor defined in 7.3.2(2) according to the number m of stabilized members;

α_H is for non-spliced compression members, the reduction factor defined in 7.3.2(2) where the height H should be taken equal to the storey height H_{st} and for spliced compression members: $\alpha_H = 1,0$.

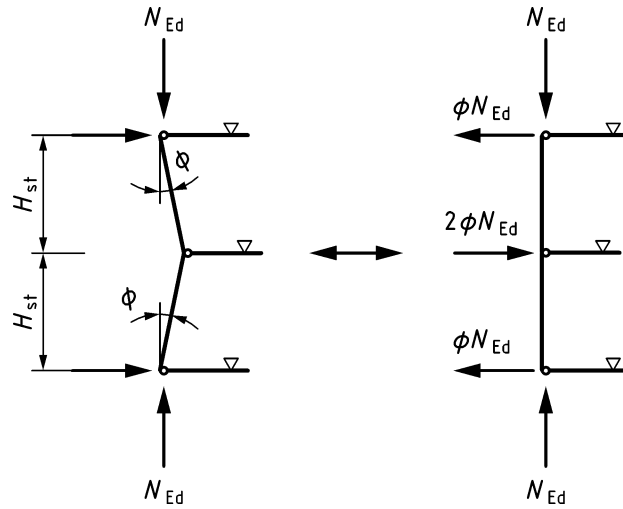


Figure 7.8 — Configuration of imperfections ϕ for horizontal forces on floor diaphragms

(2) At points where compression members are supported, it should also be verified that the bracing system is able to resist a local force Q_m according to Formula (7.17), applied to it by each beam or compression member which is supported at that point, and to transmit this force to the adjacent points at which that compression member is restrained.

$$Q_m = 2 \phi N_{Ed} \quad (7.17)$$

where

ϕ is defined in (1)B;

N_{Ed} is the design value of the compressive axial force in the stabilized member.

7.3.6 Imperfection based on elastic critical buckling modes

(1) As an alternative to 7.3.2 and to 7.3.3.1, the shape of the elastic critical buckling mode $\eta_{cr}(x)$ of the frame structure or of the verified member may be used as the unique global and local initial imperfection. The shape of the equivalent geometrical imperfection $\eta_{init}(x)$ may be expressed using Formula (7.18):

$$\eta_{init}(x) = e_{0,m} \frac{N_{cr,m}}{EI_m \left| \eta_{cr,m}'' \right|} \eta_{cr}(x) \quad (7.18)$$

where

$$e_{0,m} = \alpha_m \left(\bar{\lambda}_m - 0,2 \right) \frac{M_{Rk,m}}{N_{Rk,m}} \quad \text{for } \bar{\lambda}_m > 0,2 \quad (7.19)$$

m is an index that denotes the critical cross-section of the frame structure or of the verified member (see Note 3). Index m indicates that the value or property belongs to the critical cross-section;

$\bar{\lambda}_m = \sqrt{\frac{N_{Rk,m}}{N_{cr,m}}}$	is the relative slenderness of the member determined at the critical cross-section m;
$\alpha_{\eta,m}$	is the imperfection factor α for the relevant buckling curve at the critical cross-section m, see Table 8.2 and Table 8.3;
$N_{cr,m} = \alpha_{cr} N_{Ed,m}$	is the value of critical axial force at the cross-section m and also the critical axial force for the equivalent member;
α_{cr}	is the minimum force amplifier for the axial force configuration N_{Ed} in members to reach the elastic critical buckling load of the structure;
$M_{Rk,m}$	is the characteristic value of the bending moment resistance at the critical cross-section m, e.g. $M_{el,Rk,m}$ or $M_{pl,Rk,m}$ as relevant;
$N_{Rk,m}$	is the characteristic value of resistance to axial force of the critical cross-section;
$EI_m \eta''_{cr} $	is the absolute value of the bending moment due to $\eta_{cr,m}$ at the critical cross-section m;
$\eta_{cr}(x)$	is the shape of the relevant (first or higher) elastic critical buckling mode.

NOTE 1 The imperfection $\eta_{init}(x)$ in the shape of the elastic critical buckling mode is applicable generally for all members in compression and for frames buckling in their plane. It is especially suitable for members with cross-section characteristics and/or axial force not constant on their length and for frames containing such members.

NOTE 2 For calculating the amplifier α_{cr} , the members of the structure are considered to be loaded by axial forces N_{Ed} resulting from the first order elastic analysis of the structure for the design loads.

NOTE 3 The critical cross-section m is the cross-section with the highest utilization ratio regarding the effect of the axial force and the bending moments due to imperfections. For strongly irregular members, the position of the critical cross-section m might need to be determined using an iterative procedure.

(2) The expression $EI_m |\eta''_{cr,m}|$ in Formula (7.18) may be replaced by $M_{\eta,cr,m}^{\text{II}} (\alpha_{cr} - 1)$.

where

$M_{\eta,cr,m}^{\text{II}}$ is the bending moment at the cross-section m calculated by using the second order analysis of the structure with the imperfection in the shape of the elastic buckling mode η_{cr} .

(3) When the global analysis is carried out using the imperfection defined in (1), the cross-sections should be verified according to the criterion given in 8.2.1(7). If the criterion is applied with reference to the plastic resistance, the design value of the bending moment resistance $M_{c,Rd}$ should be limited to $1,25 M_{el,Rd}$ for both strong axis and weak axis.

7.4 Methods of analysis considering material non-linearities

7.4.1 General

(1) The internal forces and moments may be determined using either:

- a) Elastic global analysis (see 7.4.2) or
- b) Plastic global analysis (see 7.4.3).

NOTE For finite element model (FEM) analysis, see prEN 1993-1-14⁴.

(2) Elastic global analysis may be used in all cases.

(3) Plastic global analysis may be used for structures with members made of steel grades up to and including S460 and where the structure has sufficient rotation capacity at the actual locations of the plastic hinges, whether this is in the members or in the joints.

- a) Where a plastic hinge occurs in a member, the member cross-sections should be doubly symmetric or single symmetric with a plane of symmetry in the same plane as the rotation of the plastic hinge, and it should satisfy the requirements specified in 7.6.
- b) Where a plastic hinge occurs in a joint according to the structural analysis, the joint should be able to sustain the plastic resistance for a sufficient rotation, see EN 1993-1-8.

NOTE Rules for plastic global analysis based on plastic hinges for steel grades higher than S460 can be set by the National Annex. In that case, relevant rules such as those in 7.4.3 and 8.2.3 can be changed accordingly.

(4)B As a simplified method for a limited plastic redistribution of moments in continuous beams where following an elastic analysis some peak moments exceed the plastic bending resistance by 15 % maximum, the parts in excess of these peak moments may be redistributed in any member, provided, that:

- a) the internal forces and moments in the frame remain in equilibrium with the applied loads, and
- b) all the members in which the moments are reduced are of steel grades up to and including S460 and have Class 1 or Class 2 cross-sections (see 7.5), and
- c) lateral torsional buckling of the members is prevented, see 8.3.5.

7.4.2 Elastic global analysis

(1) Elastic global analysis should be based on the assumption that the stress-strain behaviour of the material is linear, whatever the stress level is.

NOTE For the choice of a semi-continuous joint model, see 7.1.2.

(2) Internal forces and moments may be determined according to elastic global analysis even if the resistance of a cross-section is based on its plastic resistance, see 8.2.

⁴ Under preparation.

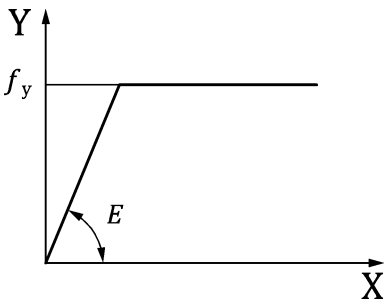
7.4.3 Plastic global analysis

(1) Plastic global analysis allows for the effects of material non-linearity in calculating the action effects of a structural system. The behaviour should be modelled by either of the two methods depending on the conditions indicated in (4) to (8):

- a) Plastic hinge method: the non-linear material behaviour is concentrated in plastified sections and/or joints as plastic hinges;
- b) Plastic zone method: the partial plastification of members in plastic zones is explicitly considered.

(2) Plastic global analysis may be used if sufficient lateral restraint is provided in the vicinity of sections where a plastic hinge or a plastic zone can develop under the design loads, see 8.3.5.

(3) For steel grades up to and including S460, the bi-linear stress-strain relationship indicated in Figure 7.9 may be used.



Key
X strain
Y stress

Figure 7.9 — Bi-linear stress-strain relationship

- (4) In case of modelling as plastic hinges, see (1) a):
- Only steel grades up to and including S460 and steels satisfying the conditions in 5.2.2 a) may be used;
 - Plastic global analysis may only be used where the members (or joints) are capable of sufficient rotation capacity to enable the required redistribution of bending moments to develop, see 7.5 and 7.6.

NOTE For steel grades higher than S460, see (8) and (9).

(5) In case of modelling as plastic hinges and if second order effects can be neglected (see 7.2.1(4) and 7.2.1(5)), rigid plastic analysis neglecting the elastic behaviour between hinges may be used. In this case joints are classified only by strength, see EN 1993-1-8.

(6) In case of modelling as plastic hinges and if second order effects have to be considered, elastic plastic analysis should be carried out. The effects of deformed geometry of the structure and the structural stability of the frame should be verified according to the principles in 7.2.

(7) When using plastic analysis, the conditions (7.1) and (7.2) should apply for the system before forming the last plastic hinge or should be checked for each individual system along the formation of the different plastic hinges, up to the design loading.

NOTE The maximum resistance of a frame with significantly deformed geometry can occur before all hinges of the first order collapse mechanism have developed.

(8) If steel grades higher than S460 are used, plastic analysis should be performed considering partial plastification in plastic zones according to the plastic zone method given in (1) b).

(9) In case of modelling members with partial plastification in plastic zones, see (1) b):

- Rules for design with finite element method are given in prEN 1993-1-14;
- Steel grades higher than S460 may be used provided that the stress-strain curves including strain limitations are considered and, if appropriate, also the effects of local plate imperfections according to prEN 1993-1-14.

(10) In case of material non-linearities, the action effects in a structure may be determined incrementally based on the design loads for each relevant design situation. In this incremental approach, each permanent or variable action should be increased proportionally.

7.5 Classification of cross-sections

7.5.1 Basis

(1) Cross-sections should be classified depending on the extent to which their resistance and rotation capacity is limited by their local buckling resistance.

7.5.2 Classification

(1) For the determination of its resistance, a cross-section should be classified according to one of the four following classes:

- Class 1 cross-sections are those which can form a plastic hinge with the rotation capacity required from plastic global analysis without reduction of the resistance.
- Class 2 cross-sections are those which can develop their plastic bending moment resistance, but have limited rotation capacity because of local buckling.
- Class 3 cross-sections are those in which the stress in the extreme compression fibre of the steel member assuming an elastic distribution of stresses can reach the yield strength, but local buckling is liable to prevent development of the plastic bending moment resistance.
- Class 4 cross-sections are those in which local buckling will occur before the attainment of yield strength in any part of the cross-section.

(2) In Class 4 cross-sections, effective widths may be used to make the necessary allowances for reductions in resistance due to the effects of local buckling, see EN 1993-1-5.

(3) Cross-sections should be classified depending on the width to thickness ratio, the yield strength and the stress distribution of the parts subject to compression.

NOTE The various compression parts in a cross-section (such as a web or flange) can, in general, be in different classes.

(4) Compression parts should include every part of a cross-section which is either totally or partially in compression, due to axial force and/or bending moment, under the load combination considered.

(5) A cross-section should be classified according to the highest (least favourable) Class of its compression parts. Exceptions are specified in 8.2.1(10) and 8.2.2.4(1).

(6) Alternatively, the classification of a cross-section may be defined by quoting both the flange classification and the web classification.

(7) The maximum width-to-thickness ratios for Class 1, 2, and 3 compression parts should be obtained from Table 7.3 to Table 7.5. They depend on the material parameter ε defined in 5.2.5(2). A part which fails to satisfy the limits for Class 3 should be taken as Class 4.

(8) For compression parts with longitudinal stiffeners, EN 1993-1-5 should be applied.

(9) Except as given in (10), Class 4 sections may be treated as Class 3 sections if the width to thickness ratios are less than the limiting proportions for Class 3 obtained from Table 7.3 to Table 7.5 when ε is

increased by $\sqrt{\frac{f_y / \gamma_{M0}}{\sigma_{\text{com,Ed}}}}$, where $\sigma_{\text{com,Ed}}$ is the maximum design compressive stress in the part taken from

first order or where necessary second order analysis.

(10) The provisions of (9) should not be used when verifying the design buckling resistance of a member using 8.3. The limiting proportions for Class 3 should then be obtained from Table 7.3 to Table 7.5.

(11) Cross-sections with a Class 3 web and Class 1 or 2 flanges may be classified as Class 2 cross-sections with an effective web in accordance with 8.2.2.4.

(12) Where the web is considered to resist shear forces only and is assumed not to contribute to the bending and axial force resistance of the cross-section, the cross-section may be designed as Class 2, 3 or 4 sections, depending only on the flange class.

NOTE For flange induced web buckling, see EN 1993-1-5.

7.6 Cross-section requirements for plastic global analysis

(1) At plastic hinge locations, the cross-section of the member should have sufficient rotation capacity.

(2) In a uniform member sufficient rotation capacity may be assumed at a plastic hinge if both the following requirements are satisfied:

- a) the member has Class 1 cross-section at the plastic hinge location (see 7.5.2 for cross-section classification);
- b) where a transverse force that exceeds 10 % of the shear resistance of the cross-section, see 8.2.6, is applied to the web at the plastic hinge location, web stiffeners should be provided within a distance along the member of $h/2$ from the plastic hinge location, where h is the height of the cross-section.

(3) Where the cross-section of the member varies along its length, the following criteria should be satisfied in addition to those given in (2):

- a) Adjacent to plastic hinge locations, the thickness of the web should not be reduced for a distance each way along the member from the plastic hinge location of at least $2h_w$, where h_w is the clear depth of the web at the plastic hinge location.
- b) Adjacent to plastic hinge locations, the compression flange should be Class 1 for a distance each way along the member from the plastic hinge location of not less than the greater of:
 - $2h_w$, where h_w is as defined in (3)a)
 - the distance to the adjacent point at which the moment in the member has fallen to 80 % of the plastic bending moment resistance at the point concerned.

- c) Elsewhere in the member the compression flange should be Class 1 or Class 2 and the web should be Class 1, Class 2 or Class 3.
- (4) Adjacent to plastic hinge locations, any fastener holes in tension should satisfy 8.2.5(4) for a distance such as defined in (3) b) each way along the member from the plastic hinge location.
- (5) For plastic design of a frame, regarding cross-section requirements, the capacity of plastic redistribution of moments may be assumed sufficient if the requirements in (2) to (4) are satisfied for all members where plastic hinges exist, can occur or have occurred under design loads.
- (6) The provisions in (2) to (5) may be disregarded if methods of plastic global analysis are used, which consider the real stress and strain behaviour along the member including the combined effect of local, member and global buckling.

Table 7.3 — Maximum width-to-thickness ratios for internal compression parts

Internal compression parts			
Key 1 Axis of bending			
	Part subject to bending	Part subject to compression	Part subject to bending and axial force
Stress distribution in parts (compression positive)			
Class 1	$c/t \leq 72 \varepsilon$	$c/t \leq 28 \varepsilon$	$\text{when } \alpha_c > 0,5 : c / t \leq \frac{126 \varepsilon}{5,5 \alpha_c - 1}$ $\text{when } \alpha_c \leq 0,5 : c / t \leq \frac{36 \varepsilon}{\alpha_c}$
Class 2	$c/t \leq 83 \varepsilon$	$c/t \leq 34 \varepsilon$	$\text{when } \alpha_c > 0,5 : c / t \leq \frac{188 \varepsilon}{6,53 \alpha_c - 1}$ $\text{when } \alpha_c \leq 0,5 : c / t \leq \frac{41,5 \varepsilon}{\alpha_c}$
Stress distribution in parts (compression positive)			

Class 3	$c/t \leq 121 \varepsilon$	$c/t \leq 38 \varepsilon$	$\begin{aligned} \text{when } \psi > -1 : c/t &\leq \frac{38 \varepsilon}{0,608 + 0,343 \psi + 0,049 \psi^2} \\ \text{when } \psi \leq -1^a : c/t &\leq 60,5 \varepsilon (1 - \psi) \end{aligned}$
For I or H sections with equal flanges, under axial force and bending moment about the main axis parallel to the flanges, the parameter α_c that defines the position of the plastic neutral axis may be calculated as follows: If $N_{Ed} \geq c t_w f_y$ $\alpha_c = 1,0$ If $N_{Ed} \leq -c t_w f_y$ $\alpha_c = 0$ In other cases: $\alpha_c = 0,5 \left(1 + \frac{N_{Ed}}{c t_w f_y} \right)$ where N_{Ed} is the design axial force taken as positive for compression and negative for tension.			
^a $\psi \leq -1$ and a compression stress of $\sigma_{com,Ed} = f_y$ applies where the tensile strain $\varepsilon_t > f_y/E$			

Table 7.4 — Maximum width-to-thickness ratios for compression parts of outstand flanges

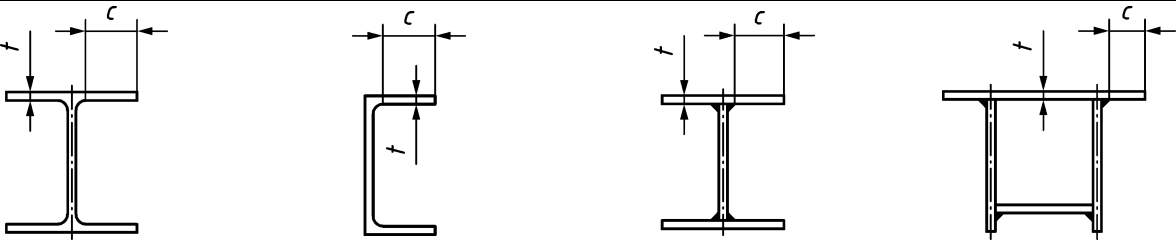
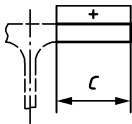
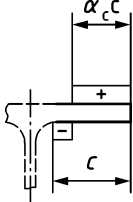
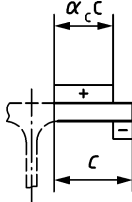
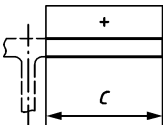
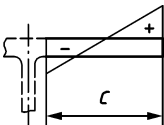
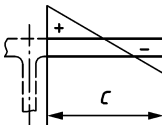
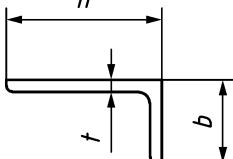
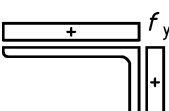
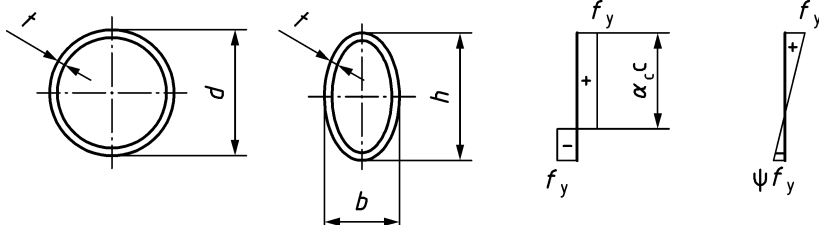
Outstand flanges			
			
Rolled sections		Welded sections	
	Part in pure compression	Part subject to bending and axial force	
Stress distribution in parts (compression positive)			
Class 1	$c/t \leq 9 \varepsilon$	$c/t \leq \frac{9 \varepsilon}{\alpha_c}$	$c/t \leq \frac{9 \varepsilon}{\alpha_c \sqrt{\alpha_c}}$
Class 2	$c/t \leq 10 \varepsilon$	$c/t \leq \frac{10 \varepsilon}{\alpha_c}$	$c/t \leq \frac{10 \varepsilon}{\alpha_c \sqrt{\alpha_c}}$
Stress distribution in parts (compression positive)			
Class 3	$c/t \leq 14 \varepsilon$	$c/t \leq 21 \varepsilon \sqrt{k_\sigma}$ For k_σ see EN 1993-1-5	

Table 7.5 — Maximum width-to-thickness ratios for compression parts of angles and circular and elliptical hollow sections

Angles Refer also to “Outstand flanges” (see sheet 2 of 3)  Does not apply to angles in continuous contact with other components			
Stress distribution across section (compression positive)			
Class 3	$\frac{h}{t} \leq 15 \varepsilon \quad \text{and} \quad \frac{b+h}{2t} \leq 11,5 \varepsilon$		
Circular and elliptical hollow sections 			
	Section in compression	Section in bending	Section under bending and axial force
Class 1	$d_e/t \leq 50 \varepsilon^2$	$d_e/t \leq 50 \varepsilon^2$	$d_e/t \leq 50 \varepsilon^2$
Class 2	$d_e/t \leq 70 \varepsilon^2$	$d_e/t \leq 70 \varepsilon^2$	$d_e/t \leq 70 \varepsilon^2$
Class 3	$d_e/t \leq 90 \varepsilon^2$	$d_e/t \leq 140 \varepsilon^2$	$d_e / t \leq \frac{2520 \varepsilon^2}{5\psi + 23}$
Equivalent diameter d_e for circular and elliptical hollow sections			
For circular hollow sections: $d_e = d$			
For elliptical hollow sections:			
In compression: $d_e = d_{e,c} = h \left[1 + \left(1 - 2,3 \left(\frac{t}{h} \right)^{0,6} \right) \left(\frac{h}{b} - 1 \right) \right]$ or, conservatively : $d_e = d_{e,c} = \frac{h^2}{b}$			
In bending about the strong axis: For $h / b \leq 1,36$: $d_e = d_{e,b,y} = \frac{b^2}{h}$			
For $h / b > 1,36$: $d_e = d_{e,b,y} = 0,4 \frac{h^2}{b}$			
In bending about the weak axis, or compression and bending about the weak axis: $d_e = d_{e,b,z} = \frac{h^2}{b}$			

In compression and bending about the strong axis, the equivalent diameter d_e may be determined by linear interpolation between the equivalent diameter for compression, $d_{e,c}$, and that for strong axis bending, $d_{e,b,y}$, as given by:

$$d_e = d_{e,b,y} + (d_{e,c} - d_{e,b,y})(2\alpha_c - 1) \quad \text{for class 1 and 2 cross-sections}$$

$$d_e = d_{e,b,y} + (d_{e,c} - d_{e,b,y})\left(\frac{\psi + 1}{2}\right) \quad \text{for class 3 and 4 cross-sections}$$

In compression and biaxial bending, the equivalent diameter d_e may be taken as described above, but with α_c and ψ determined using a modified axial force equal to $N_{Ed} + M_{z,Ed} A/W_{pl,z}$ for Class 1 and Class 2 cross-sections and $N_{Ed} + M_{z,Ed} A/W_{el,z}$ for Class 3 and Class 4 cross-sections.

8 Ultimate limit states

8.1 Partial factors

(1) The partial factors γ_{Mi} as defined in 4.4.5(1) should be applied to the characteristic values of the following resistances:

- resistance of cross-sections (whatever the class is): γ_{M0}
- resistance of members to instability assessed by member checks: γ_{M1}
- resistance of cross-sections in tension to fracture: γ_{M2}
- resistance of joints: see EN 1993-1-8

NOTE 1B The partial factors γ_{Mi} for buildings are given below unless the National Annex gives different values:

$$\gamma_{M0} = 1,00$$

$$\gamma_{M1} = 1,00 \text{ (where option a) in 5.2.1(1) is adopted)}$$

$$\gamma_{M2} = 1,25$$

NOTE 2B Annex E provides information on the basis for the calibration of the partial factors γ_{Mi} for buildings.

NOTE 3 For structures other than buildings, values of partial factors γ_{Mi} are given in other parts of EN 1993 as relevant. For structures not covered by EN 1993-2 to EN 1993-7⁵, the values of partial factors γ_{Mi} are given in EN 1993-2 unless the National Annex gives different values.

8.2 Resistance of cross-sections

8.2.1 General

(1) The design value of an action effect at each cross-section shall not exceed the corresponding design resistance. If several action effects act simultaneously, the combined effect shall not exceed the resistance for that combination.

⁵ EN 1993-7 is under preparation.

(2) Shear lag effects and local buckling effects should be determined by considering an effective width according to EN 1993-1-5. Shear buckling effects should also be taken into account according to EN 1993-1-5.

(3) The design values of resistance may be based on the elastic resistance or the plastic resistance and should be determined based on the classification of the cross-section.

(4) Design according to the elastic resistance may be carried out for all cross-sectional classes. For Class 4 cross-sections, the effective cross-sectional properties should be used to determine the elastic resistance.

(5) For elastic design, the yield criterion for the critical point of the cross-section given in criterion in Formula (8.1) may be used, unless other interaction criteria apply, see 8.2.8 to 8.2.10, and Annex B.

$$\frac{\sigma_{\text{Eq,Ed}}}{f_y / \gamma_{\text{M0}}} \leq 1,0 \quad (8.1)$$

where

$$\sigma_{\text{Eq,Ed}} = \sqrt{\sigma_{\text{x,Ed}}^2 + \sigma_{\text{z,Ed}}^2 - \sigma_{\text{x,Ed}} \sigma_{\text{z,Ed}} + 3\tau_{\text{Ed}}^2} \quad (8.2)$$

$\sigma_{\text{x,Ed}}$ is the design value of the longitudinal normal stress at the point of consideration;

$\sigma_{\text{z,Ed}}$ is the design value of the transverse normal stress at the point of consideration;

τ_{Ed} is the design value of the shear stress at the point of consideration.

NOTE Verification according to (5) can be conservative as it excludes partial plastic stress distribution.

(6) Design according to the plastic resistance should be carried out by establishing a stress distribution which is in equilibrium with the internal forces and moments without exceeding the yield strength. This stress distribution should be compatible with the associated plastic deformations.

(7) As a conservative approximation for all cross-sectional classes, a linear summation of the utilization ratios for each stress resultant may be used. For Class 1, Class 2 or Class 3 cross-sections subjected to the combination of N_{Ed} , $M_{\text{y,Ed}}$ and $M_{\text{z,Ed}}$ this method may be applied by using the criterion in Formula (8.3):

$$\frac{N_{\text{Ed}}}{N_{\text{Rd}}} + \frac{M_{\text{y,Ed}}}{M_{\text{y,Rd}}} + \frac{M_{\text{z,Ed}}}{M_{\text{z,Rd}}} \leq 1,0 \quad (8.3)$$

where

N_{Rd} , $M_{\text{y,Rd}}$ and $M_{\text{z,Rd}}$ are the design values of the resistance depending on the cross-sectional classification and including any reduction that may be caused by shear effects, see 8.2.8.

NOTE 1 For members with torsion, see condition in Formula (C.6).

NOTE 2 For Class 4 cross-sections, see 8.2.9.3(2).

(8) Where all compression parts of a cross-section are Class 1 or Class 2, the cross-section may be taken as capable of developing its full plastic resistance in bending.

(9) Where all compression parts of a cross-section are Class 3, its resistance should be based on an elastic distribution of stresses across the cross-section. Compressive stresses should be limited to the yield strength at the extreme fibres.

(10) For Class 3 cross-sections, partial plastification may be taken into account.

NOTE See 8.2.2.6 and Annex B.

(11) For Class 3 cross-sections where yielding first occurs on the tension side of the cross-section, the plastic reserves of the tension zone may be utilized by accounting for partial plastification.

8.2.2 Section properties

8.2.2.1 Gross cross-section

(1) Properties of the gross cross-section should be determined according to the nominal dimensions. Holes for fasteners may be neglected, but larger openings should be taken into account. Splice materials should not be included.

8.2.2.2 Net area

(1) The net area A_{net} of a cross-section should be taken as its gross area less appropriate deductions for all holes and other openings, see Formula (8.4).

$$A_{\text{net}} = A - \Delta A_{\text{net}} \quad (8.4)$$

where

A is the gross cross-section area;

ΔA_{net} is the section area to be deducted to take into account the holes.

(2) The net area deduction for a hole should be the cross-sectional area of the hole in the plane of the cross-section. For countersunk holes, appropriate allowance should be made for the countersunk portion.

(3) Where holes are not staggered, the total area to be deducted should be the maximum sum of the cross-sectional areas of the holes in any cross-section perpendicular to the member axis (see failure plane ② in Figure 8.1).

NOTE The maximum sum denotes the position of the critical fracture line.

(4) Where holes are staggered, the total area ΔA_{net} to be deducted should be the greater of:

a) the deduction for staggered holes (failure plane ① in Figure 8.1)

$$\Delta A_{\text{net},1} = t \left(n_1 d_0 - \sum \frac{s^2}{4p_2} \right) \quad (8.5)$$

b) the deduction for non-staggered holes given in (3) (failure plane ② in Figure 8.1)

$$\Delta A_{\text{net},2} = n_2 d_0 t \quad (8.6)$$

where

s is the staggered pitch, the spacing of the centres of two consecutive holes in the chain measured parallel to the member axis;

p_2 is the spacing of the centres of the same two holes measured perpendicular to the member axis;

t is the thickness;

n_1 is the number of holes extending in any diagonal or zig-zag line progressively across the member or part of the member, see failure plane ① in Figure 8.1;

n_2 is the number of holes in a section where the holes are not staggered; see failure plane ② in Figure 8.1;

d_0 is the diameter of the hole.

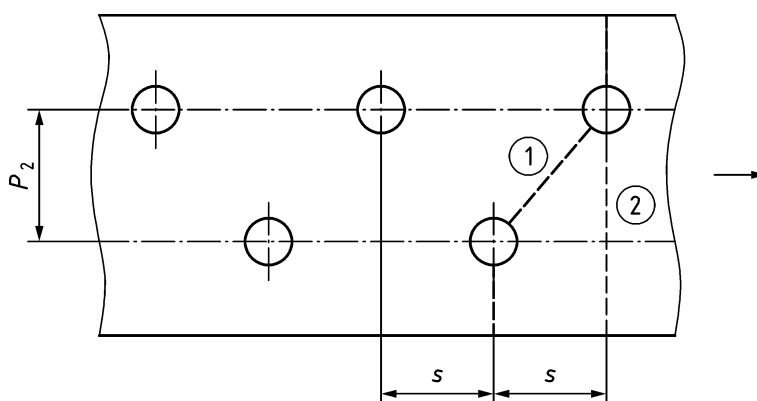


Figure 8.1 — Staggered holes and critical failure lines 1 and 2

(5) In an angle or other member with holes in more than one plane, the spacing p should be measured along the centre of thickness of the material (see Figure 8.2).

NOTE Further details on net section verifications of angles are given in EN 1993-1-8.

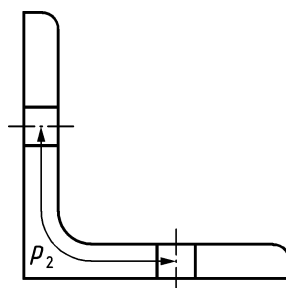


Figure 8.2 — Angles with holes in both legs

8.2.2.3 Shear lag effects

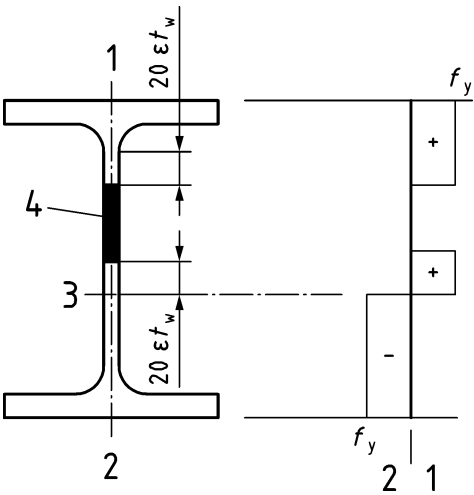
(1) Shear lag effects should be considered by calculating an effective width according to EN 1993-1-5.

(2) In Class 4 sections the interaction between shear lag and local buckling should be considered according to EN 1993-1-5.

NOTE For cold-formed members (other than structural hollow sections conforming to EN 10219 (all parts)), see EN 1993-1-3.

8.2.2.4 Effective properties of cross-sections with Class 3 webs and Class 1 or 2 flanges

(1) As an alternative to Annex B, for cross-sections with a Class 3 web and Class 1 or Class 2 flanges classified as effective Class 2 cross-sections, see 7.5.2(11), the effective cross-sectional properties should be determined replacing the portion of the web in compression by an effective part of $20\epsilon t_w$ adjacent to the compression flange and another part of $20\epsilon t_w$ adjacent to the plastic neutral axis of the effective cross-section in accordance with Figure 8.3.



- Key**
- 1 compression
 - 2 tension
 - 3 plastic neutral axis
 - 4 neglect

Figure 8.3 — Effective Class 2 web

8.2.2.5 Effective cross-section properties of Class 4 cross-sections

(1) The effective cross-section properties of Class 4 cross-sections should be based on the effective widths of the compression parts.

NOTE For cold-formed sections (other than structural hollow sections conforming to EN 10219 (all parts)), see EN 1993-1-3.

- (2) The effective widths of planar compression parts should be determined according to EN 1993-1-5.
- (3) Where a Class 4 cross-section is subjected to a compressive axial force N_{Ed} , the method given in EN 1993-1-5 should be used to determine the possible shift e_N of the centroid of the effective area A_{eff} relative to the centroid of the gross cross-section and the resulting additional moment:

$$\Delta M_{Ed} = N_{Ed} e_N \tag{8.7}$$

NOTE The sign of the additional moment depends on the effect in the combination of internal forces and moments, see 8.2.9.3(2).

- (4) For Class 4 circular or elliptical hollow sections conforming to EN 10210 (all parts) and EN 10219 (all parts), in compression, the effective cross-sectional area A_{eff} may be determined using the equivalent diameter d_e according to Table 7.5 and the thickness t as:

$$A_{\text{eff}} = A \sqrt{\frac{90 \varepsilon^2}{d_e / t}} \quad \text{for } d_e / t \leq 240 \varepsilon^2 \quad (8.8)$$

(5) For Class 4 circular or elliptical hollow sections conforming to EN 10210 (all parts) and EN 10219 (all parts), in bending, the effective section modulus W_{eff} may be determined using the equivalent diameter d_e according to Table 7.5 and the thickness t as:

$$W_{\text{eff}} = W_{\text{el}} \sqrt[4]{\frac{140 \varepsilon^2}{d_e / t}} \quad \text{for } d_e / t \leq 240 \varepsilon^2 \quad (8.9)$$

(6) Circular or elliptical hollow sections exceeding the limit of d_e/t specified in (4) and (5) should be designed in accordance with EN 1993-1-6. The tolerance quality class should be chosen on the basis of the manufacturing tolerances specified in the pertinent hollow section product standard.

NOTE EN 1993-1-6 does not make reference to cross-sectional classes and effective cross-sections, but allows for the verification of the hollow section as a shell structure depending on specified tolerance quality classes.

8.2.2.6 Section properties for the characteristic resistance

(1) The characteristic resistance to an axial force N_{Rk} and the characteristic bending moment resistance M_{Rk} should be determined using the section properties from Table 8.1, according to the class of the cross-section, as follows:

$$N_{\text{Rk}} = A_i f_y \quad (8.10)$$

$$M_{y,\text{Rk}} = W_y f_y \quad (8.11)$$

$$M_{z,\text{Rk}} = W_z f_y \quad (8.12)$$

Table 8.1 — Section properties according to the class of the cross-section

Class	1	2	3	4
Section area A_i	A	A	A	A_{eff}
Section modulus W_y	$W_{\text{pl},y}$	$W_{\text{pl},y}$	$W_{\text{el},y}^{\text{a, b}}$	$W_{\text{eff},y}^{\text{b}}$
Section modulus W_z	$W_{\text{pl},z}$	$W_{\text{pl},z}$	$W_{\text{el},z}^{\text{a, b}}$	$W_{\text{eff},z}^{\text{b}}$
^a Alternatively, $W_{\text{ep},y}$ or $W_{\text{ep},z}$ may be used, see Annex B. ^b The elastic section modulus corresponds to the extreme fibre with the maximum elastic stress.				

8.2.3 Tension

(1) The design value of the tensile axial force N_{Ed} at each cross-section shall satisfy the criterion in Formula (8.13):

$$\frac{N_{Ed}}{N_{t,Rd}} \leq 1,0 \quad (8.13)$$

where

$N_{t,Rd}$ is the design tension resistance., it is given by:

(2) The design tension resistance of the gross cross-section $N_{t,Rd}$ should be taken as:

$$N_{t,Rd} = N_{pl,Rd} = \frac{A f_y}{\gamma_{M0}} \quad (8.14)$$

(3) For cross-sections with holes, the design tension resistance $N_{t,Rd}$ should be taken as the smaller of:

- a) the design plastic resistance, $N_{pl,Rd}$, of the gross cross-section according to Formula (8.14);
- b) the design ultimate resistance of the net cross-section considering holes:

$$N_{u,Rd} = \frac{k A_{net} f_u}{\gamma_{M2}} \quad (8.15)$$

where

$k = 1,0$ for sections with smooth holes (i.e. holes without notches), for example holes fabricated by drilling or water jet cutting;

$k = 0,9$ for sections with rough holes (i.e. holes with notches), for example holes fabricated by punching or flame cutting;

$k = 0,9$ for structures subjected to fatigue.

(4) Where capacity design according to EN 1998 (see 3.1.10) is required, the design plastic resistance $N_{pl,Rd}$ of the gross cross-section should be less than the design ultimate resistance of the net cross-section considering holes $N_{u,Rd}$. Steel grades higher than S460 should not be used where capacity design is required.

(5) For category C connections according to EN 1993-1-8, the design tension resistance $N_{t,Rd}$ of the net cross-section considering holes for fasteners should be taken as $N_{net,Rd}$, from Formula (8.16):

$$N_{net,Rd} = \frac{A_{net} f_y}{\gamma_{M0}} \quad (8.16)$$

NOTE For angles connected by one leg and other asymmetrically connected members in tension, see EN 1993-1-8.

8.2.4 Compression

(1) The design value of the compressive axial force N_{Ed} at each cross-section shall satisfy the criterion in Formula (8.17):

$$\frac{N_{Ed}}{N_{c,Rd}} \leq 1,0 \quad (8.17)$$

(2) The design resistance of the cross-section for uniform compression $N_{c,Rd}$ should be taken as:

$$N_{c,Rd} = \frac{N_{Rk}}{\gamma_{M0}} \quad (8.18)$$

where

N_{Rk} is the characteristic resistance to an axial force according to 8.2.2.6.

(3) Fastener holes, except for oversized and slotted holes as defined in EN 1090-2, may be neglected in compression members if they are filled by fasteners.

(4) For asymmetric Class 4 cross-sections, the method given in 8.2.9.3 should be used to take into account the additional moment ΔM_{Ed} due to the eccentricity of the centroidal axis of the effective cross-section, see 8.2.2.5(3).

8.2.5 Bending

(1) The design value of the bending moment M_{Ed} at each cross-section shall satisfy the criterion in Formula (8.19):

$$\frac{M_{Ed}}{M_{c,Rd}} \leq 1,0 \quad (8.19)$$

where

$M_{c,Rd}$ is determined considering fastener holes, see (4) to (6).

(2) The design bending moment resistance of the cross-section for bending about one principal axis $M_{c,Rd}$, should be calculated from Formula (8.20):

$$M_{c,Rd} = \frac{M_{Rk}}{\gamma_{M0}} \quad (8.20)$$

where

M_{Rk} is the characteristic value of the bending moment resistance according to 8.2.2.6, for the appropriate axis of bending.

(3) For bending about both axes, the methods given in 8.2.9 should be used.

(4) Holes in the tension flange may be ignored if for the tension flange the condition in Formula (8.21) is satisfied:

$$\frac{A_{t,net} k f_u}{\gamma_{M2}} \geq \frac{A_t f_y}{\gamma_{M0}} \quad (8.21)$$

where

k is defined in 8.2.3(2);

$A_{t,net}$ is the net area of the tension flange;

A_t is the area of the tension flange.

NOTE The criterion in (4) provides capacity design (see 3.1.10).

(5) Holes in the tension zone of the web may be ignored if the limit given in (4) is satisfied for the complete tension zone comprising the tension flange plus the tension zone of the web.

(6) Fastener holes except for oversized and slotted holes in the compression zone of the cross-section may be neglected provided that they are filled by fasteners.

8.2.6 Shear

(1) The design value of the shear force V_{Ed} at each cross-section shall satisfy the criterion in Formula (8.22):

$$\frac{V_{Ed}}{V_{c,Rd}} \leq 1,0 \quad (8.22)$$

where

$V_{c,Rd}$ is the design shear resistance. For plastic design $V_{c,Rd}$ is the design plastic shear resistance $V_{pl,Rd}$.

For elastic design, verification of the cross-section should be made using the yield criterion in (4) and (5).

(2) In the absence of torsion the design plastic shear resistance $V_{pl,Rd}$ should be calculated from Formula (8.23):

$$V_{pl,Rd} = \frac{A_v \left(f_y / \sqrt{3} \right)}{\gamma_{M0}} \quad (8.23)$$

where

A_v is the shear area.

(3) Depending on the section, the shear area A_v may be taken as follows:

a) in doubly symmetric rolled I and H sections, for load parallel to the web:

$$A - 2bt_f + (t_w + 2r)t_f \quad \text{but not less than } \eta h_w t_w$$

b) in doubly symmetric rolled I and H sections, for load parallel to the flanges: $2bt_f$

c) in rolled channel sections, for load parallel to the web: $A - 2bt_f + (t_w + r)t_f$

d) in T-sections, for load parallel to the web

— for rolled T-sections: $A - bt_f + (t_w + 2r) \frac{t_f}{2}$

— for welded T-sections: $t_w \left(h - \frac{t_f}{2} \right)$

e) in welded I, H and box sections, for load parallel to the web: $\eta \Sigma (h_w t_w)$

f) in welded I, H, channel and box sections, for load parallel to the flanges: $A - \Sigma (h_w t_w)$

g) in rectangular hollow sections of uniform thickness:

— for load parallel to the depth: $A h / (b + h)$

— for load parallel to the width: $A b / (b + h)$

h) in circular hollow sections and tubes of uniform thickness: $2A/\pi$

i) in elliptical hollow sections of uniform thickness:

— for load parallel to the depth: $2 (h - t) t$

— for load parallel to the width: $2 (b - t) t$

where

A is the cross-sectional area;

b is the overall width;

h is the overall depth;

h_w is the depth of the web measured between the flanges, see Figure 3.1;

r is the root radius;

t_f is the flange thickness;

t_w is the web thickness (if the web thickness is not constant t_w should be taken as the minimum thickness);

t is the thickness of the hollow section;

η see EN 1993-1-5.

NOTE η can conservatively be taken as 1,0.

(4) For elastic design, the yield criterion in Formula (8.24) for the critical point of the cross-section may be used, unless shear buckling verification according to EN 1993-1-5 is required:

$$\frac{\tau_{Ed}}{\left(f_y / \sqrt{3}\right) \gamma_{M0}} \leq 1,0 \quad (8.24)$$

The design value of the shear stress τ_{Ed} may be calculated from Formula (8.25):

$$\tau_{Ed} = \frac{V_{Ed} S}{I t} \quad (8.25)$$

where

V_{Ed} is the design value of the shear force;

S is the static moment about the centroidal axis of the portion of the cross-section between the verification point at which the shear stress is calculated and the boundary of the cross-section;

I is the moment of inertia of the whole cross-section;

t is the thickness at the verification point.

NOTE Verification according to (4) can be conservative as it excludes partial plastic shear distribution, which is permitted in elastic design.

(5) For I- or H-sections, the design value of the shear stress τ_{Ed} in the web may be calculated from Formula (8.26):

$$\tau_{Ed} = \frac{V_{Ed}}{A_w} \quad \text{if } A_f / A_w \geq 0,6 \quad (8.26)$$

where

A_f is the area of one flange;

A_w is the area of the web: $A_w = h_w t_w$.

(6) The shear buckling resistance for webs without intermediate stiffeners should be determined according to EN 1993-1-5, if the web slenderness is such that the condition in Formula (8.27) applies:

$$\frac{h_w}{t_w} > 72 \frac{\varepsilon}{\eta} \quad (8.27)$$

For the value of η , see EN 1993-1-5. For the criterion in Formula (8.27), $\eta = 1,2$ may be assumed.

NOTE For shear buckling resistance, EN 1993-1-5 covers panels supported on four edges only.

(7) For circular or elliptical hollow sections of uniform thickness, the shear buckling resistance should be determined in accordance with EN 1993-1-6 (see also 8.2.2.5(6)) if:

- $(d - t) / t > 100 \varepsilon^2$ for circular hollow sections;
- $(h - t) / t > 100 \varepsilon^2$ for elliptical hollow sections loaded parallel to the depth;
- $(b - t) / t > 100 \varepsilon^2$ for elliptical hollow sections loaded parallel to the width.

(8) Fastener holes should be considered for the verification of the design shear resistance at connection zones in accordance with EN 1993-1-8 but may be neglected elsewhere.

8.2.7 Torsion

(1) The total internal torsional moment T_{Ed} at any cross-section should be considered as the sum of two contributions as indicated in Formula (8.28):

$$T_{Ed} = T_{t,Ed} + T_{w,Ed} \quad (8.28)$$

where

$T_{t,Ed}$ is the design value of the internal St. Venant torsional moment;

$T_{w,Ed}$ is the design value of the internal warping torsional moment.

(2) The values of $T_{t,Ed}$ and $T_{w,Ed}$ at any cross-section may be determined by elastic analysis, taking account of the section properties of the member, the conditions of restraint at the supports and the distribution of the actions along the member.

(3) For elastic design of members subject to a torsional moment, the criterion in 8.2.1(5) may be applied. The following stresses due to torsion should be taken into account:

- the longitudinal stresses $\sigma_{w,Ed}$ due to the bimoment B_{Ed} ;
- the shear stresses $\tau_{t,Ed}$ due to St. Venant torsion $T_{t,Ed}$;
- the shear stresses $\tau_{w,Ed}$ due to warping torsion $T_{w,Ed}$.

(4) As a simplification, in the case of a member with a hollow cross-section, the effects of torsional warping may be neglected. Also as a simplification, in the case of a member with open cross-section, such as I or H, the effects of St. Venant torsion may be neglected.

(5) For a Class 1 or Class 2 cross-section, the verification may refer to its plastic resistance and the bimoment B_{Ed} may be derived from an elastic analysis, see (2).

(6) For determining the plastic moment resistance of a cross-section due to bending and torsion, only torsion effects B_{Ed} should be derived from elastic analysis, see (3).

(7) For Class 1 or Class 2 symmetric I-cross-sections in bending about the major axis, the design plastic bending moment resistance $M_{c,B,Rd}$ reduced by the bimoment B_{Ed} may be calculated from Formula (8.29):

$$M_{c,B,Rd} = \sqrt{1 - \frac{B_{Ed}}{B_{Rd}}} M_{pl,Rd} \quad (8.29)$$

where

B_{Ed} is the design value of the bimoment provided that: $B_{Ed} \leq B_{Rd}$;

B_{Rd} is the design value of the plastic resistance to bimoment;

$$B_{Rd} = \frac{W_{pl,B} f_y}{\gamma_{M0}}$$

$W_{pl,B}$ is the plastic section modulus for the plastic resistance to bimoment.

NOTE For doubly symmetric I- and H-sections: $W_{pl,B} = 0,25 t_f b^2 (h - t_f)$.

(8) For the verification of the resistance of closed hollow sections, the design shear resistance of the individual parts of the cross-section according to EN 1993-1-5 should be taken into account if shear buckling is relevant.

(9) Where the torsional moment is combined with a shear force, the plastic shear resistance should be reduced from $V_{pl,Rd}$ to $V_{pl,T,Rd}$ and the design shear force should satisfy:

$$\frac{V_{Ed}}{V_{pl,T,Rd}} \leq 1,0 \quad (8.30)$$

in which $V_{pl,T,Rd}$ should be taken as:

— for an I or H section:

$$V_{pl,T,Rd} = \sqrt{1 - \frac{\tau_{t,Ed}}{1,25 \left(f_y / \sqrt{3} \right) / \gamma_{M0}}} V_{pl,Rd} \quad (8.31)$$

— for a channel section:

$$V_{pl,T,Rd} = \left[\sqrt{1 - \frac{\tau_{t,Ed}}{1,25 \left(f_y / \sqrt{3} \right) / \gamma_{M0}}} - \frac{\tau_{w,Ed}}{\left(f_y / \sqrt{3} \right) / \gamma_{M0}} \right] V_{pl,Rd} \quad (8.32)$$

— for a structural hollow section:

$$V_{pl,T,Rd} = \left[1 - \frac{\tau_{t,Ed}}{\left(f_y / \sqrt{3} \right) / \gamma_{M0}} \right] V_{pl,Rd} \quad (8.33)$$

8.2.8 Combined bending and shear

(1) Where bending moment and shear force are present, the effect of shear force on the bending moment resistance of the cross-section should be considered.

(2) The design value of the bending moment M_{Ed} at each cross-section shall satisfy the criterion in Formula (8.34):

$$\frac{M_{Ed}}{M_{V,Rd}} \leq 1,0 \quad (8.34)$$

where

$M_{V,Rd}$ is the design plastic bending moment resistance reduced due to the shear force V_{Ed} .

(3) The effect of the shear force V_{Ed} on the design bending moment resistance $M_{c,Rd}$ may be neglected if the condition in Formula (8.35) is satisfied:

$$V_{Ed} \leq \eta_V V_{pl,Rd} \quad (8.35)$$

Where shear buckling reduces the section resistance, see 8.2.6(6), 8.2.6(7) and EN 1993-1-5.

NOTE The value of η_V is 0,5 unless the National Annex gives a different value depending on the type of section. Modifications of ρ in Formulae (8.37), (8.40) and (8.61) can be given accordingly.

(4) In all other cases, the reduced design bending moment resistance of the cross-section to the combination of bending moment and shear force should be taken as the design bending moment resistance of the cross-section determined using a reduced yield strength $f_{y,red}$ for the cross-sectional area affected by shear:

$$f_{y,red} = (1 - \rho) f_y \quad (8.36)$$

where

$$\rho = \left(\frac{2 V_{Ed}}{V_{c,Rd}} - 1 \right)^2 \quad (8.37)$$

$V_{c,Rd}$ is taken as $V_{pl,Rd}$ obtained from 8.2.6(2).

For circular and elliptical hollow sections of uniform thickness, the reduced yield strength $f_{y,red}$ from Formula (8.36) should be applied to the full cross-section area.

(5) For calculation purposes, instead of reducing the yield strength, the plate thickness of the relevant part of the cross-section may be reduced.

(6) When torsion is present $V_{c,Rd}$ should be taken as $V_{pl,T,Rd}$ from 8.2.7.

(7) For I-sections with Class 1 and Class 2 cross-sections, if torsion is present, the reduced design plastic moment resistance $M_{B,V,Rd}$ should be taken as the design resistance of the cross-section calculated using a reduced yield strength $f_{y,red}$ as follows:

$$f_{y,red} = (1 - \rho) f_y \frac{M_{B,Rd}}{M_{c,Rd}} \quad \text{for the cross-sectional affected by shear} \quad (8.38)$$

$$f_{y,red} = f_y \frac{M_{B,Rd}}{M_{c,Rd}} \quad \text{for the rest of the cross-section} \quad (8.39)$$

where

$M_{B,Rd}$ is given in 8.2.7(7). In the absence of warping torsion: $M_{B,Rd} = M_{c,Rd}$

$$\rho = \left(\frac{2 V_{Ed}}{V_{pl,T,Rd}} - 1 \right)^2 \quad \begin{array}{l} \text{see 8.2.7, but should be taken as 0 for} \\ V_{Ed} \leq \eta_V V_{pl,T,Rd} \end{array} \quad (8.40)$$

where

η_V is given in 8.2.8(3).

(8) Alternatively, for doubly symmetric I-sections with Class 1 and Class 2 cross-sections under bending about the major axis y-y, the reduced design plastic moment resistance allowing for the shear force and the bimoment may be obtained, as follows:

$$M_{y,B,V,Rd} = \frac{\left(W_{pl,y} - \rho \frac{t_w h_w^2}{4} \right) f_y}{\gamma_{M0}} \frac{M_{y,B,Rd}}{M_{c,y,Rd}} \quad \text{but } M_{y,B,V,Rd} \leq M_{y,B,Rd} \quad (8.41)$$

where

$M_{c,y,Rd}$ is obtained from 8.2.5(2)

$M_{y,B,Rd}$ is obtained from 8.2.7(7)

When warping torsion is not present, then:

$$M_{y,B,Rd} = M_{c,y,Rd} \quad (8.42)$$

NOTE For the interaction of bending, shear and transverse loads, see 8.2.11.

8.2.9 Combined bending and axial force

8.2.9.1 Class 1 and Class 2 cross-sections

(1) Where bending moment and axial force are present, the effect of the axial force on the plastic bending moment resistance should be considered.

(2) For Class 1 and Class 2 cross-sections, the design value of the bending moment M_{Ed} at each cross-section shall satisfy the criterion in Formula (8.43):

$$\frac{M_{Ed}}{M_{N,Rd}} \leq 1,0 \quad (8.43)$$

where

$M_{N,Rd}$ is the design plastic bending moment resistance reduced due to the axial force N_{Ed} .

(3) For a rectangular solid section without fastener holes, $M_{N,Rd}$ should be taken from Formula (8.44):

$$M_{N,Rd} = M_{pl,Rd} \left[1 - \left(\frac{N_{Ed}}{N_{pl,Rd}} \right)^2 \right] \quad (8.44)$$

(4) For doubly symmetric I- and H-sections or other sections with flanges, the effect of the axial force on the plastic bending moment resistance about the y-y axis may be neglected if both the following conditions are satisfied:

$$N_{Ed} \leq 0,25 N_{pl,Rd} \quad (8.45)$$

and

$$N_{Ed} \leq \frac{0,5 h_w t_w f_y}{\gamma_{M0}} \quad (8.46)$$

For doubly symmetric I- and H-sections the effect of the axial force on the plastic bending moment resistance about the z-z axis may be neglected if condition in Formula (8.47) is satisfied:

$$N_{Ed} \leq \frac{h_w t_w f_y}{\gamma_{M0}} \quad (8.47)$$

(5) For doubly symmetric I- and H-sections where holes may be neglected $M_{N,Rd}$ may be taken as:

$$M_{N,y,Rd} = M_{pl,y,Rd} \frac{1-n}{1-0,5a} \quad \text{but } M_{N,y,Rd} \leq M_{pl,y,Rd} \quad (8.48)$$

$$\text{for } n \leq a: \quad M_{N,z,Rd} = M_{pl,z,Rd} \quad (8.49)$$

$$\text{for } n > a \quad M_{N,z,Rd} = M_{pl,z,Rd} \left[1 - \left(\frac{n-a}{1-a} \right)^2 \right] \quad (8.50)$$

where

$$n = N_{Ed} / N_{pl,Rd}$$

$$a = (A - 2bt_f) / A \quad \text{but } a \leq 0,5$$

(6) For rectangular hollow sections of uniform thickness, where holes may be neglected and for welded box sections with equal flanges and equal webs where holes may be neglected, $M_{N,Rd}$ may be taken as:

$$M_{N,y,Rd} = M_{pl,y,Rd} \frac{1-n}{1-0,5a_w} \quad \text{but } M_{N,y,Rd} \leq M_{pl,y,Rd} \quad (8.51)$$

$$M_{N,z,Rd} = M_{pl,z,Rd} \frac{1-n}{1-0,5a_f} \quad \text{but } M_{N,z,Rd} \leq M_{pl,z,Rd} \quad (8.52)$$

where

$$a_w = (A - 2bt) / A \quad \text{but } a_w \leq 0,5 \quad \text{for hollow sections;}$$

$$a_w = (A - 2bt_f) / A \quad \text{but } a_w \leq 0,5 \quad \text{for welded box sections;}$$

$$a_f = (A - 2ht) / A \quad \text{but } a_f \leq 0,5 \quad \text{for hollow sections;}$$

$$a_f = (A - 2ht_w) / A \quad \text{but } a_f \leq 0,5 \quad \text{for welded box sections.}$$

(7) For cross-sections where fastener holes may be neglected, the approximation in Formula (8.53) may be used for circular hollow sections of uniform thickness:

$$M_{N,y,Rd} = M_{N,z,Rd} = M_{pl,Rd} (1 - n^{1,7}) \quad (8.53)$$

(8) For cross-sections where fastener holes may be neglected, the approximation in Formulae (8.54) and (8.55) may be used for elliptical hollow sections of uniform thickness:

$$M_{N,y,Rd} = M_{pl,y,Rd} (1 - n^{1,7}) \quad (8.54)$$

$$M_{N,z,Rd} = M_{pl,z,Rd} (1 - n^{1,7})^{1,18} \quad (8.55)$$

(9) For bi-axial bending the criterion in Formula (8.56) should be satisfied:

$$\left(\frac{M_{y,Ed}}{M_{N,y,Rd}} \right)^{\alpha_y} + \left(\frac{M_{z,Ed}}{M_{N,z,Rd}} \right)^{\alpha_z} \leq 1 \quad (8.56)$$

in which α_y and α_z may be taken as:

— For I- and H-sections:

$$\alpha_y = 2 ; \alpha_z = 5n \text{ but } \alpha_z \geq 1$$

— For rectangular hollow sections:

$$\alpha_y = \alpha_z = \frac{1,66}{1 - 1,13n^2} \quad \text{for } n \leq 0,8$$

$$\alpha_y = \alpha_z = 6 \quad \text{for } n > 0,8$$

— For circular hollow sections:

$$\alpha_y = 2; \alpha_z = 2$$

— For elliptical hollow sections:

$$\alpha_y = 2; \alpha_z = 1,7$$

where

$$n = N_{Ed} / N_{pl,Rd}$$

α_y and α_z may conservatively be taken as 1,0.

8.2.9.2 Class 3 cross-sections

(1) In the absence of shear stress, the maximum longitudinal normal stress $\sigma_{x,Ed}$ for Class 3 cross-sections shall satisfy the criterion in Formula (8.57):

$$\sigma_{x,Ed} \leq \frac{f_y}{\gamma_{M0}} \quad (8.57)$$

where

$\sigma_{x,Ed}$ is the design value of the longitudinal normal stress due to moment and axial force taking account of fastener holes where relevant, see 8.2.3, 8.2.4 and 8.2.5.

(2) Alternatively, for Class 3 doubly symmetric I or H sections, hollow sections and doubly symmetric welded box sections, where fastener holes may be neglected, the elasto-plastic bending moment resistance may be taken into account. In this case, the rules given in Annex B should be applied.

8.2.9.3 Class 4 cross-sections

(1) In the absence of shear stress, the maximum longitudinal normal stress $\sigma_{x,Ed}$ for Class 4 cross-sections shall satisfy the criterion in Formula (8.58):

$$\sigma_{x,Ed} \leq \frac{f_y}{\gamma_{M0}} \quad (8.58)$$

where

$\sigma_{x,Ed}$ is the design value of the longitudinal normal stress due to bending moments and axial force determined using the effective cross-section properties (see 8.2.2.5), taking account of fastener holes where relevant, see 8.2.3, 8.2.4 and 8.2.5.

(2) Alternatively, the simplified criterion in Formula (8.59) may be used:

$$\frac{N_{Ed}}{N_{Rd}} + \frac{M_{y,Ed} + \Delta M_y}{M_{y,Rd}} + \frac{M_{z,Ed} + \Delta M_z}{M_{z,Rd}} \leq 1,0 \quad (8.59)$$

where

N_{Rd} is the design value of the resistance to axial force calculated with the effective area of the cross-section when subjected to uniform compression only, see 8.2.2.6(1);

$M_{y,Rd}$ and $M_{z,Rd}$ are the design values of bending resistance calculated with the effective section modulus (corresponding to the fibre with the maximum elastic stress) of the cross-section when subjected only to moment about the strong axis and the weak axis respectively, see 8.2.2.6(1);

ΔM_y and ΔM_z are the additional bending moments about the strong axis and the weak axis respectively, due to the shift of the centroidal axis when the cross-section is subjected to compression only, see 8.2.2.5(3).

NOTE The signs of N_{Ed} , $M_{y,Ed}$, $M_{z,Ed}$ and ΔM_i depend on the combination of the respective direct stresses.

8.2.10 Combined bending, shear and axial force

(1) Where bending moment, shear force and axial force are present, the effect of both shear force and axial force on the bending moment resistance of the cross-section should be taken into account.

(2) The effect of the shear force V_{Ed} on the resistances defined for bending and axial force in 8.2.9 may be neglected if the condition in Formula (8.35) is satisfied, except where shear buckling reduces the section resistance, see EN 1993-1-5 (for circular or elliptical hollow sections, see EN 1993-1-6).

(3) In all other cases, the reduced design bending moment resistance of the cross-section for combined bending moment, shear force and axial force should be determined using a reduced yield strength $f_{y,red}$ from Formula (8.60) for the cross-sectional area affected by shear when determining the combined resistance for bending moment and axial force:

$$f_{y,red} = (1 - \rho) f_y \quad (8.60)$$

where

$$\rho = \left(\frac{2V_{Ed}}{V_{c,Rd}} - 1 \right)^2 \quad (8.61)$$

$V_{c,Rd}$ is taken as $V_{pl,Rd}$ obtained from 8.2.6(2).

For circular and elliptical hollow sections of uniform thickness, the design resistance of the cross-section for combined bending moment and axial force should be determined using a reduced yield strength $f_{y,red}$ from Formula (8.60) for the full cross-section area.

(4) Instead of reducing the yield strength, the plate thickness of the relevant part of the cross-section may be reduced.

(5) Where torsion is present, $V_{c,Rd}$ should be taken as $V_{pl,T,Rd}$ obtained from 8.2.7.

8.2.11 Resistance to transverse forces

(1) The resistance of unstiffened webs against the design transverse forces $F_{z,Ed}$ should be verified using the rules in EN 1993-1-5.

(2)B As simplified approach for a structural member in a building with transverse force applied on one flange only, provided that the condition in Formula (8.62) and criterion in Formula (8.63) are fulfilled:

$$F_{z,Ed} \leq 534 \varepsilon^2 \frac{t_w^3}{d_w} \frac{f_y}{\gamma_{M0}} \quad (8.62)$$

$$M_{Ed} \leq M_{el,y,Rd} \quad (8.63)$$

the resistance of an unstiffened web to a design transverse force $F_{z,Ed}$ may be verified using criterion in Formula (8.64):

$$\frac{F_{z,Ed}}{k_w F_{z,Rd}} \leq 1,0 \quad (8.64)$$

where

$$k_w = 1,0 \quad \text{for} \quad |\sigma_{com,Ed}| \leq 0,7 \frac{f_y}{\gamma_{M0}} \quad \text{or} \quad F_{z,Ed} \text{ applied at the tension flange}$$

$$k_w = 1,7 - \frac{|\sigma_{com,Ed}|}{\frac{f_y}{\gamma_{M0}}} \quad \text{if} \quad V_{Ed} \leq F_{z,Rd} \quad \text{and} \quad F_{z,Ed} \text{ applied at the compression flange}$$

$$k_w = 3,33 \cdot \left(1 - \frac{|\sigma_{com,Ed}|}{\frac{f_y}{\gamma_{M0}}} \right) \quad \text{if} \quad V_{Ed} > F_{z,Rd} \quad \text{and} \quad F_{z,Ed} \text{ applied at the compression flange}$$

$\sigma_{com,Ed}$ is the design value of the longitudinal compressive stress σ_x in the web at the end of fillets or weld toes directly below the transverse force $F_{z,Ed}$;

d_w is the web depth between fillets or weld toes, see Figure 8.4;

$F_{z,Rd}$ is the design resistance of an unstiffened web to a transverse force:

$$F_{z,Rd} = \frac{\ell_y t_w f_y}{\gamma_{M0}} \quad (8.65)$$

ℓ_y is the effective length for the resistance to transverse forces determined according to the length of stiff bearing s_s , see Figure 8.4.

V_{Ed} is the design shear force in the section where the transverse force $F_{z,Ed}$ is applied.

M_{Ed} is the design bending moment in the section where the transverse force $F_{z,Ed}$ is applied.

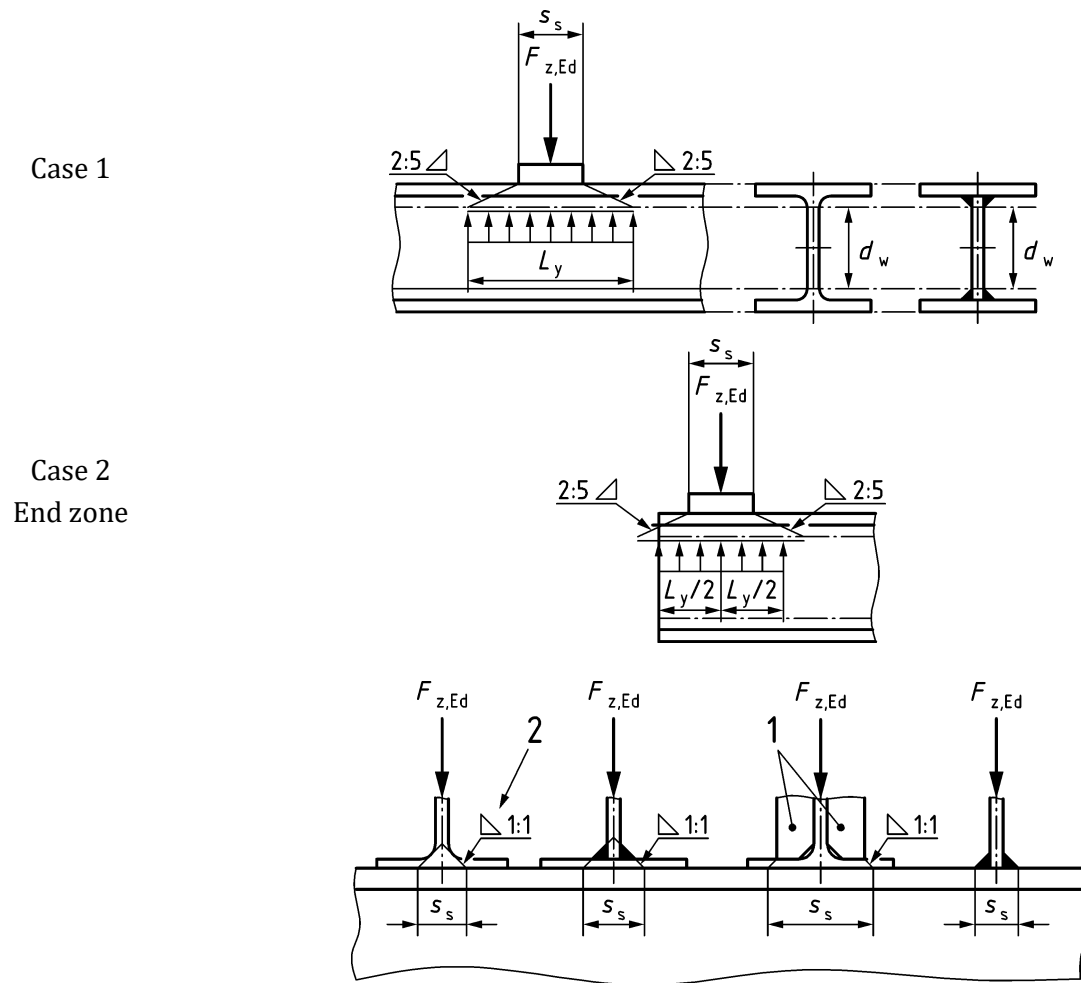


Figure 8.4 — Definition of the lengths ℓ_y , d_w and s_s

(3) In the presence of longitudinal tensile stress in the flange under the transverse compressive force, the criterion in Formula (8.1) should be satisfied with:

$$\sigma_{z,Ed} = \frac{F_{z,Ed}}{\ell_y t_w} \quad (8.66)$$

8.3 Buckling resistance of members

8.3.1 Uniform members in compression

8.3.1.1 Buckling resistance

(1) The design value of the compressive axial force N_{Ed} should satisfy the criterion in Formula (8.67):

$$\frac{N_{Ed}}{N_{b,Rd}} \leq 1,0 \quad (8.67)$$

where

N_{Ed} is the design value of the compressive axial force;

$N_{b,Rd}$ is the design buckling resistance in compression.

(2) Verification of the buckling resistance in compression may be omitted if the relative slenderness is $\bar{\lambda} \leq 0,2$, see 8.3.1.2, or if the design value of the compressive axial force is $N_{Ed} \leq 0,04 N_{cr}$.

(3) For members with asymmetric Class 4 cross-sections, the additional moment ΔM_{Ed} due to the eccentricity of the centroidal axis of the effective section should be considered, see 8.2.2.5(3). Verification should be in accordance with 8.3.3 or 8.3.4.

(4) The design buckling resistance in compression $N_{b,Rd}$ should be taken as:

$$N_{b,Rd} = \frac{\chi N_{Rk}}{\gamma_{M1}} \quad (8.68)$$

where

χ is the buckling reduction factor which should be determined as a function of the relative slenderness $\bar{\lambda}$ of the compression member, for the relevant buckling mode. See 8.3.1.3 for flexural buckling and 8.3.1.4 for torsional or torsional-flexural buckling.

N_{Rk} is the characteristic value of the resistance to compression according to 8.2.2.6.

(5) For determining the buckling resistance of members with tapered sections along the member or for non-uniform distribution of the compressive axial force, second order analysis according to 7.2.2 may be performed.

NOTE For out-of-plane buckling, see also 8.3.4.

(6) In determining A and A_{eff} holes for fasteners at the compression member ends may be neglected.

8.3.1.2 Slenderness of compression members

(1) The relative slenderness $\bar{\lambda}$ should be taken from Formula (8.69):

$$\bar{\lambda} = \sqrt{\frac{N_{Rk}}{N_{cr}}} \quad (8.69)$$

where

N_{cr} is the elastic critical force for the relevant buckling mode based on the gross cross-sectional properties:

$$N_{cr} = N_{cr,y} \quad \text{for elastic flexural buckling about y-y, leading to } \bar{\lambda}_y;$$

$$N_{cr} = N_{cr,z} \quad \text{for elastic flexural buckling about z-z, leading to } \bar{\lambda}_z;$$

$$N_{cr} = N_{cr,T} \quad \text{for torsional buckling, leading to } \bar{\lambda}_T;$$

$$N_{cr} = N_{cr,TF} \quad \text{for elastic torsional or torsional-flexural buckling, leading to } \bar{\lambda}_{TF}.$$

NOTE Formulae for the elastic critical forces can be found in the technical report prCEN/TR 1993-1-103.

(2) As an alternative, for flexural buckling, the relative slenderness $\bar{\lambda}$ may be taken as:

$$\bar{\lambda} = \frac{L_{cr}}{i} \frac{1}{\lambda_1} \quad \text{for Class 1, 2 and 3 cross-sections} \quad (8.70)$$

$$\bar{\lambda} = \frac{L_{cr}}{i} \frac{\sqrt{\frac{A_{eff}}{A}}}{\lambda_1} \quad \text{for Class 4 cross-sections} \quad (8.71)$$

where

L_{cr} is the buckling length in the buckling plane considered;

i is the radius of gyration about the relevant axis, determined using the properties of the gross cross-section;

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = 93,9\varepsilon \quad (8.72)$$

where

ε is the material parameter defined in 5.2.5(2).

8.3.1.3 Buckling reduction factor for flexural buckling

(1) For flexural buckling of members in axial compression, the value of the buckling reduction factor χ for the appropriate relative slenderness $\bar{\lambda}$ should be determined from the relevant buckling curve according to Formula (8.73):

$$\chi = \frac{1}{\Phi + \sqrt{\Phi^2 - \bar{\lambda}^2}} \quad \text{but } \chi \leq 1,0 \quad (8.73)$$

where

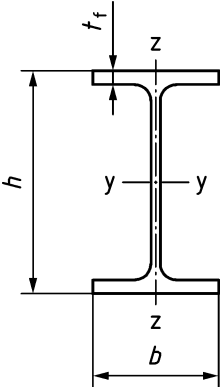
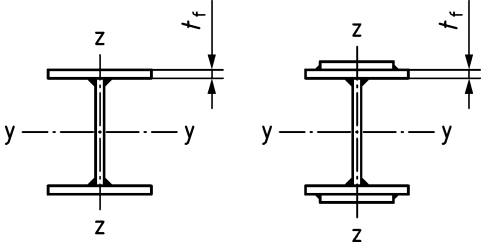
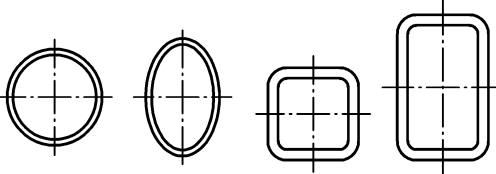
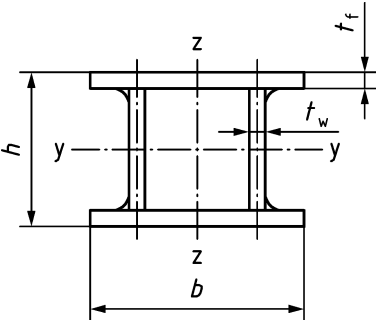
$$\Phi = 0,5 \left[1 + \alpha (\bar{\lambda} - 0,2) + \bar{\lambda}^2 \right] \quad (8.74)$$

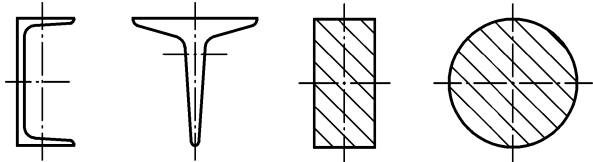
(2) The imperfection factor α corresponding to the appropriate buckling curve, see Table 8.3, should be obtained from Table 8.2.

Table 8.2 — Imperfection factors α for buckling curves

Buckling curve	a_0	a	b	c	d
Imperfection factor α	0,13	0,21	0,34	0,49	0,76

Table 8.3 — Selection of buckling curve for flexural buckling

Cross-section		Limits		Buckling about axis	Buckling curve	
					S235 S275 S355 S420	S460 up to S700 inclusive
Rolled I- or H-sections		$h/b > 1,2$	$t_f \leq 40$ mm	y-y z-z	a b	a_0 a
			$t_f > 40$ mm	y-y z-z	b c	a b
		$h/b \leq 1,2$	$t_f \leq 100$ mm	y-y z-z	b c	a b
			$t_f > 100$ mm	y-y z-z	d d	c c
Welded I-sections		$t_f \leq 40$ mm		y-y z-z	b c	b c
		$t_f > 40$ mm		y-y z-z	c d	c d
Hollow sections		hot finished		any	a	a_0
		cold-formed		any	c	c
Welded box sections		generally (except as below)		any	b	b
		thick welds: $a > 0,5t_f$ and $b/t_f < 30$, and $h/t_w < 30$		any	c	c

Cross-section		Limits	Buckling about axis	Buckling curve	
				S235 S275 S355 S420	S460 up to S700 inclusive
U-, T- and solid sections			any	c	c
L-sections		Rolled sections	any	b	a
		Welded sections $t \leq 40$ mm	any	c	c

(3) Values of the reduction factor χ for the appropriate relative slenderness $\bar{\lambda}$ may be obtained from Figure 8.5.

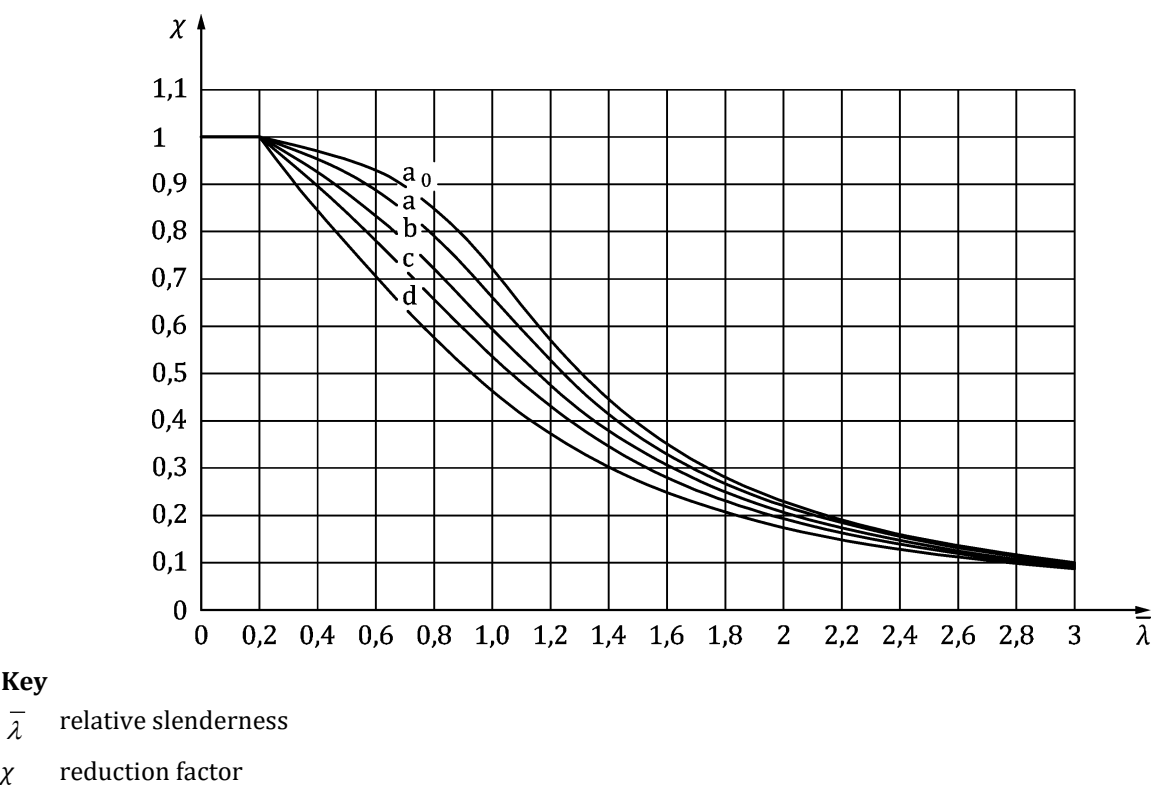


Figure 8.5 — Buckling curves

8.3.1.4 Buckling reduction factors for torsional and torsional-flexural buckling

(1) It should be considered that for members with open cross-sections, the resistance of the member to either torsional or torsional-flexural buckling can be less than its resistance to flexural buckling.

NOTE This is relevant for members with doubly symmetric cross-sections in combination with intermediate lateral restraints, as well as for cruciform and asymmetric cross-sections, e.g. channel, L, T, and I- or H- sections with single symmetry.

(2) For symmetric or asymmetric cross-sections, the buckling reduction factor χ_T for torsional buckling or χ_{TF} for torsional-flexural buckling may be determined by using the column buckling formulae and curves of Formula (8.73) and Figure 8.5, with the buckling coefficients for z-z in accordance with Table 8.3. The relative slenderness $\bar{\lambda}_T$ or $\bar{\lambda}_{TF}$ as defined in 8.3.1.2 should be used for $\bar{\lambda}$.

(3) For doubly symmetric I- and H-section members in compression with continuous or discrete intermediate lateral restraints, failing in torsional or torsional-flexural buckling (see Figure 8.6), the appropriate buckling reduction factor χ_{TF} may be more accurately determined from Formula (8.75):

$$\chi_{TF} = \frac{1}{\Phi_{TF} + \sqrt{\Phi_{TF}^2 - \bar{\lambda}_{TF}^2}} \quad \text{but } \chi_{TF} \leq 1,0 \tag{8.75}$$

where

$$\Phi_{TF} = 0,5 \left[1 + \left(\frac{\bar{\lambda}_{TF}}{\bar{\lambda}_z} \right)^2 \cdot \alpha_{TF} (\bar{\lambda}_z - 0,2) + \bar{\lambda}_{TF}^2 \right] \tag{8.76}$$

NOTE 1 For the relative slenderness $\bar{\lambda}_{TF}$ and $\bar{\lambda}_z$, see (5).

NOTE 2 For the imperfection factor α_{TF} , see (6).

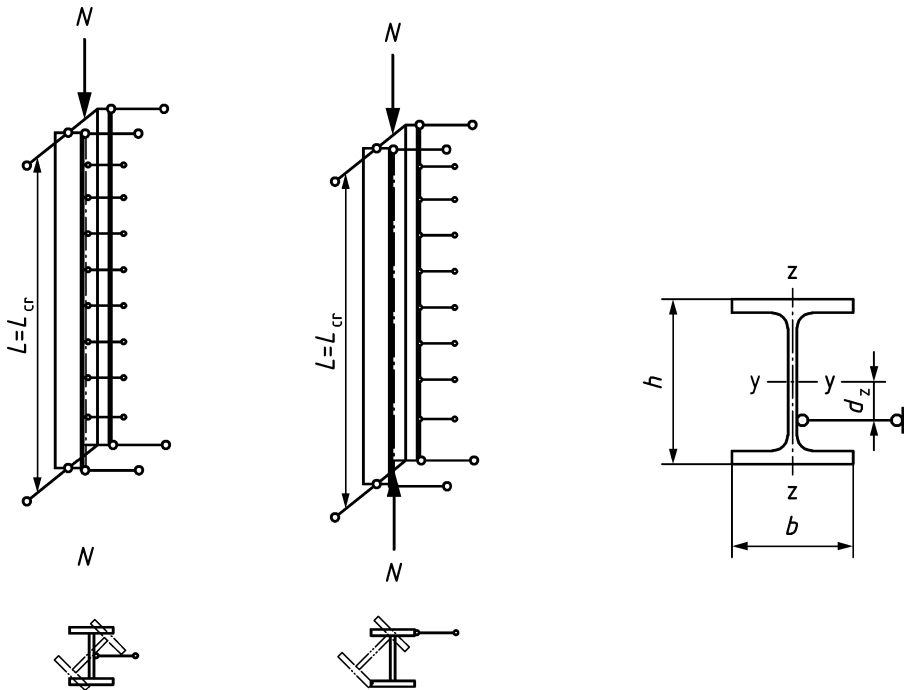


Figure 8.6 — Torsional-flexural buckling of laterally braced doubly symmetric sections – Geometrical definitions

(4) For a torsional buckling mode, the buckling reduction factor χ_T should be used; it should be determined from Formula (8.75) using the relative slenderness $\bar{\lambda}_T$ and the parameter Φ_T , instead of $\bar{\lambda}_{TF}$ and Φ_{TF} .

(5) The relative slenderness $\bar{\lambda}_{TF}$ and $\bar{\lambda}_z$ in Formulae (8.75) and (8.76) should be determined in accordance with 8.3.1.2 using the weak axis (z-z) buckling length L_{cr} of the unrestrained flange.

(6) The imperfection factor α_{TF} in Formula (8.76) should be determined from Formula (8.77):

$$\alpha_{TF} = \alpha \sqrt{\frac{A f_y (i_p^2 + d_z^2)}{6,25 G I_T}} \quad \text{but} \quad \alpha_{TF} \leq \alpha \quad (8.77)$$

where

α is the imperfection factor according to Table 8.3, for buckling about z-z axis,

i_p is the polar radius of gyration such that:

$$i_p^2 = \frac{I_y + I_z}{A}$$

d_z is the distance of the intermediate lateral supports from the shear centre, see Figure 8.6.

(7) Where individual/discrete intermediate lateral supports are provided, a buckling check for weak-axis flexural buckling between the intermediate supports should be carried out.

8.3.2 Uniform members in bending

8.3.2.1 Lateral torsional buckling resistance

(1) A laterally unrestrained member subject to a major axis bending moment should be verified against lateral torsional buckling. The design value of the bending moment M_{Ed} should satisfy the criterion in Formula (8.78):

$$\frac{M_{Ed}}{M_{b,Rd}} \leq 1,0 \quad (8.78)$$

where

M_{Ed} is the design value of the moment;

$M_{b,Rd}$ is the design value of the lateral torsional buckling resistance.

(2) Verification of the lateral torsional buckling resistance may be neglected for hollow sections and welded box sections or beams with sufficient restraint to the compression flange which are not susceptible to lateral torsional buckling.

(3) Conditions on stiffness of restraint devices of beams in buildings may be taken from Annex D.

(4) The design value of the lateral torsional buckling resistance $M_{b,Rd}$ of a laterally unrestrained beam should be taken from Formula (8.79):

$$M_{b,Rd} = \chi_{LT} \frac{M_{Rk}}{\gamma_{M1}} \quad (8.79)$$

where

χ_{LT} is the reduction factor for lateral torsional buckling, which should be determined as a function of the relative slenderness $\bar{\lambda}_{LT}$ of the member, see 8.3.2.3;

M_{Rk} is the characteristic value of the bending moment resistance, see 8.2.2.6.

(5) For determining the lateral torsional buckling resistance of beams with tapered sections, either the general method in 8.3.4 or a second order analysis according to 7.2.2 may be performed.

(6) In determining M_{Rk} , holes for fasteners at the beam end may be neglected.

8.3.2.2 Slenderness for lateral torsional buckling curves

(1) The relative slenderness for lateral torsional buckling $\bar{\lambda}_{LT}$ should be taken from Formula (8.80):

$$\bar{\lambda}_{LT} = \sqrt{\frac{M_{Rk}}{M_{cr}}} \quad (8.80)$$

where

M_{cr} is the elastic critical moment for lateral torsional buckling.

(2) M_{cr} should be based on gross cross-sectional properties and considers the loading conditions, the actual moment distribution and the lateral restraints.

NOTE Formulae for the elastic critical moment can be found in the technical report prCEN/TR 1993-1-103.

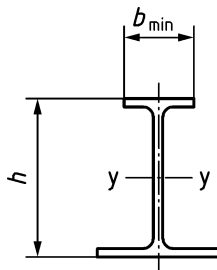
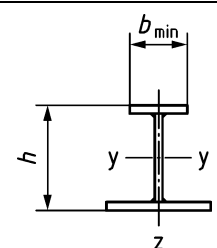
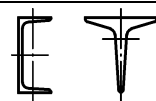
8.3.2.3 Reduction factors χ_{LT} for lateral torsional buckling

(1) Verification of lateral torsional buckling resistance may be neglected when the relative slenderness for lateral torsional buckling satisfies $\bar{\lambda}_{LT} \leq \bar{\lambda}_{LT,0}$ or when the design value of the bending moment satisfies: $M_{Ed} \leq \bar{\lambda}_{LT,0}^2 M_{cr}$.

NOTE $\bar{\lambda}_{LT,0} = 0,4$ if end-fork support conditions are assumed for the calculation of M_{cr} between lateral supports. Limitations of validity concerning the beam depth or h/b ratio and the limiting value $\bar{\lambda}_{LT,0}$ can be set by the National Annex.

(2) In general cases of prismatic members with arbitrary boundary conditions, the lateral torsional buckling reduction factor χ_{LT} may be determined by using the buckling formulae and curves of Formula (8.73) and Figure 8.5. The relative slenderness for lateral torsional buckling $\bar{\lambda}_{LT}$ should be used for the relative slenderness $\bar{\lambda}$, and the selection of buckling curves as specified in Table 8.4 should apply.

Table 8.4 — Selection of lateral torsional buckling curves for general cases

Cross-section		Limits	Lateral torsional buckling curve
Rolled I- or H-sections ^a		$h/b_{\min} \leq 2,0$	a
		$h/b_{\min} > 2,0$	b
Welded I-sections ^a		$h/b_{\min} \leq 2,0$	c
		$h/b_{\min} > 2,0$	d
Other sections (e.g. C-, U-, T-sections)		—	d
NOTE The selection of the lateral torsional buckling curve is based on the minimum flange width $b_{\min}$, irrespective of this being the tension or compression flange.			
^a This may also be applied to doubly symmetric sections, where $b_{\min} = b$.			

(3) For doubly symmetric I- and H- sections and fork boundary conditions at both ends, the lateral torsional buckling reduction factor χ_{LT} may be taken as:

$$\chi_{LT} = \frac{f_M}{\Phi_{LT} + \sqrt{\Phi_{LT}^2 - f_M \bar{\lambda}_{LT}^2}} \quad \text{but } \chi_{LT} \leq 1,0 \quad (8.81)$$

where

$$\Phi_{LT} = 0,5 \left[1 + f_M \left(\left(\frac{\bar{\lambda}_{LT}}{\bar{\lambda}_z} \right)^2 \alpha_{LT} (\bar{\lambda}_z - 0,2) + \bar{\lambda}_{LT}^2 \right) \right] \quad (8.82)$$

α_{LT} is the imperfection factor taken from Table 8.5;

$\bar{\lambda}_{LT}$ is the relative slenderness for lateral torsional buckling, as defined in 8.3.2.2;

$\bar{\lambda}_z$ is the corresponding relative slenderness for weak axis flexural buckling, as defined in 8.3.1.2, with the buckling length $L_{cr,z}$, taken as the distance between the discrete lateral restraints;

f_M is a factor that accounts for the effect of the bending moment distribution between discrete lateral restraints. It may conservatively be taken as 1,0 in cases that cannot be approximated by the diagrams in Table 8.6.

Table 8.5 — Imperfection factor α_{LT} for lateral torsional buckling of doubly symmetric I- and H-sections

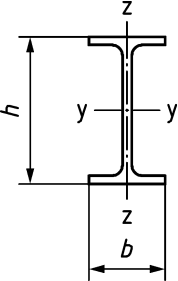
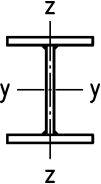
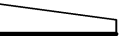
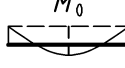
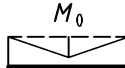
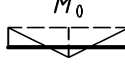
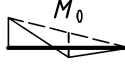
Cross-section		Limits		α_{LT}
Rolled I- or H-sections		$h/b > 1,2$	$t_f \leq 40 \text{ mm}$	$0,12 \sqrt{\frac{W_{el,y}}{W_{el,z}}}$ but : $\alpha_{LT} \leq 0,34$
			$t_f > 40 \text{ mm}$	$0,16 \sqrt{\frac{W_{el,y}}{W_{el,z}}}$ but : $\alpha_{LT} \leq 0,49$
		$h/b \leq 1,2$	—	$0,16 \sqrt{\frac{W_{el,y}}{W_{el,z}}}$ but : $\alpha_{LT} \leq 0,49$
Welded I-sections		$t_f \leq 40 \text{ mm}$		$0,21 \sqrt{\frac{W_{el,y}}{W_{el,z}}}$ but : $\alpha_{LT} \leq 0,64$
		$t_f > 40 \text{ mm}$		$0,25 \sqrt{\frac{W_{el,y}}{W_{el,z}}}$ but : $\alpha_{LT} \leq 0,76$

Table 8.6 — Factors f_M and k_c

Load case	Factor f_M	Factor k_c
$M = \text{uniform}$ 	1,0	1,0
M  ψM $-1 \leq \psi \leq +1$	$1,25 - 0,1\psi - 0,15\psi^2$	$\frac{1}{1,33 - 0,33\psi}$
	1,05	0,94
M_h  M_h 	For $0 \leq \frac{M_0}{M_h} < 2,0$: $1,0 + 1,35 \frac{M_0}{M_h} - 0,33 \left(\frac{M_0}{M_h} \right)^3$ For $\frac{M_0}{M_h} \geq 2$: 1,05	$\frac{M_0}{M_h} < 1,0$: 1,00 $\frac{M_0}{M_h} \geq 1,0$: 0,90
M_h 	For $0 \leq \frac{M_0}{M_h} < 1,47$: $1,25 + 0,5 \left(\frac{M_0}{M_h} \right)^2 - 0,275 \left(\frac{M_0}{M_h} \right)^4$ For $\frac{M_0}{M_h} \geq 1,47$: 1,05	$\frac{M_0}{M_h} < 0,5$: 0,75 $\frac{M_0}{M_h} \geq 0,5$: 0,91
	1,10	0,86
M_h  M_h 	For $0 \leq \frac{M_0}{M_h} < 2,0$: $1,0 + 1,25 \frac{M_0}{M_h} - 0,30 \left(\frac{M_0}{M_h} \right)^3$ For $\frac{M_0}{M_h} \geq 2,0$: 1,10	$\frac{M_0}{M_h} < 1,0$: 1,00 $\frac{M_0}{M_h} \geq 1,0$: 0,77
M_h 	For $0 \leq \frac{M_0}{M_h} < 1,5$: $1,25 + 0,325 \left(\frac{M_0}{M_h} \right)^2 - 0,175 \left(\frac{M_0}{M_h} \right)^4$ For $\frac{M_0}{M_h} \geq 1,5$: 1,10	$\frac{M_0}{M_h} < 0,5$: 0,75 $\frac{M_0}{M_h} \geq 0,5$: 0,82
M_0 is the free bending moment at mid-span (determined on an equivalent member with simply supported end conditions). It may be determined by subtracting the average of the two (hogging) moments at the member ends from the (sagging) moment at mid-span. For $M_0/M_h < 0$, the factors f_m and k_c may be conservatively taken as 1,0.		

8.3.2.4 Simplified method for lateral torsional buckling of beams based on buckling of the compression flange

(1) As a simplified method for lateral torsional buckling verification of mono-symmetric and doubly symmetric I- and H-sections, according to 8.3.2.1(1) for member with or without lateral restraints, the design value of bending moment M_{Ed} should satisfy the criterion in Formula (8.83):

$$\frac{M_{Ed}}{M_{b,Rd}} \leq 1,0 \quad (8.83)$$

where

M_{Ed} is the design value of the moment;

$M_{b,Rd}$ is the design value of lateral torsional buckling resistance.

NOTE The method assumes end-fork support conditions for the member ends.

(2) The design value of lateral torsional buckling resistance $M_{b,Rd}$ should be calculated from Formula (8.84) based on the flexural buckling resistance of the equivalent compression flange (the flange plus part of the web):

$$M_{b,Rd} = \chi_{c,z} W_y \frac{f_y}{\gamma_{M1}} \quad (8.84)$$

where

$\chi_{c,z}$ is the reduction factor for flexural buckling of the equivalent compression flange about the weak axis of the section determined with $\bar{\lambda}_{c,z,mod}$; see (3).

W_y $W_{pl,y}$ for Class 1 and 2 cross-sections, $W_{el,y,min}$ for Class 3 cross-sections and $W_{eff,y,min}$ for cross-sections with a Class 4 web.

NOTE This method does not apply to cross-sections with Class 4 flanges.

(3) The buckling reduction factor $\chi_{c,z}$ should be determined by using the buckling formulae and curves of Formula (8.73) and Figure 8.5. The relative slenderness of the equivalent compression flange $\bar{\lambda}_{c,z,mod}$ should be used for the relative slenderness $\bar{\lambda}$. The buckling curves to be used in (2) should be taken as follows:

- buckling curve c for hot rolled sections;
- buckling curve d for welded sections.

(4) The relative slenderness of the equivalent compression flange should be determined as follows:

$$\bar{\lambda}_{c,z} = \sqrt{\frac{A_c f_y}{N_{cr,c,z}}} \quad (8.85)$$

where

A_c is the area of the equivalent compression flange;

$$A_c = \begin{cases} A_f + \frac{1}{2} \cdot A_w & \text{for load application at compression flange} \\ A_f + \frac{1}{6} \cdot A_w & \text{for load application at shear centre} \\ A_f & \text{for load application at tension flange} \end{cases}$$

For other loadings than transverse loads the area of the equivalent compression flange should be taken as $A_c = A_f + \frac{1}{6} \cdot A_w$.

A_f is the area of the compression flange: $A_f = b \cdot t_f$;

A_w is the total web area: $A_w = h_w \cdot t_w$;

$N_{cr,c,z}$ elastic critical axial force of the equivalent compression flange for weak axis buckling of the section determined with the moment of inertia of the equivalent compression flange for weak axis buckling of the section and the length between restraints.

(5) The modified relative slenderness of the equivalent compression flange should be determined as follows:

$$\bar{\lambda}_{c,z,mod} = k_c \cdot \beta_c \cdot \bar{\lambda}_{c,z} \quad (8.86)$$

where

$$\beta_c = \sqrt{\frac{0,06 \frac{h}{t_{f,max}}}{\bar{\lambda}_{c,z} + \frac{t_{f,max}}{t_{f,min}}}} \quad \text{but} \quad \beta_c \leq 1,0 \quad (8.87)$$

k_c is a slenderness correction factor for moment distribution between restraints, see Table 8.6;

$t_{f,max}$ is the maximum thickness of the top and bottom flange of the section;

$t_{f,min}$ is the minimum thickness of the top and bottom flange of the section.

(6) The verification according to Formulae (8.83) to (8.87) should be applied for all compression flanges within a member.

8.3.3 Uniform members in bending and axial compression

(1) Unless second order analysis with member bow imperfection is carried out, in accordance with Method M4 in 7.2.2(7) or Method M5 in 7.2.2(8), the stability of members should be verified using the rules given in the following clauses.

(2) The stability of uniform members in bending and axial compression with doubly symmetric cross-sections which are not susceptible to distortional deformations should be checked according to the method described in (3) to (14). For mono-symmetric I-, H- and welded box sections, rules given in Annex C, C.1 should be used.

NOTE 1 The method given by the technical specification CEN/TS 1993-1-101 is an alternative. The applicability of CEN/TS 1993-1-101 can be set by the National Annex.

NOTE 2 In case the conditions of application expressed in (2) are not fulfilled, see 8.3.4.

(3) The resistance of the cross-section at each end of the member should satisfy the requirements given in 8.2.

(4) For members of structural systems, the buckling resistance may be verified in isolation of the system by using one of the methods in 7.2.2.

(5) Members which are subjected to combined bending and axial compression should satisfy the criteria in Formulae (8.88) and (8.89):

$$\frac{N_{Ed}}{\chi_y N_{Rk}} + k_{yy} \frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{LT} \frac{M_{y,Rk}}{\gamma_{M1}}} + k_{yz} \frac{M_{z,Ed} + \Delta M_{z,Ed}}{\gamma_{M1}} \leq 1,0 \tag{8.88}$$

$$\frac{N_{Ed}}{\chi_z N_{Rk}} + k_{zy} \frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{LT} \frac{M_{y,Rk}}{\gamma_{M1}}} + k_{zz} \frac{M_{z,Ed} + \Delta M_{z,Ed}}{\gamma_{M1}} \leq 1,0 \tag{8.89}$$

- where
- $N_{Ed}, M_{y,Ed}$ and $M_{z,Ed}$ are the design value of the compressive axial force and the design values of the maximum bending moments about the y-y and z-z axis along the member, respectively;
 - $N_{Rk}, M_{y,Rk}$ and $M_{z,Rk}$ are the characteristic values of the cross-sectional resistance to compressive axial force and bending moment about the y-y and z-z axes respectively, see Table 8.1;
 - $\Delta M_{y,Ed}, \Delta M_{z,Ed}$ are the moments due to the shift of the centroid axis according to 8.2.2.5 for Class 4 sections;
 - χ_y and χ_z are the flexural buckling reduction factors from 8.3.1. For members susceptible to torsional-flexural buckling, see (7);
 - χ_{LT} is the lateral torsional buckling reduction factor from 8.3.2. For members not susceptible to lateral torsional buckling: $\chi_{LT} = 1,0$;
 - $k_{yy}, k_{yz}, k_{zy}, k_{zz}$ are interaction factors, see (6).

NOTE The criteria in Formulae (8.88) and (8.89) are based on the modelling of simply supported single span members with end fork conditions with or without continuous lateral restraints, subjected to compressive axial forces, end moments and/or transverse loads.

(6) The interaction factors k_{yy}, k_{yz}, k_{zy} and k_{zz} to be used for the design of members with doubly symmetric cross-sections in the criteria in Formulae (8.88) and (8.89) are given in Table 8.7 and Table 8.8.

Table 8.7 — Interaction factors k_{yy} and k_{yz} for criterion in Formula (8.88)

Plastic cross-sectional properties Class 1, Class 2, Class 3 (with W_{ep} for Class 3, according to Annex B)	Elastic cross-sectional properties Class 3 with W_{el} , Class 4
For $\bar{\lambda}_y < 1,0$: $k_{yy} = C_{my} \left[1 + (\bar{\lambda}_y - 0,2) n_y \right]$	For $\bar{\lambda}_y < 1,0$: $k_{yy} = C_{my} \left(1 + 0,6 \bar{\lambda}_y n_y \right)$
For $\bar{\lambda}_y \geq 1,0$: $k_{yy} = C_{my} \left(1 + 0,8 n_y \right)$	For $\bar{\lambda}_y \geq 1,0$: $k_{yy} = C_{my} \left(1 + 0,6 n_y \right)$
$k_{yz} = 0,6 k_{zz}$, see Table 8.8	$k_{yz} = k_{zz}$, see Table 8.8
NOTE 1 See (9) for n_y .	
NOTE 2 See (10) and Table 8.9 for C_{my} .	

Table 8.8 — Interaction factors k_{zy} and k_{zz} for criterion in Formula (8.89)

	Type of section	Plastic cross-sectional properties Class 1, Class 2, Class 3 (with W_{ep} for Class 3, according to Annex B)	Elastic cross-sectional properties Class 3 with W_{el} , Class 4
k_{zy}	Not susceptible to lateral-torsional buckling	$k_{zy} = 0,6 k_{yy}$ See Table 8.7.	$k_{zy} = 0,8 k_{yy}$ See Table 8.7.
	Susceptible to lateral-torsional buckling	For $\bar{\lambda}_z < 1,0$: $k_{zy} = 1 - \frac{0,1 \bar{\lambda}_z n_z}{C_{mLT} - 0,25}$ but $k_{zy} \leq 0,6 + \bar{\lambda}_z$ for $\bar{\lambda}_z < 0,4$ For $\bar{\lambda}_z \geq 1,0$: $k_{zy} = 1 - \frac{0,1 n_z}{C_{mLT} - 0,25}$	For $\bar{\lambda}_z < 1,0$: $k_{zy} = 1 - \frac{0,05 \bar{\lambda}_z n_z}{C_{mLT} - 0,25}$ For $\bar{\lambda}_z \geq 1,0$: $k_{zy} = 1 - \frac{0,05 n_z}{C_{mLT} - 0,25}$
k_{zz}	I-sections	For $\bar{\lambda}_z < 1,0$: $k_{zz} = C_{mz} \left[1 + (2 \bar{\lambda}_z - 0,6) n_z \right]$ For $\bar{\lambda}_z \geq 1,0$: $k_{zz} = C_{mz} (1 + 1,4 n_z)$	For $\bar{\lambda}_z < 1,0$: $k_{zz} = C_{mz} (1 + 0,6 \bar{\lambda}_z n_z)$
	Rectangular, circular or elliptical hollow sections	For $\bar{\lambda}_z < 1,0$: $k_{zz} = C_{mz} \left[1 + (\bar{\lambda}_z - 0,2) n_z \right]$ For $\bar{\lambda}_z \geq 1,0$: $k_{zz} = C_{mz} [1 + 0,8 n_z]$	For $\bar{\lambda}_z \geq 1,0$: $k_{zz} = C_{mz} (1 + 0,6 n_z)$
NOTE 1 See (9) for n_z .			
NOTE 2 See (10) and Table 8.9 for C_{my} and C_{mLT} .			

(7) For members susceptible to torsional and torsional-flexural buckling, the reduction factor χ_z above and in the following should be replaced by χ_{TF} in accordance with 8.3.1.4(3).

(8) The interaction factor k_{zy} in Table 8.8 distinguishes between members that are susceptible and not susceptible to lateral torsional buckling. The following criteria may be used to assess this susceptibility:

- Structural hollow sections, welded box sections and open sections with sufficient rotation restraint (see Annex D) or narrowly spaced lateral bracing may generally be regarded as not susceptible to lateral torsional buckling. It may be assumed that open sections are not susceptible to lateral torsional buckling when the relative slenderness $\bar{\lambda}_{LT}$ fulfils the condition in Formula (8.90):

$$\bar{\lambda}_{LT} \leq \bar{\lambda}_{LT,lim} = \bar{\lambda}_{LT,0} \sqrt{1 - \bar{\lambda}_z^2 \chi_z}$$

(8.90)

where $\bar{\lambda}_{LT,0}$ is defined in 8.3.2.3(1).

— In other cases, open sections (I- and H-sections) should be regarded as susceptible to lateral torsional buckling.

(9) The parameters n_y and n_z used in Table 8.7 and 8.8 should be calculated from Formulae (8.91) and (8.92) respectively:

$$n_y = \frac{N_{Ed}}{\chi_y N_{Rk} / \gamma_{M1}}$$

(8.91)

$$n_z = \frac{N_{Ed}}{\chi_z N_{Rk} / \gamma_{M1}}$$

(8.92)

(10) The equivalent uniform moment factors C_m may be determined using the formulae in Table 8.9, for uniformly distributed loads and for a concentrated load applied at mid-span. M_s always represents the sagging moment at mid-span, which does not have to be the maximum sagging moment in the span. As the values in this table were derived for members with buckling lengths corresponding to the member length between supports, factors C_{my} , C_{mz} and C_{mLT} should be obtained by considering the bending moment diagrams between the relevant braced points, as follows:

Moment factor	Bending axis	Point braced in direction
C_{my}	y-y	z-z
C_{mz}	z-z	y-y
C_{mLT}	y-y	y-y

NOTE Figure 8.7 shows an example for the determination of the equivalent uniform moment factors for a beam-column loaded by unequal end moments about both axes and braced with different spacing in direction y and z.

(11) Both segments between the member ends and the intermediate lateral restraint (see Figure 8.7) may need to be checked, depending on length of the segment and shape of the bending diagram.

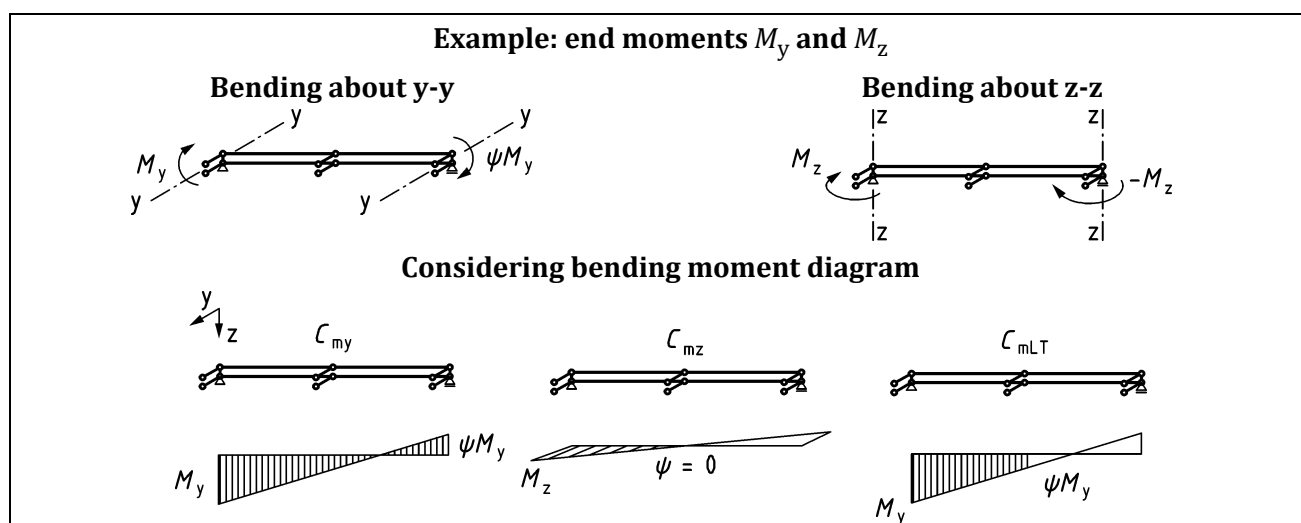


Figure 8.7 — Example illustrating the applicable bending moment diagrams for the determination of the coefficients C_{my} , C_{mz} and C_{mLT}

Table 8.9 — Equivalent uniform moment factors C_m

Moment diagram	Range		C_{my} and C_{mz} and C_{mLT}	
	$-1 \leq \psi \leq 1$		$0,6 + 0,4\psi \geq 0,4$	
Members with transverse loading		Uniform loading	Concentrated load	
$\alpha_s = M_s / M_h$	$0 \leq \alpha_s \leq 1$	$-1 \leq \psi \leq 1$	$0,2 + 0,8\alpha_s \geq 0,4$	$0,2 + 0,8\alpha_s \geq 0,4$
	$-1 \leq \alpha_s < 0$	$0 \leq \psi \leq 1$	$0,1 - 0,8\alpha_s \geq 0,4$	$-0,8\alpha_s \geq 0,4$
		$-1 \leq \psi < 0$	$0,1(1 - \psi) - 0,8\alpha_s \geq 0,4$	$-0,2\psi - 0,8\alpha_s \geq 0,4$
$\alpha_h = M_h / M_s$	$0 \leq \alpha_h \leq 1$	$-1 \leq \psi \leq 1$	$0,95 + 0,05\alpha_h$	$0,90 + 0,10\alpha_h$
	$-1 \leq \alpha_h < 0$	$0 \leq \psi \leq 1$	$0,95 + 0,05\alpha_h$	$0,90 + 0,10\alpha_h$
		$-1 \leq \psi < 0$	$0,95 + 0,05\alpha_h (1 + 2\psi)$	$0,90 + 0,10\alpha_h (1 + 2\psi)$

(12) If the global analysis is carried out according to the method EM in 7.2.2(9) and the member (e.g. in an unbraced frame) can fail in a sway buckling mode, the factors C_m should be taken as $C_{my} = 0,9$ or $C_{mz} = 0,9$ for any bending moment diagram.

(13) The sign of the hogging moment M_h and the mid-span sagging moment M_s should be considered when calculating α_s or α_h in Table 8.9. Thereby:

- α_s is relevant if the absolute value of M_h is larger than M_s ;
- α_h is relevant if the absolute value of M_s is larger than M_h .

(14) For members subjected to compression and bending about the y-y axis only, some or all of the following simplifications may be considered:

- The terms containing k_{zz} and k_{yz} are omitted;
- The factor k_{yy} may be calculated by: $k_{yy} = C_{my} (1 + 0,8n_y)$;
- The factor k_{zy} may be taken equal to 1,0 for members susceptible to lateral torsional buckling;
- The equivalent uniform bending moment factors, C_{my} and C_{mLT} , may be taken equal to 1,0.

8.3.4 General method for lateral and lateral torsional buckling of structural components

(1) If the methods given in 8.3.1, 8.3.2 and 8.3.3 do not apply, the following method may be used when dealing with lateral and lateral torsional buckling for structural components such as:

- structural components with mono symmetric cross-sections, built-up or not, uniform or not, with complex support conditions or not, or;
- plane frames or subframes composed of such members;

which are subject to compression and/or uni-axial bending in the plane, but which do not contain rotated plastic hinges.

NOTE The field and limits of application of the method given in 8.3.4 can be set by the National Annex.

(2) For the verification of the out-of-plane buckling resistance, structural components conforming to (1) should satisfy criterion in Formula (8.93):

$$\frac{1}{\chi_{op} \alpha_{ult,k} / \gamma_{M1}} \leq 1,0 \quad (8.93)$$

where

$\alpha_{ult,k}$ is the minimum load amplifier of the design value of the loading to reach the characteristic resistance of the most critical cross-section of the structural component considering its in plane behaviour including instability, i.e. all effects due to in-plane geometrical deformation and imperfections of members, global and local, where relevant but without taking flexural buckling out of plane or lateral torsional buckling into account;

χ_{op} is the reduction factor for the relative slenderness $\bar{\lambda}_{op}$, see (3), to take account of lateral and lateral torsional buckling.

(3) The global relative slenderness $\bar{\lambda}_{op}$ for the structural component should be determined from:

$$\bar{\lambda}_{op} = \sqrt{\frac{\alpha_{ult,k}}{\alpha_{cr,op}}} \quad (8.94)$$

where

$\alpha_{ult,k}$ is defined in (2);

$\alpha_{cr,op}$ is the minimum amplifier for the design value of the in-plane loading to reach the elastic critical load of the structural component with regards to lateral or lateral torsional buckling without accounting for in-plane flexural buckling.

(4) In determining $\alpha_{cr,op}$ and $\alpha_{ult,k}$ Finite Element analysis may be used (see prEN 1993-1-14).

(5) The reduction factor χ_{op} should be determined as the minimum value of:

χ for lateral buckling according to 8.3.1;

χ_{LT} for lateral torsional buckling according to 8.3.2;

each determined for the global relative slenderness $\bar{\lambda}_{op}$.

NOTE For example, where $\alpha_{ult,k}$ is determined by the cross-section check $\frac{1}{\alpha_{ult,k}} = \frac{N_{Ed}}{N_{Rk}} + \frac{M_{y,Ed}}{M_{y,Rk}}$ this method leads to:

$$\frac{N_{Ed}}{\chi_{op} N_{Rk} / \gamma_{M1}} + \frac{M_{y,Ed}}{\chi_{op} M_{y,Rk} / \gamma_{M1}} \leq 1 \quad (8.95)$$

8.3.5 Lateral torsional buckling of members with plastic hinges in buildings

8.3.5.1 General

(1)B Building structures may be designed using plastic analysis, provided that lateral torsional buckling in the frame is prevented by the following means:

- Inclusion of effective restraint at locations of “rotated” plastic hinges, see 8.3.5.2, and;
- Ensuring that the length of the segment between such a restraint and the next lateral restraint is not greater than the stable length, see 8.3.5.3, and;
- Ensuring the out-of-plane stability of all other segments.

(2)B Where the plastic hinge is “not-rotated” under all ultimate limit state load combinations, restraints may be omitted for such a plastic hinge.

8.3.5.2 Restraints at rotated plastic hinges

(1)B At each rotated plastic hinge location, the cross-section should have an effective lateral and torsional restraint with appropriate resistance to lateral forces and torsion induced by local plastic deformations of the member at that location.

(2)B Effective lateral and torsional restraint should be provided:

- For members carrying either moment or moment and axial force, by lateral restraint to both flanges. This can be provided by lateral restraint to one flange and a stiff torsional restraint to the cross-section preventing the lateral displacement of the compression flange relative to the tension flange, see Figure 8.8.

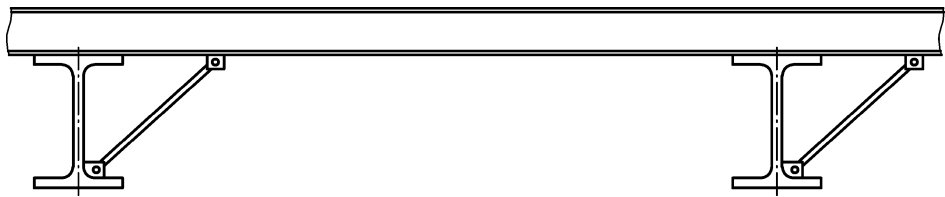
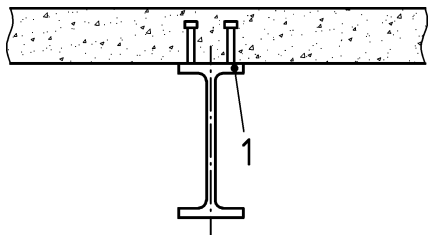


Figure 8.8 — Typical stiff torsional restraint

- For members carrying either moment alone or moment and tensile axial force in which the compression flange is in contact with a floor slab, by lateral and torsional restraint to the compression flange (e.g. by connecting it to the slab, see Figure 8.9).



Key
1 compression flange

Figure 8.9 — Typical lateral and torsional restraint by a slab to the compression flange

- For cross-sections that are more slender than rolled I- and H-sections, the distortion of the cross-section should be prevented at the plastic hinge location (e.g. by means of a web stiffener also connected to the compression flange with a stiff joint from the compression flange into the slab).

(3)B At each plastic hinge location, the connection (e.g. bolts) of the compression flange to the resisting element at that point (e.g. purlin), and any intermediate element (e.g. diagonal brace) should be designed to resist a local force of at least 2,5 % of $N_{f,Ed}$ (defined in 8.3.5.2(5)B) transmitted by the flange in its plane and perpendicular to the web plane, without any combination with other loads.

(4)B Restraints should be provided within a distance of the plastic hinge of $h/2$ along the length of the member, where h is the overall depth of the cross-section at the plastic hinge location.

(5)B In addition to the design requirements of 7.3.5, bracing systems should be designed to resist the effects of local forces Q_m applied at each stabilized member at the plastic hinge locations, where:

$$Q_m = 1,5 \alpha_m \frac{N_{f,Ed}}{100} \tag{8.96}$$

where

$N_{f,Ed}$ is the axial force in the compressed flange of the stabilized member at the plastic hinge location;

α_m is according to 7.3.2(2).

NOTE B For combination with external loads, see also 7.3.5.

8.3.5.3 Stable length of segment

(1) The lateral torsional buckling verification of segments between restraints may be performed by checking that the length between restraints is not greater than the stable length.

(2) For uniform beam segments with I- or H- cross-sections with $\frac{h}{t_f} \leq 40\varepsilon$ under linear moment diagram and without significant axial compression, the stable length L_{st} may be taken from:

$$L_{st} = 35 \varepsilon i_z \quad \text{for } 0,625 \leq \psi \leq 1 \quad (8.97)$$

$$L_{st} = (60 - 40\psi) \varepsilon i_z \quad \text{for } -1 \leq \psi \leq 0,625 \quad (8.98)$$

where

ε is the material parameter defined in 5.2.5(2).

$$\psi = \frac{M_{Ed,min}}{M_{pl,Rd}} \quad = \text{ratio of end moments in the segment.}$$

(3) Where a rotated plastic hinge location occurs immediately adjacent to one end of a haunch, the tapered segment need not be treated as a segment adjacent to a plastic hinge location if the following criteria are satisfied:

- a) the restraint at the plastic hinge location is within a distance $h/2$ along the length of the tapered segment, not the uniform segment;
- b) the compression flange of the haunch remains elastic throughout its length.

8.4 Uniform built-up compression members

8.4.1 Assumptions and constructional details

8.4.1.1 Assumptions

(1) Uniform built-up compression members with hinged ends that are laterally supported should be designed according to the following model, see Figure 8.10:

- a) The member should be considered as a column with a bow imperfection as defined in 7.3.3.1(4), for buckling in the plane of the lacing or battens;
- b) The elastic deformations of lacings or battens, see Figures 8.10, should be taken into account by a continuous (smeared) shear stiffness S_V of the columns.

NOTE For other end conditions, appropriate modifications can be performed.

(2) The model of a uniform built-up compression member applies where:

- a) The lacings or battens consist of equal modules with parallel chords;
- b) The minimum number of modules in a member is three.

NOTE This assumption allows the structure to be regular and smears the discrete structure to a continuum.

(3) The design procedure is applicable to built-up members with lacings in two planes, see Figure 8.11.

(4) The chords may be solid members or may themselves be laced or battened in the perpendicular plane.

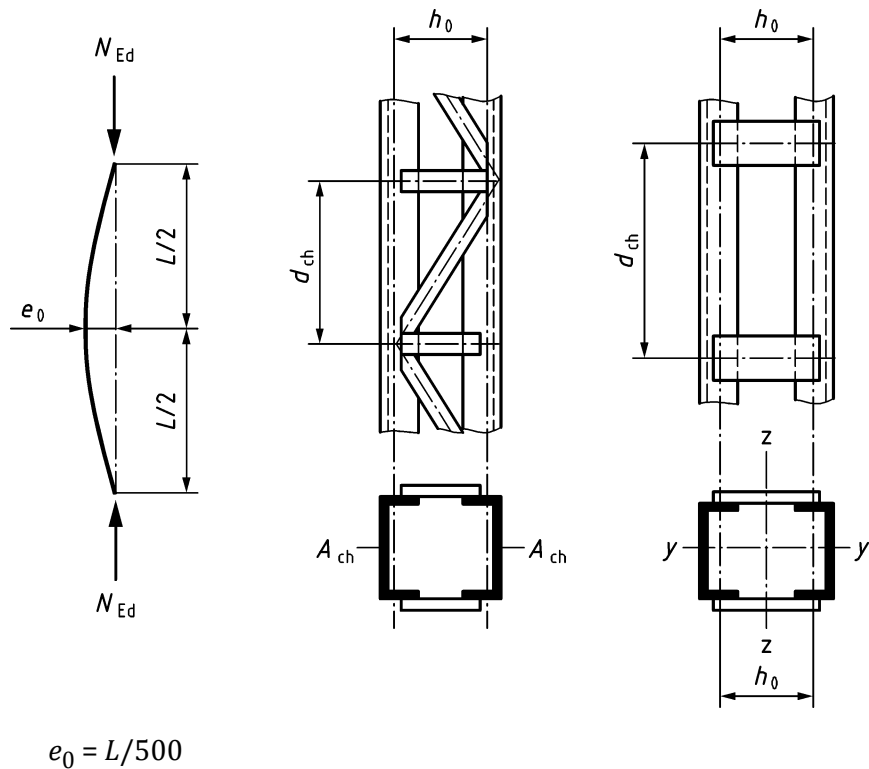


Figure 8.10 — Uniform built-up columns with lacings and battens

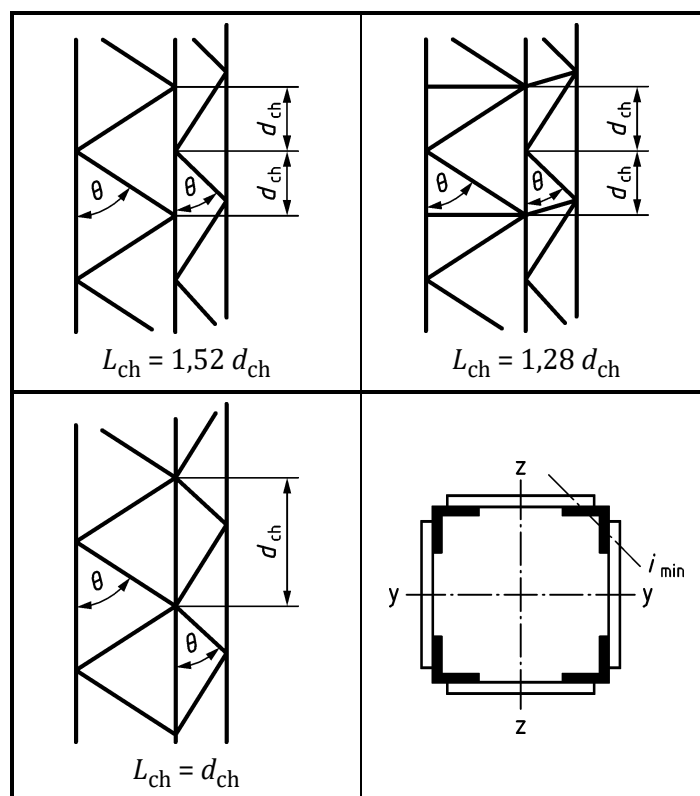


Figure 8.11 — Lacings on four sides and buckling length L_{ch} of chords

8.4.1.2 Arrangement of single lacing systems

- (1) Single lacing systems in opposite faces of built-up members with two parallel laced planes may be arranged as shown in Figure 8.12(a).
- (2) When the single lacing systems in opposite faces of built-up members with two parallel laced planes are mutually opposed in direction, as shown in Figure 8.12(b), the resulting torsional effects in the built-up member should be taken into account.
- (3) Tie panels should be provided at the ends of lacing systems, at points where the lacing is interrupted and at joints with other members.

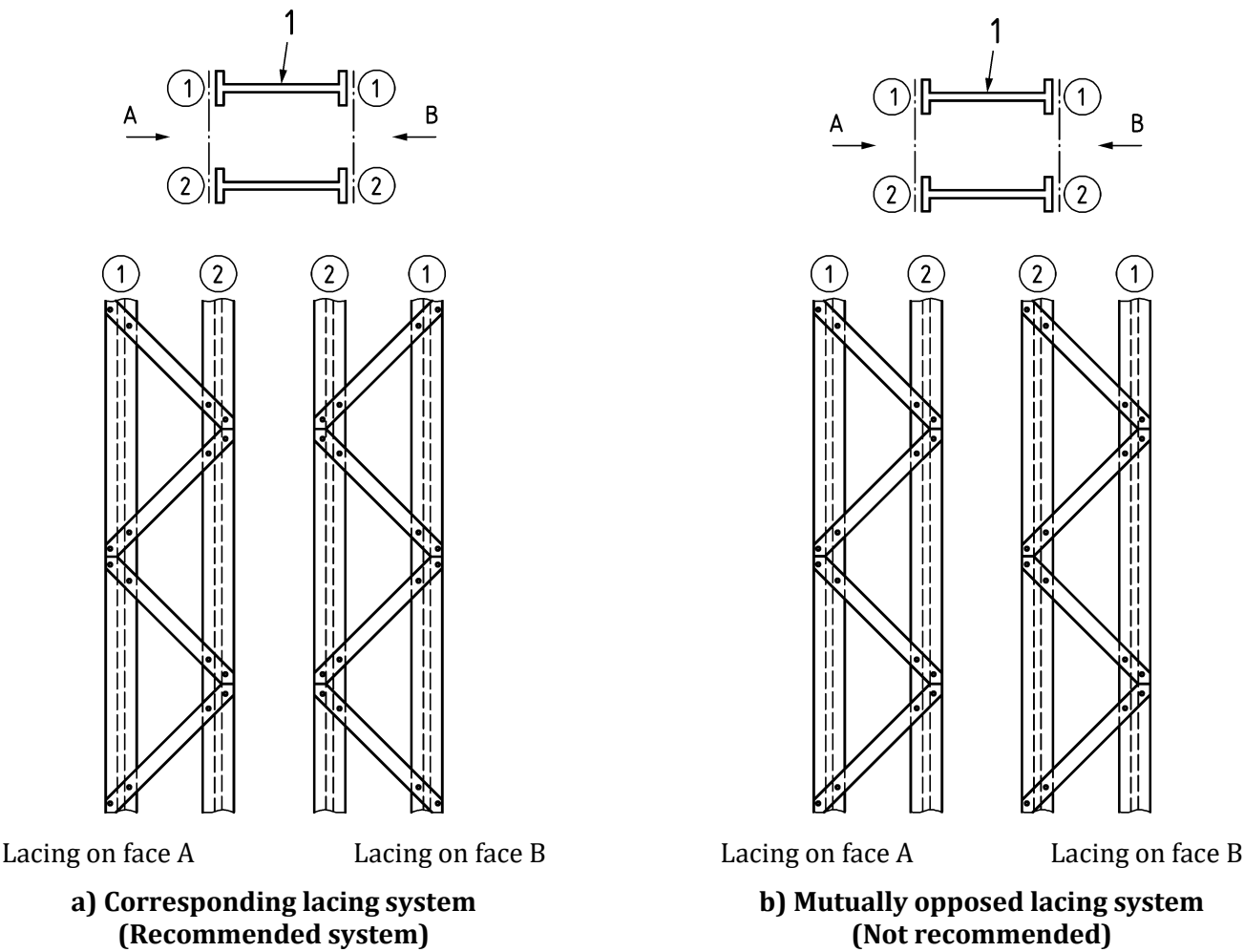


Figure 8.12 — Single lacing system on opposite faces of a built-up member with two parallel laced planes

8.4.1.3 Arrangement of battens

- (1) Battens should be provided at each end of a member.
- (2) Where parallel planes of battens are provided, the battens in each plane should be arranged opposite to each other.
- (3) Battens should be provided at intermediate points where loads are applied or lateral restraint is provided.

8.4.2 Design forces for components

- (1) Checks should be performed for chords using the design value of the chord forces $N_{ch,Ed}$ derived from the design value of the compressive axial force N_{Ed} and the design value of the bending moment M_{Ed} at mid span of the built-up member.
- (2) For a member with two identical chords, the design value of the chord force $N_{ch,Ed}$ should be determined from Formula (8.99):

$$N_{ch,Ed} = 0,5N_{Ed} + \frac{M_{Ed}h_0A_{ch}}{2I_{eff}} \quad (8.99)$$

where

$$M_{Ed} = \frac{N_{Ed}e_0 + M_{Ed}^I}{1 - \frac{N_{Ed}}{N_{cr,V}}} \quad (8.100)$$

$N_{cr,V}$ is the critical force of the built-up member, including the effect of its shear stiffness:

$$N_{cr,V} = \frac{1}{\frac{L^2}{\pi^2 EI_{eff}} + \frac{1}{S_{vb}}} \quad (8.101)$$

N_{Ed} is the design value of the compressive axial force to the built-up member;

M_{Ed} is the design value of the maximum bending moment in the built-up member considering second order effects;

M_{Ed}^I is the design value of the maximum bending moment in the built-up member without second order effects;

h_0 is the distance between the centroids of chords;

A_{ch} is the cross-sectional area of one chord;

I_{eff} is the effective moment of inertia of the built-up member, see 8.4.3 and 8.4.4;

S_{vb} is the shear stiffness of the lacings or battened panel, see 8.4.3 and 8.4.4.

(3) The checks for the lacings of laced built-up members or for the frame moments and shear forces of the battened panels of battened built-up members should be performed for the end panel taking account of the shear force in the built-up member given by Formula (8.102):

$$V_{Ed} = V_{Ed}^I + \frac{\pi}{L} (M_{Ed} - M_{Ed}^I) \quad (8.102)$$

where

V_{Ed}^I is the design value of the shear force at the end, due to transverse loads, without second order effects.

8.4.3 Resistance of components of laced compression members

(1) The chords and diagonal lacings subject to compression should be designed for buckling and secondary moments may be neglected.

(2) For chords, the design value of the compressive axial force in the chord at mid-length of the built-up member should satisfy the criterion in Formula (8.103):

$$\frac{N_{ch,Ed}}{N_{b,Rd}} \leq 1,0$$

(8.103)

where

$N_{ch,Ed}$ is the design compressive axial force in the chord at mid-length of the built-up member according to 8.4.2(2) and

$N_{b,Rd}$ is the design value of the buckling resistance in compression of the chord taking the buckling length L_{ch} from Figure 8.11.

(3) The shear stiffness S_{Vb} of the lacings should be taken from Figure 8.13.

(4) The effective moment of inertia of laced built-up members may be taken as:

$$I_{eff} = 0,5h_0^2 A_{ch}$$

(8.104)

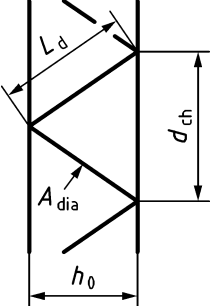
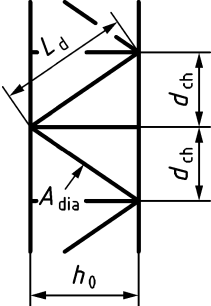
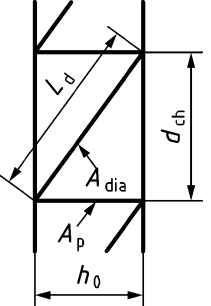
System			
S_{Vb}	$\frac{n_p E A_{dia} d_{ch} h_0^2}{2 L_d^3}$	$\frac{n_p E A_{dia} d_{ch} h_0^2}{L_d^3}$	$\frac{n_p E A_{dia} d_{ch} h_0^2}{L_d^3 \left(1 + \frac{A_{dia} h_0^3}{A_p L_d^3} \right)}$
n_p is the number of planes of lacings A_{dia} and A_p refer to the cross-sectional area of the bracings (diagonals and posts)			

Figure 8.13 — Shear stiffness of lacings of built-up members

8.4.4 Resistance of components of battened compression members

(1) The chords and the battens and their joints to the chords should be checked for the actual moments and forces in an end panel and at mid-span as indicated in Figure 8.14.

(2) For simplicity, the maximum chord forces $N_{ch,Ed}$ may be combined with the maximum shear force V_{Ed} .

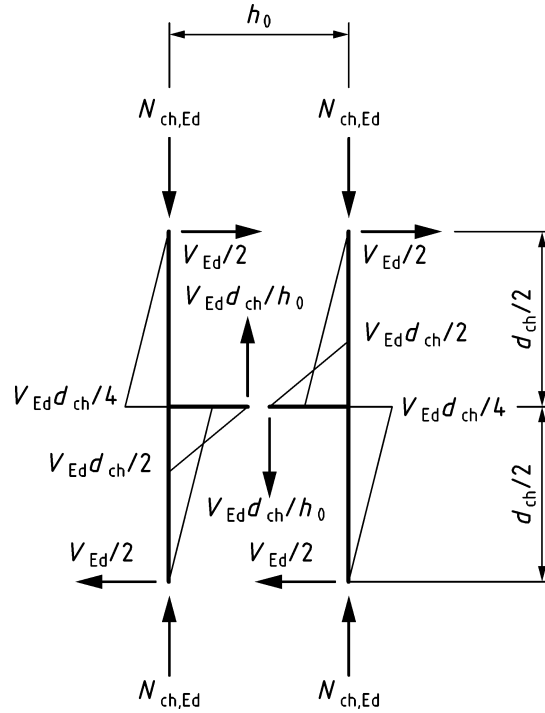


Figure 8.14 — Moments and forces in an end panel of a battened built-up member

(3) The shear stiffness S_{Vb} should be taken from Formula (8.105):

$$S_{Vb} = \frac{24EI_{ch}}{d_{ch}^2 \left[1 + \frac{2I_{ch}}{n_p I_b} \frac{h_0}{d_{ch}} \right]} \leq \frac{2\pi^2 EI_{ch}}{d_{ch}^2} \quad (8.105)$$

(4) The effective moment of inertia of battened built-up members may be taken from Formula (8.106):

$$I_{eff} = 0,5h_0^2 A_{ch} + 2\mu I_{ch} \quad (8.106)$$

where

I_{ch} is the in-plane moment of inertia of one chord;

I_b is the in-plane moment of inertia of one batten;

μ is the efficiency factor from Table 8.10;

n_p is the number of planes of battens.

Table 8.10 — Efficiency factor μ

Criterion	Efficiency factor μ
$\lambda \geq 150$	0
$75 < \lambda < 150$	$\mu = 2 - \frac{\lambda}{75}$
$\lambda \leq 75$	1,0

where

$$\lambda = \frac{L}{i_0}; \quad i_0 = \sqrt{\frac{I_1}{2A_{ch}}}; \quad I_1 = 0,5h_0^2 A_{ch} + 2I_{ch}$$

8.4.5 Closely spaced built-up members

- (1) If the conditions in Table 8.11 are met, built-up compression members with chords in contact or closely spaced and connected through packing plates, see Figure 8.15, or star batteded angle members connected by pairs of battens in two perpendicular planes, see Figure 8.16 should be checked for buckling as a single integral member and the effect of shear stiffness may be ignored ($S_{vb} = \infty$).
- (2) If the conditions in Table 8.11 are not met, closely spaced built-up compression members should be checked by taking into account the effect of shear stiffness in accordance with 8.4.2 and 8.4.4 or any other appropriate method.

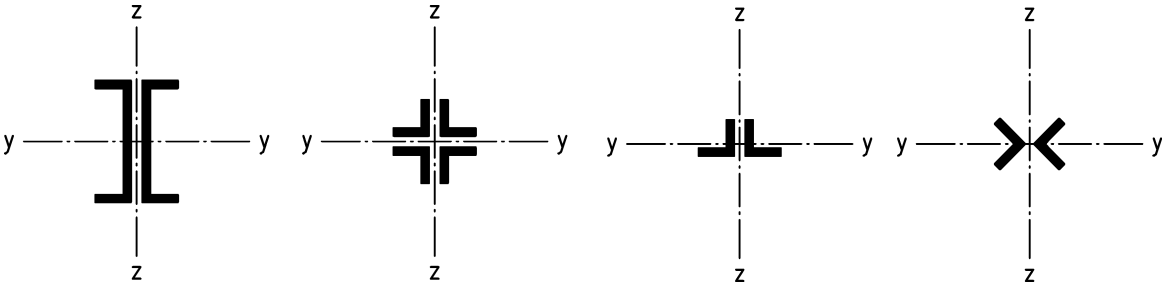


Figure 8.15 — Closely spaced built-up members

Table 8.11 — Maximum spacing for interconnections in closely spaced built-up or star batteded angle members

Type of built-up member	Maximum spacing between interconnections ^a
Members according to Figure 8.15 connected by bolts or welds	$15 i_{min}$
Members according to Figure 8.16 connected by pair of battens	$70 i_{min}$

i_{min} is the minimum radius of gyration of one chord or one angle.

^a Centre-to-centre distance of interconnections.

- (3) The shear forces to be transmitted by the battens should be determined in accordance with 8.4.4(1).

(4) In the case of unequal-leg angles, see Figure 8.16, buckling about the y-y axis should be verified using a radius of gyration about the y-y axis i_y which may be approximated from Formula (8.107):

$$i_y = \frac{i_0}{1,15} \quad (8.107)$$

where i_0 is the minimum radius of gyration of the built-up member calculated with reference to the 0-0 axis (axis parallel to the longer leg of the angle section).

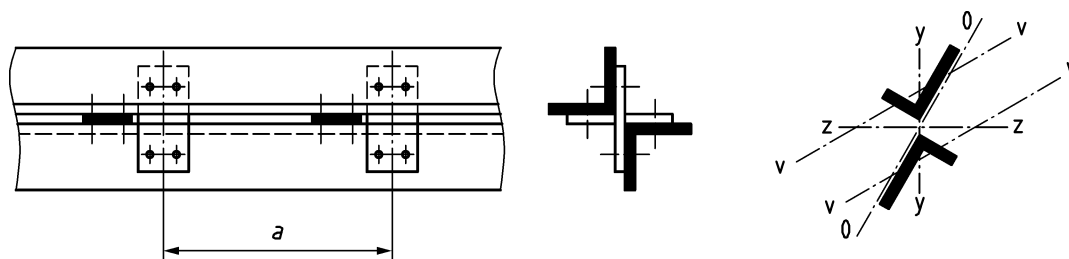


Figure 8.16 — Star-battened angle members

9 Serviceability limit states

9.1 General

- (1) A steel structure shall be designed and constructed such that all relevant serviceability criteria are satisfied.
- (2) The basic requirements for serviceability limit states given in EN 1990:—, 5.4 shall be followed.
- (3) Any serviceability limit states and the associated loading and analysis models should be as specified by the relevant authority or, where not specified, agreed for a specific project by the relevant parties.
- (4) Where plastic global analysis is used for the ultimate limit state, the effects of plastic redistribution of forces and moments at the serviceability limit state should be considered.

9.2 Deformations and dynamic effects for buildings

- (1)B With reference to EN 1990, the limits for deformations for buildings should be as specified by the relevant authority or, where not specified, agreed for a specific project by the relevant parties.
- (2)B With reference to EN 1990, the vibrations of building structures on which the public can walk should be limited to avoid significant discomfort to users, and limits should be as specified by the relevant authority or, where not specified, agreed for a specific project by the relevant parties.

NOTE B Limits for deformation and vibration of floors can be set by the National Annex.

- (3)B Deformations and dynamic effects may be calculated considering the restraint, damping and stiffness of non-structural elements.

10 Fatigue

- (1) Fatigue verification according to EN 1993-1-9 shall be carried out for structures exposed to variable actions, except in the cases given in (2) and (3).
- (2) Fatigue verification may be omitted for structures subjected to static and quasi-static actions only.

NOTE 1 Such actions are for example:

- a) Imposed loads for buildings according to EN 1991-1-1;
- b) Snow action according to EN 1991-1-3;
- c) Thermal action according to EN 1991-1-5.

NOTE 2 Wind action on buildings does not normally lead to fatigue, for details see EN 1991-1-4.

(3) For members without constructional details subjected to secondary stresses, the fatigue verification may be omitted if either the condition in Formula (10.1) or the condition in Formula (10.2) is fulfilled:

$$\Delta\sigma_d \leq \frac{\Delta\sigma_D}{\gamma_{Mf}} \quad (10.1)$$

$$N \leq 5 \times 10^6 \left(\frac{\Delta\sigma_D / \gamma_{Mf}}{\Delta\sigma_d} \right)^3 \quad (10.2)$$

where

N is the expected number of stress cycles during the design service life;

γ_{Mf} is the partial factor for fatigue;

$$\Delta\sigma_D = 26 \text{ N} / \text{mm}^2$$

$\Delta\sigma_d$ is the stress range determined according to elastic theory in N/mm^2 caused by variable design loads at ultimate limit state, excluding the actions mentioned in (2):

$$\Delta\sigma_d = |\sigma_{\max} - \sigma_{\min}|$$

$\sigma_{\max}$ and $\sigma_{\min}$ are the maximum and the minimum stresses, respectively.

NOTE 1 The conditions refer to a verification for the most unfavourable detail category 36 according to EN 1993-1-9 with constant amplitude fatigue limit $\Delta\sigma_D$ under constant stress spectrum $\Delta\sigma_d$. Since the design stresses at ultimate limit state are used in contrast to the rules of EN 1993-1-9, the conditions are safe-sided unless secondary stresses occur that are generally neglected at ultimate limit state, such as stresses due to secondary bending moments in truss joints or local stresses due to wheel load. Where change of load direction occurs, the sign of the stresses $\sigma_{\max}$ and $\sigma_{\min}$ is relevant.

NOTE 2 The value of the partial factor γ_{Mf} is 1,35 unless the National Annex gives a different value. The value accounts for no inspections and high failure consequences. See EN 1993-1-9.

Annex A **(normative)**

Selection of execution class

A.1 Use of this Annex

(1) This normative Annex contains provisions for the selection of an appropriate execution class.

A.2 Scope and field of application

(1) This normative Annex covers the selection process of an appropriate execution class for steel structures executed to EN 1090-2 and EN 1090-4.

A.3 Execution class

(1) Execution class (EXC) is defined as a classified set of requirements specified for the execution of the works as a whole, of an individual component or of a detail of a component.

(2) In order to specify requirements for the execution of steel structures to EN 1090-2 and EN 1090-4, the execution class, EXC1, EXC2, EXC3 or EXC4, should be specified prior to the commencement of execution. The execution requirements are progressively more demanding from EXC1 up to EXC4.

NOTE EN 1993 is based on the assumption that it is used in conjunction with EN 1090-2 and EN 1090-4. EN 1993-1-9, EN 1993-2 and EN 1993-3 give supplementary requirements to EN 1090-2 for the execution of structures or components or details subject to fatigue actions. In addition to EN 1090-2, EN 1993-5 refers to other European Standards for the execution of piling works.

(3) Execution class EXC2 should be applied unless the selection process gives another class.

A.4 Selection process

(1) The selection of the execution class should be based on the following three factors:

- the required reliability;
- the type of loading for which the structure, component or detail is designed;
- the type of structure, component or detail.

(2) In terms of reliability management, the selection of execution class should be based on the required consequence class (CC).

NOTE 1 The concept of consequence class is defined in EN 1990.

NOTE 2 The National Annex can set the choice in terms of the type of the structure.

(3) In terms of the type of loading applied to a steel structure or component or detail, the selection of execution class should be based on whether the structure or component or detail is designed for static actions, quasi-static actions, fatigue actions or seismic actions according to EN 1998-1.

NOTE The selection of execution class (EXC) based on type of loading is as given in Table A.1 unless the National Annex specifies differently.

Table A.1 (NDP) — Selection of execution class based on the type of loading (EXC)

Consequence class (CC)	Type of loading				
	Static Quasi-static	Seismic			Fatigue ^b
		DC1 ^a	DC2 ^a	DC3 ^a	
CC3	EXC3 ^c	EXC3 ^c	EXC3 ^c	EXC3 ^c	EXC3 ^c
CC2	EXC2	EXC2	EXC2	EXC3 ^d	EXC3
CC1	EXC1	EXC2 ^e	EXC2	EXC2	EXC2
^a Seismic ductility classes (DC's) are defined in prEN 1998-1-1. ^b See EN 1993-1-9. ^c EXC4 may be considered for special cases, including those typically covered by CC4 of EN 1990:—. ^d Only the primary seismic resisting system falls in EXC3; the gravity load resisting system may fall in EXC2. ^e If the seismic action index is not greater than 2,5m/s² (low seismic action class, see prEN 1998-1-1), the execution class of structures in DC1 may be EXC1. NOTE Structures in consequence class CC0 are not covered by this Annex, see EN 1990:—, 4.3.					

(4) If the required execution class for particular components and/or details is different from that applicable to the structure in general, then these components and/or details should be clearly identified.

(5) If EXC1 is selected for a structure, then EXC2 should apply to the following types of component:

- a) welded components manufactured from steel products of grade S355 and above;
- b) welded components essential for structural integrity that are assembled by welding on the construction site;
- c) welded components of CHS lattice girders requiring end profile cuts;
- d) components with hot forming during manufacturing or receiving thermic treatment during manufacturing.

NOTE The National Annex can give different choices of execution class for types of components or details.

A.5 Execution class and partial factors

(1) Specification of a higher execution class for the execution of a structure or component or detail should not be used to justify the use of lower partial factors for resistance in the design of that structure or component or detail.

Annex B (normative)

Design of semi-compact sections

B.1 Scope and field of application

(1) This Annex provides alternative rules for the design of semi-compact (Class 3) doubly symmetric I- or H-sections, rectangular hollow sections, doubly symmetric box sections, circular hollow sections and elliptical hollow sections against mono- and bi-axial bending and axial force.

B.2 Elasto-plastic section modulus

(1) The elasto-plastic section modulus W_{ep} for doubly symmetric cross-sections should be determined from an interpolation between the plastic section modulus W_{pl} and the elastic section modulus W_{el} about one principal axis of a cross-section as follows:

$$W_{ep,y} = W_{pl,y} - (W_{pl,y} - W_{el,y}) \beta_{ep,y} \quad (B.1)$$

$$W_{ep,z} = W_{pl,z} - (W_{pl,z} - W_{el,z}) \beta_{ep,z} \quad (B.2)$$

where the values of $\beta_{ep,y}$ and $\beta_{ep,z}$ depend on the material parameter ε and the width-to-thickness ratios as defined in Table 7.3 to Table 7.5. They should be taken as:

— For I- or H- sections, rolled or welded:

$$\beta_{ep,y} = \max \left\{ \frac{\frac{c}{t_f} - 10\varepsilon}{4\varepsilon}; \frac{\frac{c}{t_w} - 83\varepsilon}{38\varepsilon}; 0 \right\} \quad \text{but } \beta_{ep,y} \leq 1,0 \quad (B.3)$$

$$\beta_{ep,z} = \max \left\{ \frac{\frac{c}{t_f} - 10\varepsilon}{6\varepsilon}; 0 \right\} \quad \text{but } \beta_{ep,z} \leq 1,0 \quad (B.4)$$

— For rectangular hollow sections or doubly symmetric welded box sections:

$$\beta_{ep,y} = \max \left\{ \frac{\frac{c}{t_f} - 34\varepsilon}{4\varepsilon}; \frac{\frac{c}{t_w} - 83\varepsilon}{38\varepsilon}; 0 \right\} \quad \text{but } \beta_{ep,y} \leq 1,0 \quad (B.5)$$

$$\beta_{ep,z} = \max \left\{ \frac{\frac{c}{t_w} - 34\varepsilon}{4\varepsilon}; 0 \right\} \quad \text{but } \beta_{ep,z} \leq 1,0 \quad (\text{B.6})$$

NOTE 1 The width c is defined in Table 7.3 to Table 7.5 for the appropriate compression part.

NOTE 2 $t_f = t_w = t$ for rectangular hollow sections.

— For circular or elliptical hollow sections:

$$\beta_{ep,y} = \beta_{ep,z} = \max \left\{ \frac{\frac{d_e}{t} - 70\varepsilon^2}{70\varepsilon^2}; 0 \right\} \quad \text{but } \beta_{ep,y} = \beta_{ep,z} \leq 1,0 \quad (\text{B.7})$$

where

d_e is the equivalent diameter according to Table 7.5.

B.3 Resistance of cross-sections

(1) In the combined presence of bending moment and axial force, the design value of the bending moment M_{Ed} at each cross-section should satisfy criterion in Formula (B.8):

$$\frac{M_{Ed}}{M_{N,ep,Rd}} \leq 1,0 \quad (\text{B.8})$$

where

$M_{N,ep,Rd}$ is the design elasto-plastic bending moment resistance reduced due to the axial force.

(2) The design reduced elasto-plastic bending moment resistance $M_{N,ep,Rd}$ should be taken as:

— For doubly symmetric I- or H-sections, rolled or welded:

$$M_{N,ep,y,Rd} = M_{ep,y,Rd} (1 - n) \quad (\text{B.9})$$

$$M_{N,ep,z,Rd} = M_{ep,z,Rd} (1 - n^2) \quad (\text{B.10})$$

— For rectangular hollow sections, doubly symmetric welded box sections, circular hollow sections or elliptical hollow sections:

$$M_{N,ep,y,Rd} = M_{ep,y,Rd} (1 - n) \quad (\text{B.11})$$

$$M_{N,ep,z,Rd} = M_{ep,z,Rd} (1 - n) \quad (\text{B.12})$$

where

$$M_{ep,y,Rd} = \frac{W_{ep,y} f_y}{\gamma_{M0}} \quad (B.13)$$

$$M_{ep,z,Rd} = \frac{W_{ep,z} f_y}{\gamma_{M0}} \quad (B.14)$$

$$n = \frac{N_{Ed}}{N_{pl,Rd}} \quad (B.15)$$

(3) For bi-axial bending, the following criterion should be satisfied:

$$\left(\frac{M_{y,Ed}}{M_{N,ep,y,Rd}} \right)^{\alpha_y} + \left(\frac{M_{z,Ed}}{M_{N,ep,z,Rd}} \right)^{\alpha_z} \leq 1,0 \quad (B.16)$$

In which α_y and α_z should be taken as:

— For doubly symmetric I- or H-sections, rolled or welded:

$$\alpha_y = 2; \quad \alpha_z = 5n \quad \text{but} \quad \alpha_z \geq 1,0$$

— For rectangular hollow sections and doubly symmetric welded box sections:

$$\alpha_y = \alpha_z = \frac{1,66}{1 - 1,13n^2} \quad \text{when } n \leq 0,8$$

$$\alpha_y = \alpha_z = 6,0 \quad \text{when } n > 0,8$$

$$\text{but: } \alpha_y = \alpha_z \leq 2 + 4 \left[1 - \max \{ \beta_{ep,y}; \beta_{ep,z} \} \right]^4$$

α_y and α_z may conservatively be taken as 1,0.

— For circular hollow sections:

$$\alpha_y = 2; \quad \alpha_z = 2$$

— For elliptical hollow sections:

$$\alpha_y = 2; \quad \alpha_z = 1,7$$

B.4 Buckling resistance of members

(1) The stability of members in bending or bending and axial force should be verified using the rules of 8.3.2 or 8.3.3, respectively. For these checks, the elasto-plastic section modulus W_{ep} as well as the interaction factors k_{yy} , k_{yz} , k_{zz} and k_{zy} from 8.3.3 for plastic design may be used.

Annex C
(normative)

Additional rules for uniform members with mono-symmetric cross-sections
and for members in bending, axial compression and torsion

C.1 Additional rules for uniform members with mono-symmetric cross-section

- (1) This Clause gives additional rules for the application of the criteria in Formulae (8.88) and (8.89) in 8.3.3 to the design against buckling of members with mono-symmetric I-, H- and welded box sections, which have flanges of different size and are symmetric about the z-z axis.
- (2) This section applies to members in bending and axial compression with Class 1, 2, 3 and 4 cross-sections.
- (3) The elastic critical moment M_{cr} , the relative slenderness $\bar{\lambda}_{LT}$ for lateral torsional buckling and the corresponding reduction factor χ_{LT} relate to the compression flange due to bending about y-y axis, which can be either the smaller or the larger flange, see Figure C.1. In members with bending moment diagrams of changing sign along the member length, two separate sets of values of M_{cr} , $\bar{\lambda}_{LT}$ and χ_{LT} should be determined for each maximum absolute value of the bending moment $M_{y,Ed}$ along the member. In this case, M_{cr} may be determined by multiplying the respective value of $M_{y,Ed,i}$ with the critical load amplifier α_{cr} for lateral torsional buckling in the member.

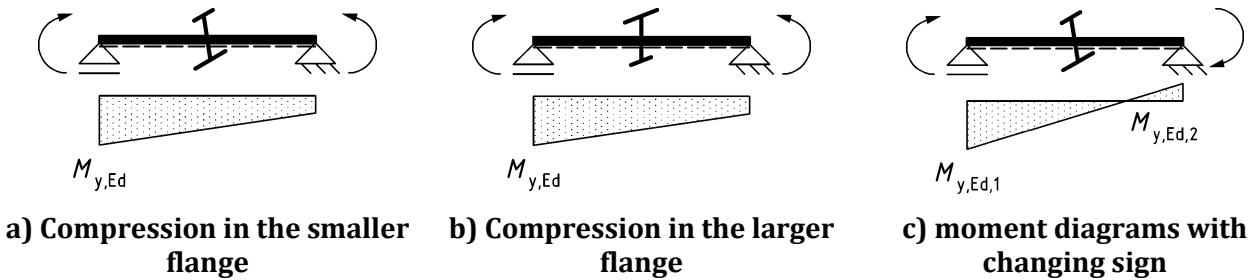


Figure C.1 — Direction of bending with respect to the section shape in mono-symmetric sections

- (4) For elastic resistance, the characteristic value of the bending moment resistance $M_{y,Rk}$ should be determined based on the first yield of the compression flange.
- (5) Irrespective of the susceptibility to lateral torsional buckling of the member, criterion in Formula (8.88) in 8.3.3 should be applied using the modified factor k_{yy} defined in Table C.1.

Table C.1 — Modified interaction factors k_{yy} in criterion in Formula (8.88) for mono-symmetric sections

Plastic cross-sectional properties Class 1, Class 2		Elastic cross-sectional properties Class 3, Class 4	
For $\bar{\lambda}_y < 1,0$:	$k_{yy} = C_{my} \left[1 + 2 \left(\bar{\lambda}_y - 0,2 \right) n_y \right]$	For $\bar{\lambda}_y < 1,0$:	$k_{yy} = C_{my} \left(1 + \bar{\lambda}_y n_y \right)$
For $\bar{\lambda}_y \geq 1,0$:	$k_{yy} = C_{my} \left(1 + 1,6 n_y \right)$	For $\bar{\lambda}_y \geq 1,0$:	$k_{yy} = C_{my} \left(1 + n_y \right)$

(6) In cases with bending moment diagrams for $M_{y,Ed}$ of changing sign along the member length, see Figure C.1 c), the criteria in Formulae (8.88) and (8.89) should be evaluated for both maximum absolute values of $M_{y,Ed}$. The equivalent uniform moment factors C_{my} and C_{mLT} of Table 8.9 should be calculated by setting negative values of ψ , α_h resp. α_s to 0,0 and disregarding the portion of opposite sign of the moment diagram. Alternatively, C_{my} and C_{mLT} may conservatively be set equal to 1,0.

(7) For members which are not susceptible to lateral torsional buckling, χ_{LT} should be set equal to 1,0 in the criteria in Formulae (8.88) and (8.89). The factor χ_{TF} for torsional-flexural buckling should be used instead of χ_z in criterion in Formula (8.89) if the moment of inertia $I_{z,fl}$ about z-z of both flanges differs by more than 50%. $I_{z,fl}$ should be calculated by considering the flanges as isolated plates or sections.

(8) For members which are susceptible to lateral torsional buckling, the buckling factor χ_z should be replaced by χ_{TF} for torsional-flexural buckling in criterion in Formula (8.89) when the bending moment $M_{y,Ed}$ causes compression in the smaller flange. The factor χ_z for flexural buckling about z-z axis should be used in criterion in Formula (8.89) when the bending moment $M_{y,Ed}$ causes compression in the larger flange. In this case, the resistance to torsional-flexural buckling should be verified by an additional condition, using χ_{TF} in criterion in Formula (8.89) and omitting the term with $M_{y,Ed}$.

C.2 Additional rules for uniform members in bending, axial compression and torsion

(1) This Clause should be used to check the buckling resistance of a simply supported, uniform I-section member in bending, axial compression and torsion, with equal flanges or with unequal flanges provided that the ratio of the moments of inertia about z-z axis of the flanges is higher or equal to 0,2 and smaller or equal to 5,0. It is intended to be used in combination with the rules in 8.3 and C.1.

(2) Members with Class 1, 2 and 3 cross-sections and a ratio of the maximum bimoment along the member B_{Ed}/B_{Rk} smaller or equal to 0,3 are covered by the rules given in C.2, where B_{Ed} and B_{Rk} are defined in (3).

(3) Unless second order analysis is carried out using imperfections, the stability of uniform members with Class 1, 2 and 3 cross-sections that are subjected to combined axial compression, bending and torsion should be checked according to the criteria in Formulae (C.1) and (C.2):

$$\frac{N_{Ed}}{\chi_y N_{Rk}} + k_{yy} \frac{M_{y,Ed}}{\chi_{LT} \frac{M_{y,Rk}}{\gamma_{M1}}} + k_{yz} \frac{M_{z,Ed}}{\frac{M_{z,Rk}}{\gamma_{M1}}} + k_w k_{zw} k_\alpha \frac{B_{Ed}}{\frac{B_{Rk}}{\gamma_{M1}}} \leq 1,0 \quad (C.1)$$

$$\frac{N_{Ed}}{\chi_z N_{Rk}} + k_{zy} \frac{M_{y,Ed}}{\chi_{LT} \frac{M_{y,Rk}}{\gamma_{M1}}} + k_{zz} \frac{M_{z,Ed}}{\frac{M_{z,Rk}}{\gamma_{M1}}} + k_w k_{zw} k_\alpha \frac{B_{Ed}}{\frac{B_{Rk}}{\gamma_{M1}}} \leq 1,0 \quad (C.2)$$

where

N_{Ed} , $M_{y,Ed}$, $M_{z,Ed}$ are the design values of the compressive axial force and the maximum moments about the y-y and z-z axis along the member, respectively;

N_{Rk} , $M_{y,Rk}$, $M_{z,Rk}$ are the characteristic values of the cross-sectional resistance to compressive axial force and bending moment about its y-y and z-z axis respectively;

B_{Ed}	is the maximum design value of the bimoment along the member;
B_{Rk}	is the characteristic value of the bimoment resistance;
χ_y and χ_z	are the flexural buckling reduction factors according to 8.3.1. The reduction factor χ_z should be replaced by χ_{TF} taken from 8.3.1.4 whenever torsional-flexural buckling is relevant;
χ_{LT}	is the reduction factor for lateral torsional buckling according to 8.3.2;
k_{yy} , k_{yz} , k_{zy} and k_{zz}	are the interaction factors according to 8.3.3. For mono-symmetric I-sections, the additional rules of C.1 should be applied;

$$k_w = 0,7 - \frac{0,2 B_{Ed}}{\frac{B_{Rk}}{\gamma_{M1}}} \quad (C.3)$$

$$k_{zw} = 1,0 - \frac{M_{z,Ed}}{\frac{M_{z,Rk}}{\gamma_{M1}}} \quad (C.4)$$

$$k_\alpha = \frac{1}{1 - \frac{M_{y,Ed}}{M_{cr}}} \quad \text{but } k_\alpha \leq 2,0 \quad (C.5)$$

M_{cr} is the elastic critical moment about the y-y-axis.

(4) For Class 3 cross-sections with unequal flanges, the characteristic value of the bending moment resistance $M_{z,Rk}$ and the bimoment resistance B_{Rk} should be determined for the smaller flange.

(5) In cases with bending moment diagrams for $M_{y,Ed}$ and $M_{z,Ed}$ of changing sign along the member length, see Figure C.1 c), the criteria in Formulae (C.1) and (C.2) should be calculated for both maximum absolute values of $M_{y,Ed}$ and $M_{z,Ed}$. The equivalent uniform moment factors C_m may be determined using the formulae in Table 8.9.

(6) The additional term for warping moments may be disregarded for the interaction criteria in Formulae (C.1) and (C.2) if the following condition in Formula (C.6) is fulfilled:

$$k_\alpha \frac{B_{Ed}}{B_{Rk} / \gamma_{M1}} \leq 0,07 \quad (C.6)$$

Or conservatively,

$$\frac{B_{Ed}}{B_{Rk} / \gamma_{M1}} \leq 0,035 \quad (C.7)$$

(7) In addition, the resistance of the cross-section of the member should satisfy the requirements given in 8.2.

NOTE As a simplification, the linear plastic interaction expressed in criterion in Formula (C.8) can be used for members with Class 1 and 2 cross-sections:

$$\frac{N_{Ed}}{N_{Rd}} + \frac{M_{y,Ed}}{M_{y,Rd}} + \frac{M_{z,Ed}}{M_{z,Rd}} + \frac{B_{Ed}}{B_{Rd}} \leq 1,0 \quad (C.8)$$

where

$$B_{Rd} = \frac{W_{pl,B} f_y}{\gamma_{M0}} \quad (C.9)$$

$W_{pl,B}$ is the plastic section modulus for the plastic resistance to bimoment, see 8.2.7(7).

(8) For cross-sections of class 3, Annex B may be considered.

Annex D (normative)

Continuous restraint of beams in buildings

D.1 Scope and field of application

(1)B This Annex provides conditions on stiffness of restraint devices of beams in buildings. Two cases are considered: continuous lateral restraint by diaphragm effect (see D.2) or continuous torsional restraint (see D.3).

D.2 Continuous lateral restraints

(1)B In buildings, if trapezoidal sheeting according to EN 1993-1-3 is connected to a beam and the condition expressed by Formula (D.1) is met, the beam at the connection may be regarded as being laterally restrained in the plane of the sheeting.

$$S_v \geq \left(EI_w \frac{\pi^2}{L^2} + GI_T + EI_z \frac{\pi^2}{L^2} 0,25h^2 \right) \frac{70}{h^2} \quad (D.1)$$

where

S_v is the shear stiffness (per unit of beam length) provided by the sheeting to the beam, regarding its deformation in the plane of the sheeting to be connected to the beam at the bottom at each rib;

I_w is the warping constant;

I_T is the torsion constant;

I_z is the moment of inertia of the cross-section about the minor axis of the cross-section;

L is the beam span;

h is the depth of the beam.

If the sheeting is connected to a beam at every second rib only, S_v should be substituted by $0,20 S_v$.

(2)B The condition in Formula (D.1) may also be used to determine the lateral stability of beam flanges used in combination with other types of cladding than trapezoidal sheeting, provided that the connections are of suitable design.

D.3 Continuous torsional restraints

(1)B A beam may be considered as sufficiently restrained from torsional deformations if the condition in Formula (D.2) is satisfied:

$$C_D > \frac{M_{pl,Rk}^2}{EI_z} K_\theta K_v$$

(D.2)

where

- C_D

rotational stiffness (per unit of beam length) provided to the beam by the stabilizing continuum (e.g. roof structure) and the connections;
- K_v

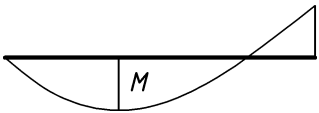
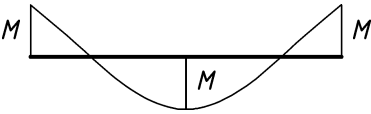
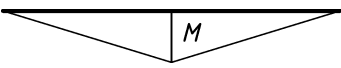
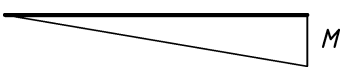
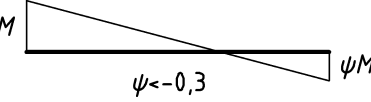
= 0,35 for elastic resistance of the cross-section;
- K_v

= 1,00 for plastic resistance of the cross-section;
- K_θ

factor for considering the moment distribution, depending on the type of restraint, the type of section and the ratio of the height h to the width b of the cross-section, according to Table D.1;
- $M_{pl,Rk}$

characteristic value of the plastic bending moment resistance of the beam.

Table D.1 — Factor K_θ for considering the moment distribution and the type of restraint

Case	Moment distribution	Type of section	Without lateral restraint		With lateral restraint	
			$h/b \leq 2,0$	$h/b > 2,0$	$h/b \leq 2,0$	$h/b > 2,0$
1		Rolled	11,0	21,4	0	0
		Welded	36,0	60,5	0	0
2		Rolled	8,8	15,5	0,073	0,16
		Welded	29,9	51,7	0,23	0,63
3		Rolled	8,0	17,4	0,085	0,19
		Welded	27,8	48,6	0,28	0,74
4		Rolled	3,2	7,7	0	0
		Welded	9,7	19,3	0	0
5		Rolled	0,73	1,4	0,60	0,75
		Welded	1,5	2,7	1,0	1,8
6		Rolled	0,44	0,77	0,35	0,49
		Welded	0,87	1,5	0,56	0,99

(2)B The rotational stiffness provided by the stabilizing continuum to the beam may be determined from:

$$\frac{1}{C_D} = \frac{1}{C_{D,C}} + \frac{1}{C_{D,B}} + \frac{1}{C_{D,A}} \quad (D.3)$$

where

$C_{D,C}$ rotational stiffness (per unit of the beam length) provided by the stabilizing continuum to the beam assuming a stiff connection to the member;

$C_{D,B}$ rotational stiffness (per unit of the beam length) deduced from an analysis of the distortional deformations of the beam cross-sections, where the flange in compression is the free one; where the compression flange is the connected one or where distortional deformations of the cross-sections may be neglected (e.g. for usual rolled profiles)
 $C_{D,B} = \infty$;

$C_{D,A}$ rotational stiffness (per unit of the beam length) of the connection between the beam and the stabilizing continuum.

NOTE B For more information on the determination of the rotational stiffness, see EN 1993-1-3.

Annex E (informative)

Basis for the calibration of partial factors

E.1 Use of this informative annex

(1) This Annex provides additional information to that given in 8.1 on the basis for the calibration of the partial factors γ_{Mi} for buildings. The information in this Annex is not meant for direct use in design.

NOTE National choice on the application of this Informative Annex is given in the National Annex. If the National Annex contains no information, this annex does not apply.

E.2 Scope and field of application

(1) This Informative Annex provides statistical data of material and dimensional properties as used for the calibration of the partial factors γ_{Mi} for buildings.

E.3 Calibration

(1) The values of the partial factors for buildings in 8.1 are calibrated considering a reliability index taken equal to 3,8, for a reference period of 50 years for variable actions and material properties and a resistance-side weighting factor of $\alpha_R = 0,8$.

(2) The design values of resistances R_d in EN 1993-1-1 are defined as the ratio between the nominal resistance and the partial factor γ_{Mi} . The nominal resistance is evaluated by using nominal values for all basic variables, see Formula E.1.

$$R_d = \frac{R_k}{\gamma_{Mi}} = \frac{R(\underline{X}_n)}{\gamma_{Mi}} \quad (E.1)$$

(3) EN 1990:—, Annex D, “Design assisted by testing” is applied for the calibration of the partial factors γ_{Mi} for buildings. The assumed scatter bands (mean values, coefficients of variation) for material and dimensional properties are given in Table E.1 and Table E.2 respectively. For dimensional properties not specifically mentioned in Table E.2, it is assumed that the mean values are equal to the nominal values and the standard deviations are equal to half of the interval between the nominal value and the lower bound of the applicable tolerance interval in EN 1090-2 or other relevant product standards.

(4) Scatter bands of parameter values X from production may generally be assumed to be in line with the assumptions made for the calibration of γ_{Mi} values for buildings, if the characteristic value X_k and the design value X_d determined from the production statistics both match or exceed the corresponding reference values $X_{5\%}$ and $X_{0,12\%}$ in Table E.1 and Table E.2. The values X_k and X_d for the production statistics may be determined by the procedures given in EN 1990:—, Annex D. They refer to non-exceedance probabilities of 5 % and 0,12 % respectively.

NOTE The values in Table E.1 and Table E.2 represent the materials and products currently available on the European market satisfying the relevant European product standards.

Table E.1 — Assumed variability of material properties

Parameter	Steel grade	Mean value X_m	Coefficient of variation	Upper reference value $X_{5\%}$	Lower reference value $X_{0,12\%}$
Yield strength, f_y	S235, S275	$1,25 R_{eH,min}^a$	5,5 %	$1,14 R_{eH,min}^a$	$1,06 R_{eH,min}^a$
	S355, S420	$1,20 R_{eH,min}^a$	5,0 %	$1,11 R_{eH,min}^a$	$1,03 R_{eH,min}^a$
	S460	$1,15 R_{eH,min}^a$	4,5 %	$1,07 R_{eH,min}^a$	$1,00 R_{eH,min}^a$
	Above S460	$1,10 R_{eH,min}^a$	3,5 %	$1,04 R_{eH,min}^a$	$1,00 R_{eH,min}^a$
Ultimate tensile strength, f_u	S235, S275	$1,20 R_{m,min}^a$	5,0 %	$1,11 R_{m,min}^a$	$1,03 R_{m,min}^a$
	S355, S420	$1,15 R_{m,min}^a$	4,0 %	$1,08 R_{m,min}^a$	$1,02 R_{m,min}^a$
	S460 and above	$1,10 R_{m,min}^a$	3,5 %	$1,04 R_{m,min}^a$	$1,00 R_{m,min}^a$
Modulus of elasticity, E	All steel grades	210 000 N/mm ²	3,0 %	200 000 N/mm ²	192 000 N/mm ²
^a $R_{eH,min}$ and $R_{m,min}$ are the minimum yield strength R_{eH} and the lower bound of the ultimate tensile strength R_m , according to the applicable product standard, e.g. EN 10025 (all parts).					

Table E.2 — Assumed variability of dimensional properties

Dimension type	Parameter	Mean value X_m	Coefficient of variation	Upper reference value $X_{5\%}$	Lower reference value $X_{0,12\%}$
Outer dimensions of cross-section	Depth h	$1,0 h_{nom}^a$	0,9 %	$0,98 h_{nom}^a$	$0,97 h_{nom}^a$
	Width b	$1,0 b_{nom}^a$	0,9 %	$0,98 b_{nom}^a$	$0,97 b_{nom}^a$
	Outer diameter d of circular hollow section	$1,0 d_{nom}^a$	0,5 %	$0,99 d_{nom}^a$	$0,98 d_{nom}^a$
Thickness	Rolled and welded I- and H-sections: flange thickness t_f	$0,98 t_{f,nom}^a$	2,5 %	$0,95 t_{f,nom}^a$	$0,91 t_{f,nom}^a$
	Rolled and welded I- and H-sections: web thickness t_w	$1,0 t_{w,nom}^a$	2,5 %	$0,96 t_{w,nom}^a$	$0,93 t_{w,nom}^a$
	Hot rolled (seamless) or welded structural hollow sections (acc. to EN 10210 (all parts)): wall thickness t	$0,99 t_{nom}^a$	2,5 %	$0,95 t_{nom}^a$	$0,92 t_{nom}^a$
	Cold-formed sections made from coils or plates (acc. to EN 10219 (all parts)): wall thickness t	$0,99 t_{nom}^a$	2,5 %	$0,95 t_{nom}^a$	$0,92 t_{nom}^a$
	All other welded sections made from heavy plates: thickness t	$0,99 t_{nom}^a$	2,5 %	$0,95 t_{nom}^a$	$0,92 t_{nom}^a$
^a Nominal dimensions according to the applicable product standard or specification.					

Bibliography

References contained in recommendations (i.e. through “should” clauses)

The following documents are referred to in the text in such a way that some or all of their content, although not requirements strictly to be followed, constitutes highly recommended choices or course of action of this document. Subject to national regulation and/or any relevant contractual provisions, alternative standards could be used/adopted where technically justified. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 1993 (all parts), *Eurocode 3: Design of steel structures*

EN 1998 (all parts), *Eurocode 8: Design of structures for earthquake resistance*

EN 10025 (all parts), *Hot rolled products of structural steels*

EN 10149 (all parts), *Hot rolled flat products made of high yield strength steels for cold forming*

EN 10164, *Steel products with improved deformation properties perpendicular to the surface of the product — Technical delivery conditions*

EN 10210 (all parts), *Hot finished structural hollow sections of non-alloy and fine grain steels*

EN 10219 (all parts), *Cold formed welded structural hollow sections of non-alloy and fine grain steels*

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EN 10365, *Hot rolled steel channels, I and H sections — Dimensions and masses*

Other references

The following documents are those not included in the above categories but are cited informatively in the document, for example in notes.

EN ISO 1461, *Hot dip galvanized coatings on fabricated iron and steel articles — Specifications and test methods (ISO 1461)*

EN ISO 12944 (all parts), *Paints and varnishes — Corrosion protection of steel structures by protective paint systems*

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prCEN/TR 1993-1-103⁶, *Eurocode 3 — Design of steel structures — Part 1-103: Elastic critical buckling of members*

⁶ Under preparation.

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